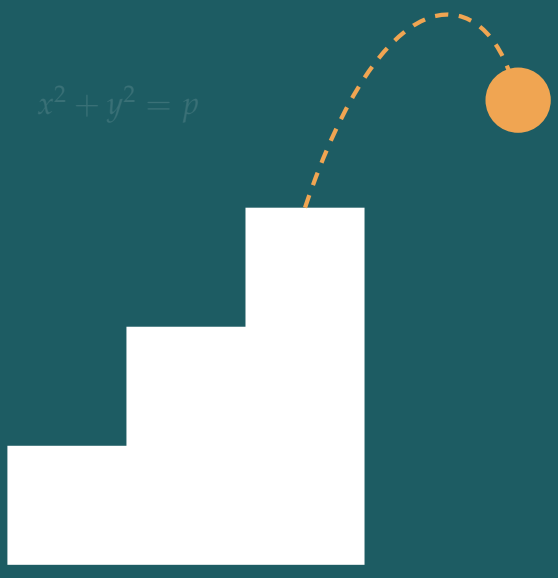
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP I, II AND III

2013 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP I, II AND III

2013 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

2013 Worked Solutions

Worked solutions to all 39 questions of the 2013 STEP papers (STEP I, II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

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WORKED SOLUTION

STEP I 2013, Question 1

Pure • Equations with Square Roots

STEP I 2013, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s1-q1

The key idea

Each equation is a quadratic in disguise. Call the square root y, write everything else in terms of y, and you get an ordinary quadratic in y. The one thing to watch is that a square root is never negative, so $y \geq 0$: that condition is what throws away the spurious root each time.

Solution

Part (i)

With $y = \sqrt{x}$, $y \geq 0$, we have $x = y^2$, and the equation becomes

$$y^2 + 3y - \tfrac{1}{2} = 0 \quad \Longleftrightarrow \quad 2y^2 + 6y - 1 = 0.$$

By the quadratic formula,

$$y = \frac{-6 \pm \sqrt{36+8}}{4} = \frac{-3 \pm \sqrt{11}}{2}.$$

The minus sign gives a negative value, which is impossible for a square root. Since $\sqrt{11} > 3$, the plus sign gives $y = \frac{1}{2}(\sqrt{11} - 3) > 0$, so this is the only option. Then

$$x = y^2 = \frac{11 - 6\sqrt{11} + 9}{4} = \frac{10 - 3\sqrt{11}}{2}.$$

Every step is reversible once we know $y \geq 0$, so this is indeed the (only) real root.

Part (ii)(a)

Here the square root is of $x+2$, so we put $y = \sqrt{x+2}$, $y \geq 0$, giving $x = y^2 - 2$. The equation becomes

$$y^2 - 2 + 10y - 22 = 0 \quad \Longleftrightarrow \quad y^2 + 10y - 24 = 0 \quad \Longleftrightarrow \quad (y+12)(y-2) = 0.$$

We reject $y = -12$ because $y \geq 0$, so $y = 2$ and $x = 2^2 - 2 = 2$.

Check: $2 + 10\sqrt{4} - 22 = 0$. The only real root is $x = 2$.

Ask yourself What would have happened if we had kept $y = -12$?

It gives $x = 144 - 2 = 142$, and then $142 + 10\sqrt{144} - 22 = 240 \neq 0$. The value $y = -12$ is a root of the related equation $x - 10\sqrt{x+2} - 22 = 0$, which is what you'd be solving if you squared things up carelessly. The condition $y \geq 0$ is exactly what separates the two.

Part (ii)(b)

This time no substitution is suggested, so we look for one. The expression under the root is

$$2x^2 - 8x - 3 = 2(x^2 - 4x) - 3,$$

and $x^2 - 4x$ also appears outside the root. That's the link.

Ask yourself What should y be here: $x^2 - 4x$, or the whole square root?

Taking y to be the square root makes the equation a quadratic with no roots left in it. Let $y = \sqrt{2x^2 - 8x - 3}$, with $y \geq 0$. Then $y^2 = 2(x^2 - 4x) - 3$, so

$$x^2 - 4x = \frac{y^2 + 3}{2}.$$

Substituting this into the equation,

$$\begin{aligned} \frac{y^2+3}{2} + y - 9 = 0 \quad &\iff \quad y^2 + 2y - 15 = 0 \\ &\iff \quad (y+5)(y-3) = 0. \end{aligned}$$

Since $y \geq 0$, we need $y = 3$. Then

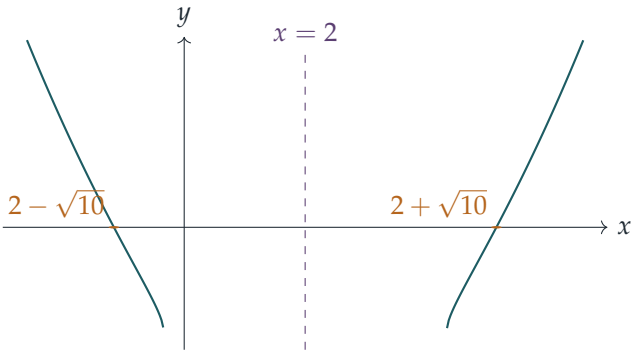
$$2x^2 - 8x - 3 = 9 \quad \iff \quad x^2 - 4x - 6 = 0 \quad \iff \quad x = 2 \pm \sqrt{10}.$$

Both values make $2x^2 - 8x - 3$ equal to 9, which is positive, so the square root is defined and equals 3. Hence both are genuine roots: the real roots are

$$x = 2 + \sqrt{10} \quad \text{and} \quad x = 2 - \sqrt{10}.$$

The pair is symmetric about $x = 2$, as it should be, because the equation depends on x only through $x^2 - 4x = (x-2)^2 - 4$.

The graph of the left-hand side shows this. It is defined only where $2x^2 - 8x - 3 \geq 0$, that is, for $x \leq 2 - \sqrt{5.5}$ or $x \geq 2 + \sqrt{5.5}$, and it crosses zero once on each branch.



Checks

- Part (i): $x = \frac{1}{2}(10 - 3\sqrt{11}) \approx 0.02506$, so $\sqrt{x} \approx 0.1583$ and $x + 3\sqrt{x} \approx 0.02506 + 0.47494 = 0.5$.
- Part (ii)(b): $x = 2 + \sqrt{10} \approx 5.1623$ gives $x^2 - 4x = 6$ and $2x^2 - 8x - 3 = 9$, so the left-hand side is $6 + 3 - 9 = 0$. The rejected value $y = -5$ would need $x^2 - 4x = 14$, and then the left-hand side is $14 + 5 - 9 = 10 \neq 0$.
- A numerical scan of each left-hand side over its whole domain finds exactly the roots above and no others.

Common slips

- Keeping the negative root of the quadratic in y. A square root is never negative, so that root does not give a solution.
- Squaring both sides straight away in (ii)(b). This gives a quartic, which is much harder, and it brings in extra roots that you would then have to test.
- In (ii)(a), forgetting that $x = y^2 - 2$, not y^2.

Beyond STEP

Introducing a new variable so that a hard equation becomes a familiar one is one of the most useful habits in mathematics. In (ii)(b) the equation depends only on $x^2 - 4x$, so it is a function of $(x - 2)^2$. Spotting a symmetry like this, and working with the quantity it leaves unchanged, comes up again in Galois theory, where you solve polynomial equations by finding expressions in the roots that are fixed by symmetries. Allenby's *Rings, Fields and Groups* is a gentle route into these ideas. For more problems of this kind, Siklos's *Advanced Problems in Mathematics* is excellent (and free).

WORKED SOLUTION

STEP I 2013, Question 2

Pure • Floor Function Equations

STEP I 2013, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s1-q2

The key idea

The function $\lfloor x \rfloor$ is constant on each interval $[k, k+1)$, so on that interval $f(x) = k/x$ is just a piece of a rectangular hyperbola. The whole graph is a string of these pieces, one for each integer k. Solving $f(x) = c$ then comes down to asking which pieces reach height c. On the piece for k the only candidate is $x = k/c$, and it counts precisely when it lands in $[k, k+1)$.

Solution

Part (i)

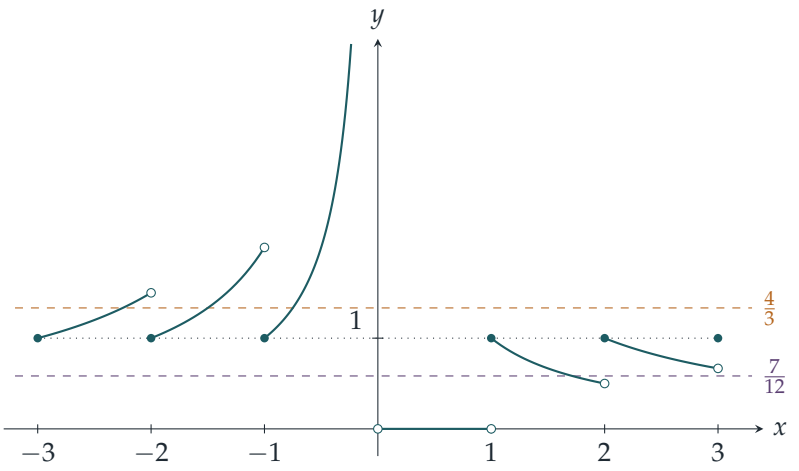
Fix an integer k and look at $k \le x < k+1$ (with $x \neq 0$), where $f(x) = k/x$.

- $k = 0$, i.e. $0 < x < 1$: here $f(x) = 0$.
- $k \ge 1$: $f(x) = k/x$ decreases from $f(k) = 1$ towards $k/(k+1)$, which it never reaches. So this piece takes exactly the values in $\left(\frac{k}{k+1}, 1\right]$.
- $k \le -1$: write $k = -m$ with $m \ge 1$. Then x and k are both negative, so $f(x) = m/|x| > 0$. As x increases from $-m$ towards $-m+1$, $|x|$ decreases, so f increases from $f(-m) = 1$. For $m \ge 2$ it tends to $\frac{m}{m-1}$, which is never reached, so this piece takes exactly the values in $\left[1, \frac{m}{m-1}\right)$. For $m = 1$ we have $f(x) = -1/x \to +\infty$ as $x \to 0^-$, so this piece takes the values in $[1, \infty)$.

For $-3 \le x \le 3$ the pieces are

x	$[-3,-2)$	$[-2,-1)$	$[-1,0)$	$(0,1)$	$[1,2)$	$[2,3)$
$f(x)$	$-3/x$	$-2/x$	$-1/x$	0	$1/x$	$2/x$
values	$[1, \frac{3}{2})$	$[1,2)$	$[1,\infty)$	0	$(\frac{1}{2}, 1]$	$(\frac{2}{3}, 1]$

and finally $f(3) = 1$. Every piece starts at height 1 at its left-hand (integer) end, where the point is included, and has an open end on the right.



Part (ii)

On the piece where $\lfloor x \rfloor = k$, the equation $f(x) = c$ (with $c \neq 0$) says $k/x = c$, so $x = k/c$. This is a genuine root exactly when $k \leq k/c < k+1$. So for each c we only have to decide which pieces reach height c, using the ranges from part (i).

$f(x) = \frac{7}{12}$. Since $0 < \frac{7}{12} < 1$, only pieces with $k \geq 1$ can help, and we need $\frac{k}{k+1} < \frac{7}{12}$, i.e. $12k < 7k + 7$, i.e. $k < \frac{7}{5}$. So only $k = 1$ works, giving $x = 1/\frac{7}{12}$:

$$x = \tfrac{12}{7}.$$

On the graph, the line $y = \frac{7}{12}$ meets only the piece over $[1, 2)$, since the piece over $[2, 3)$ stays above $\frac{2}{3}$.

$f(x) = \frac{17}{24}$. Now we need $\frac{k}{k+1} < \frac{17}{24}$, i.e. $24k < 17k + 17$, i.e. $k < \frac{17}{7}$, so $k = 1$ or $k = 2$. The roots are $x = k/\frac{17}{24}$:

$$x = \tfrac{24}{17} \quad \text{and} \quad x = \tfrac{48}{17}.$$

(Indeed $2 \leq \frac{48}{17} \approx 2.82 < 3$.)

$f(x) = \frac{4}{3}$. Since $\frac{4}{3} > 1$, only the negative pieces can help. The piece with $k = -m$ reaches $\frac{4}{3}$ when $\frac{4}{3} < \frac{m}{m-1}$ (or $m = 1$), i.e. $4m - 4 < 3m$, i.e. $m < 4$. So $m = 1, 2, 3$, and the roots are $x = -m/\frac{4}{3} = -\frac{3m}{4}$:

$$x = -\tfrac{3}{4}, \quad x = -\tfrac{3}{2}, \quad x = -\tfrac{9}{4}.$$

The case $m = 4$ just fails: the piece over $[-4, -3)$ only gets up to values below $\frac{4}{3}$.

Part (iii)

Here $0 < \frac{9}{10} < 1$, so the roots come from the pieces with $k \geq 1$ and $\frac{k}{k+1} < \frac{9}{10}$, i.e. $10k < 9k + 9$, i.e. $k < 9$. So $k = 1, 2, \ldots, 8$ each give a root $x = \frac{10k}{9}$. The largest is the one with $k = 8$:

$$x = \tfrac{80}{9} \approx 8.89,$$

which does lie in $[8, 9)$.

The final part

First, the values of c that give no roots or infinitely many.

- $c < 0$: no roots, because $f(x) \geq 0$ for $x > 0$ and $f(x) > 0$ for $x < 0$.
- $c = 0$: every $x \in (0, 1)$ is a root, so there are infinitely many.
- $c = 1$: every nonzero integer is a root, so again infinitely many.

> **Ask yourself** How do we count roots for a general c, without drawing the graph?

Use the ranges from part (i): each piece contributes at most one root, so we just count the pieces whose range contains c.

Case $0 < c < 1$**.** The pieces that reach c are those with $k \geq 1$ and $\frac{k}{k+1} < c$. Since $\frac{k}{k+1}$ increases with k, these are $k = 1, 2, \ldots$ up to the last k with $\frac{k}{k+1} < c$. So there are exactly n roots precisely when the nth piece reaches c but the $(n+1)$th does not:

$$\frac{n}{n+1} < c \leq \frac{n+1}{n+2}.$$

(For $0 < c \leq \frac{1}{2}$ no piece reaches c, and there are no roots.)

Case $c > 1$**.** The pieces that reach c are those with $k = -m$, where either $m = 1$ or $\frac{m}{m-1} > c$. Since $\frac{m}{m-1} = 1 + \frac{1}{m-1}$ decreases as m increases, these are $m = 1, 2, \ldots$ up to the last m with $\frac{m}{m-1} > c$. So there are exactly n roots precisely when the piece for $m = n$ reaches c but the piece for $m = n + 1$ does not:

$$\frac{n+1}{n} \leq c < \frac{n}{n-1} \qquad (n \geq 2),$$

and, for $n = 1$, simply $c \geq 2$ (the piece over $[-1, 0)$ always reaches c, and the piece over $[-2, -1)$ doesn't when $c \geq 2$).

Answer. For $n \geq 1$, the equation $f(x) = c$ has exactly n roots if and only if

$$\frac{n}{n+1} < c \leq \frac{n+1}{n+2} \qquad \text{or} \qquad \frac{n+1}{n} \leq c < \frac{n}{n-1},$$

where for $n = 1$ the second condition is read as $c \geq 2$. Notice where the equality signs go. Both come from open ends of pieces: $c = \frac{n+1}{n+2}$ is not reached by the piece over $[n+1, n+2)$, and $c = \frac{n+1}{n}$ is not reached by the piece over $[-(n+1), -n)$. So both of these boundary values still give exactly n roots.

Checks

- Substitute back: $\lfloor \frac{12}{7} \rfloor = 1$ and $1/\frac{12}{7} = \frac{7}{12}$; $\lfloor \frac{48}{17} \rfloor = 2$ and $2/\frac{48}{17} = \frac{17}{24}$; $\lfloor -\frac{9}{4} \rfloor = -3$ and $-3/(-\frac{9}{4}) = \frac{4}{3}$; $\lfloor \frac{80}{9} \rfloor = 8$ and $8/\frac{80}{9} = \frac{9}{10}$.
- The general answer agrees with the specific cases: $\frac{2}{3} < \frac{17}{24} \leq \frac{3}{4}$ gives $n = 2$; $\frac{4}{3} \leq \frac{4}{3} < \frac{3}{2}$ gives $n = 3$; $\frac{8}{9} < \frac{9}{10} \leq \frac{9}{10}$ gives $n = 8$. The last two sit exactly on an endpoint of the range, which tests the placing of $\leq$ and $<$.
- A boundary value: $c = 2$ should give one root. Indeed $x = -\frac{1}{2}$ works, while $x = -1$ (from $k = -2$) has $\lfloor -1 \rfloor = -1 \neq -2$.
- Try $c = \frac{1}{2}$: the conditions give no n, and indeed $x = k/c = 2k$ never satisfies $\lfloor 2k \rfloor = k$ for $k \neq 0$.

Common slips

- Forgetting that $\lfloor x \rfloor$ rounds *down* for negative x, so that $\lfloor -2.25 \rfloor = -3$, not -2. This loses the negative branch for $c > 1$ entirely.
- Getting the endpoints wrong: each piece includes its left-hand (integer) end, at height 1, but not its right-hand end. This decides whether $\leq$ or $<$ appears in the final conditions.
- Leaving out $c = 0$ and $c = 1$, where there are infinitely many roots, or $c < 0$ and $0 < c \leq \frac{1}{2}$, where there are none.

Beyond STEP

Functions such as $\lfloor x \rfloor / x$ are the simplest examples of functions that are continuous on pieces but jump at isolated points. Being careful about which endpoints are attained is exactly the habit you need for suprema and infima in a first analysis course. Here $\sup f$ over $x > 0$ is 1 and is attained, while the infimum over $x \geq 1$ is $\frac{1}{2}$ and is not. The same function also shows that $\lfloor x \rfloor / x \to 1$ as $x \to \infty$, a first taste of the estimates used in analytic number theory. Abbott's *Understanding Analysis* is a good place to read more about suprema, limits and continuity.

WORKED SOLUTION

STEP I 2013, Question 3

Pure • Vector Algebra

STEP I 2013, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s1-q3

The key idea

Everything here is linear, so the cleanest move every time is to write each point out as a combination of $\mathbf{x}$, $\mathbf{y}$ and $\mathbf{z}$ and then subtract. Two points are distinct exactly when the difference of their position vectors is non-zero, and that difference always turns out to be a simple factor times a vector like $\mathbf{x}-\mathbf{y}$. Geometrically, $X*Y$ is a point on the line XY, and when $0<\lambda<1$ it lies on the segment between them.

Solution

Throughout, two points are the same exactly when their position vectors are equal.

Part (i)

By definition $Y*X$ has position vector $\lambda\mathbf{y}+(1-\lambda)\mathbf{x}$, so

$$\begin{aligned}\big(\lambda\mathbf{x}+(1-\lambda)\mathbf{y}\big)-\big(\lambda\mathbf{y}+(1-\lambda)\mathbf{x}\big) &= (2\lambda-1)\mathbf{x}-(2\lambda-1)\mathbf{y}\\ &= (2\lambda-1)(\mathbf{x}-\mathbf{y}).\end{aligned}$$

Since X and Y are distinct, $\mathbf{x}-\mathbf{y}\neq\mathbf{0}$, so this difference is zero if and only if $2\lambda-1=0$. Hence $X*Y$ and $Y*X$ are distinct unless $\lambda=\frac{1}{2}$, when both are the midpoint of XY.

Part (ii)

Expanding each side,

$$\begin{aligned}(X*Y)*Z:&\quad \lambda\big(\lambda\mathbf{x}+(1-\lambda)\mathbf{y}\big)+(1-\lambda)\mathbf{z}=\lambda^2\mathbf{x}+\lambda(1-\lambda)\mathbf{y}+(1-\lambda)\mathbf{z},\\ X*(Y*Z):&\quad \lambda\mathbf{x}+(1-\lambda)\big(\lambda\mathbf{y}+(1-\lambda)\mathbf{z}\big)=\lambda\mathbf{x}+\lambda(1-\lambda)\mathbf{y}+(1-\lambda)^2\mathbf{z}.\end{aligned}$$

The $\mathbf{y}$ terms agree, and subtracting,

$$\begin{aligned}\text{difference} &= (\lambda^2-\lambda)\mathbf{x} + \big((1-\lambda)-(1-\lambda)^2\big)\mathbf{z}\\ &= -\lambda(1-\lambda)\mathbf{x} + \lambda(1-\lambda)\mathbf{z}\\ &= \lambda(1-\lambda)(\mathbf{z}-\mathbf{x}).\end{aligned}$$

This is non-zero exactly when each factor is non-zero. So $(X*Y)*Z$ and $X*(Y*Z)$ are distinct if and only if

$$\lambda \neq 0, \qquad \lambda \neq 1 \qquad \text{and} \qquad Z \neq X.$$

Notice that Y plays no part. The excluded values make sense: with $\lambda = 1$ the operation always returns the first point, and with $\lambda = 0$ it always returns the second, so both bracketings give X or Z respectively.

Part (iii)

Write $\mu = 1-\lambda$ to keep things tidy. The right-hand side is

$$\begin{aligned}(X*Z)*(Y*Z): \quad &\lambda(\lambda\mathbf{x}+\mu\mathbf{z}) + \mu(\lambda\mathbf{y}+\mu\mathbf{z})\\ &= \lambda^2\mathbf{x} + \lambda\mu\,\mathbf{y} + \mu(\lambda+\mu)\mathbf{z}\\ &= \lambda^2\mathbf{x} + \lambda\mu\,\mathbf{y} + \mu\,\mathbf{z},\end{aligned}$$

since $\lambda+\mu = 1$. This is exactly the expansion of $(X*Y)*Z$ found in part (ii), so

$$(X*Y)*Z = (X*Z)*(Y*Z)$$

for all points X, Y, Z, as required.

? **Ask yourself** What should the "corresponding result" for $X*(Y*Z)$ look like?

In part (iii) the outer $*Z$ is distributed over the bracket on the left. The mirror image is to distribute $X*$ over the bracket on the right, which suggests trying $(X*Y)*(X*Z)$. Indeed,

$$\begin{aligned}(X*Y)*(X*Z): \quad &\lambda(\lambda\mathbf{x}+\mu\mathbf{y}) + \mu(\lambda\mathbf{x}+\mu\mathbf{z})\\ &= \lambda(\lambda+\mu)\mathbf{x} + \lambda\mu\,\mathbf{y} + \mu^2\mathbf{z}\\ &= \lambda\mathbf{x} + \lambda\mu\,\mathbf{y} + \mu^2\mathbf{z},\end{aligned}$$

which is the expansion of $X*(Y*Z)$ from part (ii). So, for any points X, Y, Z,

$$X*(Y*Z) = (X*Y)*(X*Z).$$

Part (iv)

We claim that P_n has position vector

$$\mathbf{p}_n = \lambda^n\mathbf{x} + (1-\lambda^n)\mathbf{y}.$$

For $n = 1$ this is the definition of $X * Y$. If it holds for $n-1$, then

$$\begin{aligned}\mathbf{p}_n = \lambda\mathbf{p}_{n-1} + (1-\lambda)\mathbf{y} &= \lambda^n\mathbf{x} + \big(\lambda - \lambda^n + 1 - \lambda\big)\mathbf{y} \\ &= \lambda^n\mathbf{x} + (1-\lambda^n)\mathbf{y},\end{aligned}$$

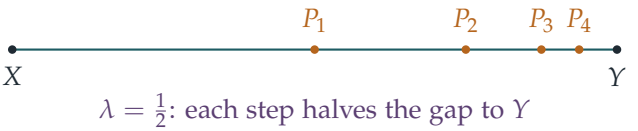
so by induction the formula holds for all $n \geq 1$.

Now measure along the segment:

$$\mathbf{p}_n - \mathbf{x} = (1-\lambda^n)(\mathbf{y}-\mathbf{x}), \qquad \mathbf{y} - \mathbf{p}_n = \lambda^n(\mathbf{y}-\mathbf{x}).$$

Both are positive multiples of the same non-zero vector $\mathbf{y} - \mathbf{x}$, because $0 < \lambda < 1$ gives $0 < \lambda^n < 1$. So P_n lies strictly between X and Y, and

$$XP_n : P_nY = (1-\lambda^n) : \lambda^n.$$



$\lambda = \frac{1}{2}$: each step halves the gap to Y

Each step multiplies the remaining distance to Y by λ, so the points march towards Y, and $P_n \to Y$ as $n \to \infty$.

Checks

- In part (i), take $\mathbf{x} = \mathbf{0}$, $\mathbf{y} = \mathbf{1}$ along a line and $\lambda = \frac{1}{3}$: $X * Y$ is at $\frac{2}{3}$ and $Y * X$ at $\frac{1}{3}$, which differ by $\frac{1}{3} = |2\lambda - 1|$.
- In part (ii), with $Z = X$ both bracketings should agree for any λ; our difference $\lambda(1-\lambda)(\mathbf{z}-\mathbf{x})$ vanishes then, as it should.
- In part (iv) with $\lambda = \frac{1}{2}$ and $n = 3$ the ratio is $\frac{7}{8} : \frac{1}{8} = 7 : 1$, and indeed P_3 is at $\frac{7}{8}$ of the way from X to Y (halving the gap three times). With $n = 1$ the ratio $(1-\lambda) : \lambda$ is just the definition of $X * Y$.

Common slips

- Forgetting $Z \neq X$ in part (ii): the answer is not just a condition on λ.
- Giving the ratio the wrong way round in part (iv). Since $\mathbf{p}_n$ carries weight λ^n on $\mathbf{x}$, it is *close* to X when λ^n is large, so the shorter piece XP_n goes with $1 - \lambda^n$.

Beyond STEP

The point $\lambda\mathbf{x} + (1-\lambda)\mathbf{y}$ is a convex (or affine) combination, the building block of convex sets, barycentric coordinates and linear interpolation in computer graphics. Part

(ii) says $*$ is not associative, while part (iii) says it is self-distributive; operations with this property are studied in algebra under names such as racks, quandles and medial algebras. For the linear-algebra viewpoint, see Strang, *Introduction to Linear Algebra*.

WORKED SOLUTION

STEP I 2013, Question 4

Pure • Integration by Substitution

STEP I 2013, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s1-q4

The key idea

In part (i) each integrand is "a function of something, times the derivative of that something": $\sec^2 x$ is the derivative of $\tan x$, and $\sec x \tan x$ is the derivative of $\sec x$. In part (ii) the extra factor of x or x^2 calls for integration by parts. The trigonometric factor is the one we integrate, and part (i) has just shown us how to do that.

Solution

Part (i)

For the first integral, substitute $u = \tan x$, so $\mathrm{d}u = \sec^2 x \,\mathrm{d}x$. When $x = 0$, $u = 0$, and when $x = \frac{1}{4}\pi$, $u = 1$. Hence

$$\int_0^{\frac{1}{4}\pi} \tan^n x \sec^2 x \,\mathrm{d}x = \int_0^1 u^n \,\mathrm{d}u = \left[\frac{u^{n+1}}{n+1}\right]_0^1 = \frac{1}{n+1},$$

using $n > 0$ so that $0^{n+1} = 0$.

Ask yourself For the second integral, which function's derivative is hiding in $\sec^n x \tan x$?

Since $\dfrac{\mathrm{d}}{\mathrm{d}x} \sec x = \sec x \tan x$, we split off one factor of $\sec x$:

$$\sec^n x \tan x = \sec^{n-1} x \,(\sec x \tan x).$$

Now substitute $u = \sec x$, so $\mathrm{d}u = \sec x \tan x \,\mathrm{d}x$, with $u = 1$ at $x = 0$ and $u = \sqrt{2}$ at $x = \frac{1}{4}\pi$. Since $n > 0$ we may divide by n, and

$$\int_0^{\frac{1}{4}\pi} \sec^n x \tan x \,\mathrm{d}x = \int_1^{\sqrt{2}} u^{n-1} \,\mathrm{d}u = \left[\frac{u^n}{n}\right]_1^{\sqrt{2}} = \frac{(\sqrt{2})^n - 1}{n},$$

as required.

Part (ii)

The first integral. Call it I_1. We integrate by parts, differentiating x and integrating $\sec^4 x \tan x$. The substitution $u = \sec x$ from part (i) gives the antiderivative

$$\int \sec^4 x \tan x \, \mathrm{d}x = \tfrac{1}{4} \sec^4 x + C.$$

(Its definite integral over $[0, \frac{1}{4}\pi]$ is $\frac{1}{4}(4-1) = \frac{3}{4}$, the case $n = 4$ of part (i).) So

$$I_1 = \left[\tfrac{1}{4} x \sec^4 x\right]_0^{\frac{1}{4}\pi} - \frac{1}{4}\int_0^{\frac{1}{4}\pi} \sec^4 x \, \mathrm{d}x = \frac{\pi}{4}\cdot\frac{4}{4} - \frac{1}{4}\int_0^{\frac{1}{4}\pi} \sec^4 x \, \mathrm{d}x,$$

because $\sec^4 \frac{1}{4}\pi = (\sqrt{2})^4 = 4$. For the remaining integral, write $\sec^2 x = 1 + \tan^2 x$:

$$\int_0^{\frac{1}{4}\pi} \sec^4 x \, \mathrm{d}x = \int_0^{\frac{1}{4}\pi} \sec^2 x \, \mathrm{d}x + \int_0^{\frac{1}{4}\pi} \tan^2 x \sec^2 x \, \mathrm{d}x = 1 + \frac{1}{3} = \frac{4}{3},$$

where the first integral is $[\tan x]_0^{\frac{1}{4}\pi} = 1$ and the second is the case $n = 2$ of part (i). Therefore

$$\int_0^{\frac{1}{4}\pi} x \sec^4 x \tan x \, \mathrm{d}x = \frac{\pi}{4} - \frac{1}{3}.$$

The second integral. Call it I_2. Again we differentiate the power of x and integrate $\sec^2 x \tan x$. One antiderivative is $\frac{1}{2}\sec^2 x$, but $\frac{1}{2}\tan^2 x$ works just as well (the two differ by the constant $\frac{1}{2}$), and it is the better choice.

? **Ask yourself** Why prefer $\frac{1}{2}\tan^2 x$ to $\frac{1}{2}\sec^2 x$?

After integrating by parts we will be left with $x \tan^2 x$, and $\tan^2 x = \sec^2 x - 1$ splits into pieces we can integrate. Also, $\tan^2 x$ vanishes at 0, which keeps the boundary terms tidy. (Using $\frac{1}{2}\sec^2 x$ gives the same answer in the end.)

So, with $\tan \frac{1}{4}\pi = 1$,

$$I_2 = \left[\tfrac{1}{2} x^2 \tan^2 x\right]_0^{\frac{1}{4}\pi} - \int_0^{\frac{1}{4}\pi} x \tan^2 x \, \mathrm{d}x = \frac{\pi^2}{32} - \int_0^{\frac{1}{4}\pi} x \tan^2 x \, \mathrm{d}x.$$

Now

$$\int_0^{\frac{1}{4}\pi} x \tan^2 x \, \mathrm{d}x = \int_0^{\frac{1}{4}\pi} x \sec^2 x \, \mathrm{d}x - \int_0^{\frac{1}{4}\pi} x \, \mathrm{d}x = \int_0^{\frac{1}{4}\pi} x \sec^2 x \, \mathrm{d}x - \frac{\pi^2}{32}.$$

Integrating by parts once more, with $\tan x$ as the antiderivative of $\sec^2 x$, and using $\int \tan x \, \mathrm{d}x = \ln \sec x$ (valid since $\sec x > 0$ on this interval):

$$\begin{aligned}\int_0^{\frac{1}{4}\pi} x \sec^2 x \, \mathrm{d}x &= \left[x \tan x\right]_0^{\frac{1}{4}\pi} - \int_0^{\frac{1}{4}\pi} \tan x \, \mathrm{d}x \\ &= \frac{\pi}{4} - \left[\ln \sec x\right]_0^{\frac{1}{4}\pi} = \frac{\pi}{4} - \ln\sqrt{2} = \frac{\pi}{4} - \frac{1}{2}\ln 2.\end{aligned}$$

Putting the pieces together,

$$I_2 = \frac{\pi^2}{32} - \left(\frac{\pi}{4} - \frac{1}{2}\ln 2 - \frac{\pi^2}{32}\right),$$

that is,

$$\int_0^{\frac{1}{4}\pi} x^2 \sec^2 x \tan x \,\mathrm{d}x = \frac{\pi^2}{16} - \frac{\pi}{4} + \frac{1}{2}\ln 2.$$

Checks

- Part (i) at $n = 1$: both formulas give $\int_0^{\frac{1}{4}\pi} \tan x \sec^2 x \,\mathrm{d}x = \frac{1}{2}$ and $\int_0^{\frac{1}{4}\pi} \sec x \tan x \,\mathrm{d}x = \sqrt{2} - 1$, which you can confirm directly from $\left[\frac{1}{2}\tan^2 x\right]$ and $[\sec x]$.
- Signs and sizes: both integrands in part (ii) are positive on $(0, \frac{1}{4}\pi)$, and so are the answers, $\frac{\pi}{4} - \frac{1}{3} \approx 0.4521$ and $\frac{\pi^2}{16} - \frac{\pi}{4} + \frac{1}{2}\ln 2 \approx 0.1780$. The second is smaller, as it should be, since there $x^2 \sec^2 x \le x \sec^4 x$ (because $x < 1 < \sec^2 x$).
- A rough bound: $x \sec^4 x \tan x \le \frac{\pi}{4} \sec^4 x \tan x$, so $I_1 \le \frac{\pi}{4} \cdot \frac{3}{4} \approx 0.589$. Our 0.452 is consistent with this. Simpson's rule with a few strips on a calculator gets close to both values.

Common slips

- Substituting $u = \sec x$ in the second integral of part (i) without first splitting off $\sec x \tan x$, which leaves the power as n rather than $n - 1$.
- Forgetting that $\sec \frac{1}{4}\pi = \sqrt{2}$, so that $\sec^4 \frac{1}{4}\pi = 4$ and $\sec^2 \frac{1}{4}\pi = 2$.
- Losing the boundary term $\left[\frac{1}{2}x^2 \tan^2 x\right]$, or getting $\int \tan x \,\mathrm{d}x$ wrong by a sign (it is $\ln \sec x = -\ln \cos x$).

Beyond STEP

The integrals in part (i) are the first steps of the standard theory of integrating $\sec^m x \tan^n x$, where reduction formulae let you lower the powers step by step. Integration by parts with a polynomial factor, as in part (ii), generalises to the "tabular" method and, much later, to the idea of moving derivatives from one function to another that underlies weak solutions of differential equations. For a careful account of why substitution and integration by parts work, see Abbott, *Understanding Analysis*; for many more problems in this style, try Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP I 2013, Question 5

Pure • Conics and Rotation

STEP I 2013, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s1-q5

The key idea

The only term that changes with k is the xy term, and it controls what kind of curve we get. With no xy term we can complete the square in x and y separately. When the quadratic part factorises, the whole equation may factorise into two straight lines. And when the quadratic part is a perfect square, $(x+y)^2$, rotating the axes by 45° turns $x+y$ into a single coordinate, leaving a familiar parabola.

Notice too that setting $x=0$ or $y=0$ kills the xy term, so the axis intercepts are the same for every k: $x^2+3x=0$ gives $x=0$ or $x=-3$, and $y^2+y=0$ gives $y=0$ or $y=-1$. Every locus passes through $(0,0)$, $(-3,0)$ and $(0,-1)$.

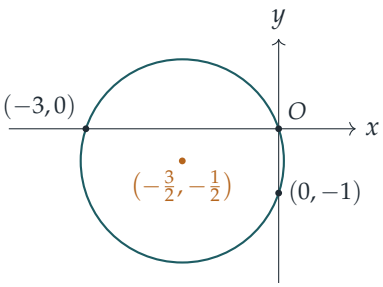
Solution

Part (i)

With $k=0$, completing the square in each variable gives

$$\left(x+\tfrac{3}{2}\right)^2+\left(y+\tfrac{1}{2}\right)^2=\tfrac{9}{4}+\tfrac{1}{4}=\tfrac{5}{2}.$$

So the locus is the circle with centre $\left(-\frac{3}{2},-\frac{1}{2}\right)$ and radius $\sqrt{5/2}\approx 1.58$. It meets the x-axis at $(0,0)$ and $(-3,0)$, and the y-axis at $(0,0)$ and $(0,-1)$. (The origin and $(-3,0)$ are the ends of a chord, so the centre lies on its perpendicular bisector $x=-\frac{3}{2}$, as it should.)



Part (ii)

With $k = \frac{10}{3}$, multiply through by 3:

$$3x^2 + 3y^2 + 10xy + 9x + 3y = 0.$$

The quadratic part factorises as

$$3x^2 + 10xy + 3y^2 = (3x + y)(x + 3y),$$

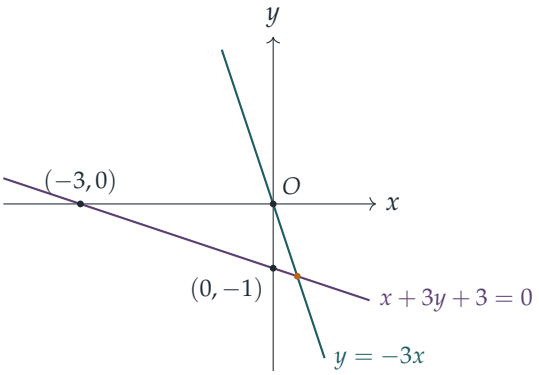
and the linear part is $9x + 3y = 3(3x + y)$, which shares the factor $3x + y$. So

$$(3x + y)(x + 3y) + 3(3x + y) = (3x + y)(x + 3y + 3) = 0.$$

A product is zero exactly when one factor is, so the locus is the pair of straight lines

$$y = -3x \qquad \text{and} \qquad x + 3y + 3 = 0.$$

The first passes through the origin (its only intersection with either axis). The second meets the x-axis at $(-3, 0)$ and the y-axis at $(0, -1)$. Solving the two together, $x - 9x + 3 = 0$, so they cross at $\left(\frac{3}{8}, -\frac{9}{8}\right)$.



Part (iii)

With $k = 2$ the equation is

$$(x + y)^2 + 3x + y = 0.$$

For $\theta = 45°$, $\cos\theta = \sin\theta = \frac{1}{\sqrt{2}}$, so

$$X = \frac{x+y}{\sqrt{2}}, \qquad Y = \frac{y-x}{\sqrt{2}}.$$

Thus $x + y = \sqrt{2}\,X$ and $y - x = \sqrt{2}\,Y$, and adding and subtracting,

$$x = \frac{X-Y}{\sqrt{2}}, \qquad y = \frac{X+Y}{\sqrt{2}}.$$

Ask yourself Why do we need x and y in terms of X and Y, rather than the other way round?

We know P lies on the curve, and we want the equation satisfied by Q. So we must substitute for x and y in the equation of P's locus, which needs them in terms of X and Y. (Because the map is a rotation, the inverse is just the rotation the other way, so this is easy.)

The linear part becomes

$$3x + y = \frac{3X - 3Y + X + Y}{\sqrt{2}} = \frac{4X - 2Y}{\sqrt{2}} = 2\sqrt{2}\,X - \sqrt{2}\,Y,$$

and $(x+y)^2 = 2X^2$. So the locus of Q is

$$2X^2 + 2\sqrt{2}\,X - \sqrt{2}\,Y = 0 \quad\Longleftrightarrow\quad \sqrt{2}\,Y = 2X^2 + 2\sqrt{2}\,X = \left(\sqrt{2}\,X + 1\right)^2 - 1,$$

as required. Every step is reversible, so this is exactly the locus of Q.

This is a parabola in the (X, Y) plane, opening in the direction of increasing Y. It is symmetric about the line where $\sqrt{2}\,X + 1 = 0$, that is $X = -\frac{1}{\sqrt{2}}$, and its vertex is at $X = -\frac{1}{\sqrt{2}}$, $Y = -\frac{1}{\sqrt{2}}$.

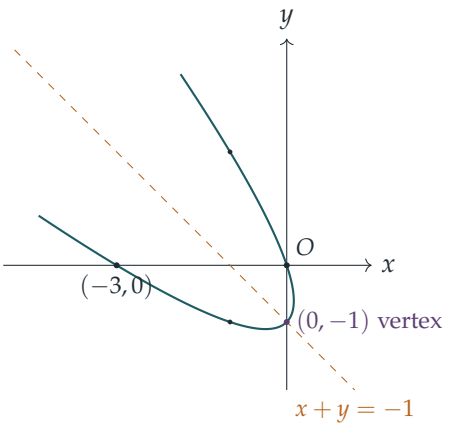
Now we undo the rotation. Q came from P by a clockwise rotation of 45°, so the locus of P is this parabola rotated 45° *anticlockwise*. In terms of x and y:

- the line of symmetry $X = -\frac{1}{\sqrt{2}}$ is $\dfrac{x+y}{\sqrt{2}} = -\dfrac{1}{\sqrt{2}}$, that is
$$x + y = -1;$$
- the vertex is at $x = \dfrac{X-Y}{\sqrt{2}} = 0$, $y = \dfrac{X+Y}{\sqrt{2}} = -1$, that is $(0, -1)$;
- the parabola opens in the direction of increasing $Y = \frac{y-x}{\sqrt{2}}$, which is the direction $(-1, 1)$: up and to the left.

The intercepts are $(0, 0)$, $(-3, 0)$ and $(0, -1)$, as noted at the start; $(0, -1)$ is the vertex. The tangent there is perpendicular to the line of symmetry, so it has gradient 1: the curve crosses the y-axis at the vertex rather than touching it. As a check on the symmetry,

reflecting in $x+y=-1$ sends (a,b) to $(-1-b,-1-a)$, so the origin goes to $(-1,-1)$ and $(-3,0)$ goes to $(-1,2)$; both satisfy $(x+y)^2+3x+y=0$.

For sketching it helps to write $u=x+y$: then $3x+y=-u^2$ and solving gives $x=-\frac{1}{2}(u^2+u)$, $y=\frac{1}{2}(u^2+3u)$.



The locus of P is a parabola with vertex $(0,-1)$ and line of symmetry $x+y=-1$.

Checks

- Substitute the intercepts $(0,0)$, $(-3,0)$, $(0,-1)$ into all three equations: each gives 0, whatever k is.
- In (ii), expand $(3x+y)(x+3y+3)$ and divide by 3 to recover the original equation with $k=\frac{10}{3}$. The crossing point $\left(\frac{3}{8},-\frac{9}{8}\right)$ satisfies both $3x+y=0$ and $x+3y+3=0$.
- In (iii), take $u=1$: the point $(-1,2)$ is on P's locus. Then $X=\frac{1}{\sqrt{2}}$, $Y=\frac{3}{\sqrt{2}}$, and $\sqrt{2}\,Y=3=(1+1)^2-1$.

Common slips

- In (iii), substituting the formulas for X,Y into the equation instead of expressing x,y in terms of X,Y.
- Rotating the parabola back the wrong way, giving the line of symmetry as $y-x=-1$ instead of $x+y=-1$.
- In (ii), missing that the linear terms share the factor $3x+y$, and treating the locus as a conic of some other kind.

Beyond STEP

Every equation $ax^2 + bxy + cy^2 + dx + ey + f = 0$ is a conic, and its type is decided by the discriminant $b^2 - 4ac$ of the quadratic part: negative gives an ellipse (here a circle, $k = 0$), zero a parabola ($k = 2$), positive a hyperbola, which can degenerate into a pair of lines ($k = \frac{10}{3}$). In a first-year linear algebra course you learn to remove the xy term systematically by rotating to the eigenvectors of the symmetric matrix of the quadratic form. Strang's *Introduction to Linear Algebra* explains this clearly.

WORKED SOLUTION

STEP I 2013, Question 6

Pure • Binomial Coefficients and Fibonacci

STEP I 2013, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s1-q6

The key idea

Pascal's rule is the whole engine. Each B_n is a sum of binomial coefficients, and if we split every term of B_{n+2} with Pascal's rule, the pieces regroup into exactly the terms of B_{n+1} and B_n. The only care needed is at the ends of the sums, where Pascal's rule doesn't apply and a lone 1 has to be matched up by hand. Because B_n is defined differently for even and odd n, we do the two cases separately.

Solution

Pascal's rule

Since $(1+x)^{n+1} = (1+x)(1+x)^n$, the coefficients of x^r on the two sides agree. On the left the coefficient is $\binom{n+1}{r}$. On the right,

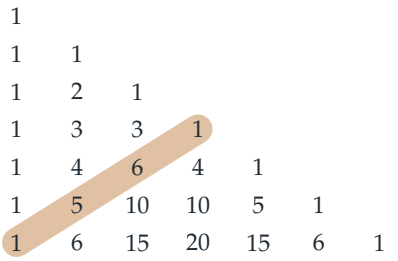
$$(1+x)\sum_{s=0}^{n}\binom{n}{s}x^s = \sum_{s=0}^{n}\binom{n}{s}x^s + \sum_{s=0}^{n}\binom{n}{s}x^{s+1},$$

and for $1 \le r \le n$ the first sum contributes $\binom{n}{r}$ (from $s=r$) and the second contributes $\binom{n}{r-1}$ (from $s=r-1$). Hence

$$\binom{n}{r} + \binom{n}{r-1} = \binom{n+1}{r} \qquad (1 \le r \le n).$$

The recurrence

It helps to see what B_n is: it adds up a "shallow diagonal" of Pascal's triangle, $\binom{n}{0} + \binom{n-1}{1} + \binom{n-2}{2} + \cdots$, stopping at the last term whose top is at least its bottom. With the triangle left-justified, the terms making up $B_6 = 1+5+6+1 = 13$ lie on a straight line:



(The amber band picks out $\binom{6}{0}, \binom{5}{1}, \binom{4}{2}, \binom{3}{3}$.)

Even case, $n = 2m$ $(m \geq 0)$. We need $B_{2m+2} = B_{2m+1} + B_{2m}$. By definition

$$B_{2m+2} = \sum_{j=0}^{m+1} \binom{2m+2-j}{j}.$$

The end terms are $j = 0$, giving $\binom{2m+2}{0} = 1$, and $j = m+1$, giving $\binom{m+1}{m+1} = 1$. For $1 \leq j \leq m$ we use Pascal's rule with top $2m+1-j$ and bottom j; this is allowed because $j \leq m$ gives $j \leq 2m+1-j$. So

$$B_{2m+2} = \underbrace{1 + \sum_{j=1}^{m} \binom{2m+1-j}{j}}_{\text{(a)}} + \underbrace{\sum_{j=1}^{m} \binom{2m+1-j}{j-1} + 1}_{\text{(b)}}.$$

In (a), the 1 is $\binom{2m+1}{0}$, the missing $k = 0$ term, so (a) is $\sum_{k=0}^{m} \binom{2m+1-k}{k} = B_{2m+1}$.

Ask yourself What should the second sum become, and where does the spare 1 go?

Put $i = j - 1$ in (b): the sum becomes $\sum_{i=0}^{m-1} \binom{2m-i}{i}$, which is B_{2m} with its last term $i = m$ missing. That last term is $\binom{m}{m} = 1$, exactly the spare 1. So (b) $= B_{2m}$, and $B_{2m+2} = B_{2m+1} + B_{2m}$. (When $m = 0$ the sums are empty and this reads $2 = 1 + 1$.)

Odd case, $n = 2m+1$ $(m \geq 0)$. We need $B_{2m+3} = B_{2m+2} + B_{2m+1}$. Now

$$B_{2m+3} = \sum_{k=0}^{m+1} \binom{2m+3-k}{k}.$$

Only the $k = 0$ term, $1 = \binom{2m+2}{0}$, is left alone. For $1 \leq k \leq m+1$, Pascal's rule with top $2m+2-k$ and bottom k is allowed, since $k \leq m+1$ gives $k \leq 2m+2-k$. Therefore

$$\begin{aligned} B_{2m+3} &= \binom{2m+2}{0} + \sum_{k=1}^{m+1} \binom{2m+2-k}{k} + \sum_{k=1}^{m+1} \binom{2m+2-k}{k-1} \\ &= \sum_{j=0}^{m+1} \binom{2m+2-j}{j} + \sum_{i=0}^{m} \binom{2m+1-i}{i} \qquad (i = k-1) \\ &= B_{2m+2} + B_{2m+1}. \end{aligned}$$

This time no end terms are left over.

Every $n \geq 0$ is of the form $2m$ or $2m+1$, so $B_{n+2} - B_{n+1} = B_n$ for all $n = 0, 1, 2, \ldots$, as required.

Relation with the Fibonacci numbers

Directly, $B_0 = \binom{0}{0} = 1$ and $B_1 = \binom{1}{0} = 1$. The Fibonacci numbers start $F_1 = 1$, $F_2 = 1$. So $B_0 = F_1$ and $B_1 = F_2$, and both sequences obey the same recurrence "each term is the sum of the previous two". By induction (if $B_n = F_{n+1}$ and $B_{n+1} = F_{n+2}$ then $B_{n+2} = F_{n+2} + F_{n+1} = F_{n+3}$),

$$B_n = F_{n+1} \qquad \text{for all } n \geq 0.$$

The B sequence is the Fibonacci sequence shifted by one place: $1, 1, 2, 3, 5, 8, 13, \ldots$.

Checks

- Compute a few terms straight from the definition: $B_2 = 1 + 1 = 2$, $B_3 = 1 + 2 = 3$, $B_4 = 1 + 3 + 1 = 5$, $B_5 = 1 + 4 + 3 = 8$. These are $F_3, \ldots, F_6$.
- A larger one: $B_9 = \binom{9}{0} + \binom{8}{1} + \binom{7}{2} + \binom{6}{3} + \binom{5}{4} = 1 + 8 + 21 + 20 + 5 = 55 = F_{10}$.
- Pascal's rule spot check: $\binom{5}{2} + \binom{5}{1} = 10 + 5 = 15 = \binom{6}{2}$.

Common slips

- Applying Pascal's rule to the end terms such as $\binom{m+1}{m+1}$, which would produce $\binom{m}{m+1}$, outside the range $1 \leq r \leq n$ for which the rule was proved. Peel off the end terms first.
- Treating only one parity. The definitions of B_{2m} and B_{2m+1} differ, and the recurrence links an even-indexed term to odd-indexed ones, so both cases are needed.
- Getting the index shift wrong: the answer is $B_n = F_{n+1}$, not $B_n = F_n$ (check $B_0 = 1 \neq F_0$).

Beyond STEP

The identity $F_{n+1} = \sum_k \binom{n-k}{k}$ has a neat counting proof: both sides count the ways of tiling a $1 \times n$ strip with squares and dominoes (k is the number of dominoes, and choosing their positions among the $n - k$ tiles gives $\binom{n-k}{k}$). Proving an identity by counting the same set in two ways is a core technique in undergraduate combinatorics, and so is the generating-function view, in which $\sum F_n x^n = x/(1 - x - x^2)$. For more, try Wilf, *generatingfunctionology*, or Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP I 2013, Question 7

Pure • First-Order Differential Equations

STEP I 2013, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s1-q7

The key idea

In (i) and (ii) the right-hand side depends only on the ratio y/x, so writing $y = ux$ turns the equation into one in which the variables separate. Part (iii) breaks that pattern, because x^2/y is not a function of y/x alone. The cure is to ask what power of x the solution "should" behave like: balancing the two terms suggests $y = u\,x^{3/2}$, and then everything separates again.

Solution

Part (i)

Put $y = ux$, so that by the product rule $\dfrac{\mathrm{d}y}{\mathrm{d}x} = u + x\dfrac{\mathrm{d}u}{\mathrm{d}x}$. Since $x/y = 1/u$ and $y/x = u$, the equation becomes

$$u + x\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{u} + u \quad \Longrightarrow \quad x\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{u}.$$

The u terms cancel, which is what makes the substitution work. Separating the variables (we may divide by x since $x > 0$),

$$\int u\,\mathrm{d}u = \int \frac{1}{x}\,\mathrm{d}x \quad \Longrightarrow \quad \tfrac{1}{2}u^2 = \ln x + C,$$

with $\ln x$ rather than $\ln|x|$ because $x > 0$. When $x = 1$, $y = 2$, so $u = y/x = 2$, giving $2 = 0 + C$. Hence

$$u^2 = 4 + 2\ln x.$$

Now $y > 0$ and $x > 0$ force $u > 0$, so we take the positive root, $u = \sqrt{4 + 2\ln x}$, and

$$y = x\sqrt{4 + 2\ln x}.$$

This needs $4 + 2\ln x > 0$ (we need $y > 0$, not merely $y \geq 0$), that is $\ln x > -2$, or $x > \mathrm{e}^{-2}$. This is the required result.

Part (ii)

The right-hand side is again a function of y/x, so we use the same substitution $y = ux$:

$$u + x\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{u} + 2u \quad \Longrightarrow \quad x\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{u} + u = \frac{1+u^2}{u}.$$

This time the u terms do not cancel, but the variables still separate:

$$\int \frac{u}{1+u^2}\,\mathrm{d}u = \int \frac{1}{x}\,\mathrm{d}x \quad \Longrightarrow \quad \tfrac{1}{2}\ln(1+u^2) = \ln x + C,$$

where the left-hand integral is of the form $\frac{1}{2}\int f'/f$ and $1+u^2 > 0$ needs no modulus. At $x = 1$ we have $u = 2$, so $C = \frac{1}{2}\ln 5$. Multiplying by 2 and exponentiating,

$$1 + u^2 = 5x^2, \qquad \text{so} \qquad u = \sqrt{5x^2 - 1}$$

(positive root, as $u > 0$). Therefore

$$y = x\sqrt{5x^2 - 1} \qquad \left(x > \tfrac{1}{\sqrt{5}}\right),$$

the restriction coming from $5x^2 - 1 > 0$ with $x > 0$.

Part (iii)

Now $y = ux$ gives $x^2/y = x/u$, which still contains x on its own, so the equation does not separate. We want a substitution under which both terms on the right scale the same way.

Ask yourself If y behaves like a power x^k, which value of k makes $\dfrac{x^2}{y}$ and $\dfrac{2y}{x}$ the same size?

We need x^{2-k} and x^{k-1} to match, so $2 - k = k - 1$, giving $k = \frac{3}{2}$. (The left-hand side $\mathrm{d}y/\mathrm{d}x$ then also behaves like $x^{1/2}$, so all three terms agree.) This suggests the substitution $y = u\,x^{3/2}$, for which

$$\frac{\mathrm{d}y}{\mathrm{d}x} = x^{3/2}\frac{\mathrm{d}u}{\mathrm{d}x} + \tfrac{3}{2}u\,x^{1/2}, \qquad \frac{x^2}{y} = \frac{x^{1/2}}{u}, \qquad \frac{2y}{x} = 2u\,x^{1/2}.$$

Dividing the whole equation by $x^{1/2}$ (which is positive),

$$x\frac{\mathrm{d}u}{\mathrm{d}x} + \tfrac{3}{2}u = \frac{1}{u} + 2u \quad \Longrightarrow \quad x\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{u} + \frac{u}{2} = \frac{2+u^2}{2u}.$$

Separating,

$$\int \frac{2u}{2+u^2}\,\mathrm{d}u = \int \frac{1}{x}\,\mathrm{d}x \quad \Longrightarrow \quad \ln(2+u^2) = \ln x + C.$$

At $x = 1$, $u = y/x^{3/2} = 2$, so $C = \ln 6$, and hence $2 + u^2 = 6x$. Taking the positive root, $u = \sqrt{6x-2}$, so

$$y = x^{3/2}\sqrt{6x-2} = \sqrt{6x^4 - 2x^3} \qquad \left(x > \tfrac{1}{3}\right),$$

where $x > \frac{1}{3}$ is needed for $6x - 2 > 0$.

An alternative for (iii). Multiplying the equation by $2y$ gives, with $z = y^2$,

$$\frac{\mathrm{d}z}{\mathrm{d}x} - \frac{4}{x}z = 2x^2,$$

a linear equation with integrating factor x^{-4}. Then $\left(x^{-4}z\right)' = 2x^{-2}$, so $x^{-4}z = -2/x + 6$ using $z(1) = 4$, giving $y^2 = 6x^4 - 2x^3$ as before. The same trick ($z = y^2$) also solves (i) and (ii).

Checks

- Each answer gives $y = 2$ at $x = 1$: $1 \cdot \sqrt{4}$, $1 \cdot \sqrt{4}$ and $\sqrt{6-2}$.
- Differentiate the answer to (ii) and substitute back. From $y^2 = 5x^4 - x^2$ we get $2yy' = 20x^3 - 2x$, so $y' = (10x^3 - x)/y$. Meanwhile $\dfrac{x}{y} + \dfrac{2y}{x} = \dfrac{x^2 + 2y^2}{xy} = \dfrac{10x^4 - x^2}{xy}$, which is the same.
- At $x = 2$ the three solutions give $2\sqrt{4 + 2\ln 2} \approx 4.642$, $2\sqrt{19} \approx 8.718$ and $4\sqrt{5} \approx 8.944$. As expected, the larger coefficient of y/x in (ii) makes y grow faster than in (i).

Common slips

- Forgetting the u in $\dfrac{\mathrm{d}y}{\mathrm{d}x} = u + x\dfrac{\mathrm{d}u}{\mathrm{d}x}$, so that the equation in u comes out wrong.
- Applying the initial condition to y instead of u: at $x = 1$ the two agree here, but in (iii) you need $u = y/x^{3/2}$ in general.
- Leaving out the domain. The square roots must be real and positive, which gives $x > \mathrm{e}^{-2}$, $x > 1/\sqrt{5}$ and $x > \frac{1}{3}$.

Beyond STEP

Equations whose right-hand side depends only on y/x are called *homogeneous*, and the substitution $y = ux$ always separates them. Part (iii) is a scaling symmetry in disguise: the equation is unchanged under $x \mapsto \lambda x$, $y \mapsto \lambda^{3/2} y$, and such a scaling symmetry always suggests a substitution that makes the equation separable. Equations of the form $y' + P(x)y = Q(x)y^n$, which become linear under $z = y^{1-n}$, are called Bernoulli equations; all three parts here are of this type with $n = -1$. Boyce and DiPrima, *Elementary Differential Equations*, covers both families in its chapter on first-order equations.

WORKED SOLUTION

STEP I 2013, Question 8

Pure • Composite Functions

STEP I 2013, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s1-q8

The key idea

A composite pq means "do q first, then p", and it is only defined where the output of q lands inside the domain of p. So the domain comes from q (trimmed if necessary), and the range is found by asking which values p actually receives. The algebra is light; the marks are in being careful with $\sqrt{x^2} = |x|$ and with the domains. In part (iii) the identity $(x + \sqrt{x^2-1})(x - \sqrt{x^2-1}) = 1$ does most of the work.

Solution

Part (i)

- $\mathrm{cb}(x) = \mathrm{c}(\ln x) = 2\ln x$. The domain is that of b, namely $x > 0$, and c accepts every real number. As $\ln x$ takes every real value, so does $2\ln x$: the range is $\mathbb{R}$.
- $\mathrm{ab}(x) = (\ln x)^2$, with domain $x > 0$. Since $\ln x$ takes every real value, its square takes every value in $[0, \infty)$ (with 0 at $x = 1$): the range is $y \geq 0$.
- $\mathrm{da}(x) = \sqrt{x^2} = |x|$, not x, because the square root sign means the non-negative root. Here $x^2 \geq 0$ always, so the domain is all of $\mathbb{R}$, and the range is $y \geq 0$.
- $\mathrm{ad}(x) = (\sqrt{x})^2 = x$, but only for $x \geq 0$, since d must act first. So the domain is $x \geq 0$ and the range is $y \geq 0$.

Notice that da and ad have the same range but different domains and different formulae: they are different functions.

Part (ii)

For fg: $\mathrm{g}(x) = \sqrt{x^2+1} \geq 1$ for every real x, so $\mathrm{g}(x)$ always lies in the domain of f. Then

$$\mathrm{fg}(x) = \sqrt{(x^2+1)-1} = \sqrt{x^2} = |x|, \qquad \text{domain } \mathbb{R}, \quad \text{range } y \geq 0.$$

For gf: f acts first, so we need $|x| \geq 1$; and g accepts any input. Then

$$\mathrm{gf}(x) = \sqrt{(x^2-1)+1} = |x|, \qquad \text{domain } |x| \geq 1, \quad \text{range } y \geq 1.$$

Again the two formulae agree, but the functions differ: the domain of gf excludes $-1 < x < 1$, and so its range starts at 1.

Part (iii)

The basic fact is

$$\left(x + \sqrt{x^2-1}\right)\left(x - \sqrt{x^2-1}\right) = x^2 - (x^2-1) = 1,$$

so $\mathrm{h}(x)\,\mathrm{k}(x) = 1$ for $x \geq 1$.

The graph of h ($x \geq 1$).

- $\mathrm{h}(1) = 1$.
- For $x > 1$, $\mathrm{h}'(x) = 1 + \dfrac{x}{\sqrt{x^2-1}} > 0$, so h is increasing; and $\mathrm{h}'(x) \to \infty$ as $x \to 1^+$, so the curve leaves $(1,1)$ with a vertical tangent.
- As $x \to \infty$, $\mathrm{h}(x) \to \infty$. For the asymptote,

  $$\mathrm{h}(x) - 2x = \sqrt{x^2-1} - x = \frac{-1}{\sqrt{x^2-1}+x} \to 0^-,$$

  so $y = 2x$ is an asymptote, approached from below.

So h increases continuously from 1 without bound: its range is $[1, \infty)$.

The graph of k, branch $x \geq 1$.

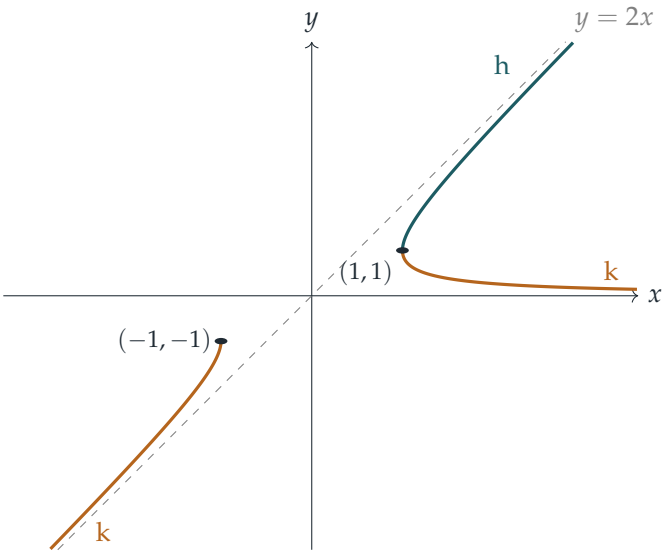
- $\mathrm{k}(1) = 1$ and, by the basic fact, $\mathrm{k}(x) = 1/\mathrm{h}(x)$.
- Since h is positive and increasing to infinity, k is positive and decreasing, with $\mathrm{k}(x) \to 0^+$. So $y = 0$ is an asymptote, approached from above.
- $\mathrm{k}'(x) = 1 - \dfrac{x}{\sqrt{x^2-1}} \to -\infty$ as $x \to 1^+$: a vertical tangent at $(1,1)$ again, where this branch joins the graph of h.

The graph of k, branch $x \leq -1$.

- $\mathrm{k}(-1) = -1$.
- For $x < -1$ we have $x < 0$, so $\mathrm{k}'(x) = 1 - \dfrac{x}{\sqrt{x^2-1}} > 1 > 0$: k is increasing. As $x \to -1^-$, $\mathrm{k}'(x) \to +\infty$, so there is a vertical tangent at $(-1,-1)$.

- Both x and $-\sqrt{x^2-1}$ are negative, so $\mathrm{k}(x) \to -\infty$ as $x \to -\infty$. Writing $x = -u$ with $u \geq 1$,
$$\mathrm{k}(x) - 2x = u - \sqrt{u^2-1} = \frac{1}{u + \sqrt{u^2-1}} \to 0^+,$$
so $y = 2x$ is an asymptote of this branch too, approached from above.

This branch takes every value in $(-\infty, -1]$.



(A neat way to see all this at once: put $x = \cosh t$ with $t \geq 0$. Then $\sqrt{x^2-1} = \sinh t$, so $\mathrm{h}(x) = e^t$ and $\mathrm{k}(x) = e^{-t}$; for $x = -\cosh t$, $\mathrm{k}(x) = -e^t$.)

The composite kh. The domain of h is $x \geq 1$, and every value $\mathrm{h}(x)$ satisfies $\mathrm{h}(x) \geq 1$, which lies in the domain of k. So kh is defined on all of $x \geq 1$.

? **Ask yourself** Which part of the graph of k does kh actually use?

Only the branch $x \geq 1$, since h never outputs anything below 1. Moreover h takes *every* value in $[1, \infty)$, so kh takes exactly the values of k on $[1, \infty)$. On that branch k decreases continuously from $\mathrm{k}(1) = 1$ towards 0, never reaching 0 (because $\mathrm{k} = 1/\mathrm{h} > 0$). Hence

$$\mathrm{kh}: \quad \text{domain } x \geq 1, \qquad \text{range } 0 < y \leq 1.$$

In the hyperbolic picture, $\mathrm{kh}(\cosh t) = \mathrm{k}(e^t)$, which is a decreasing function of t starting at 1 when $t = 0$.

Checks

- Spot values: $\mathrm{h}(2) = 2 + \sqrt{3} \approx 3.732$ and $\mathrm{k}(2) = 2 - \sqrt{3} \approx 0.268$; their product is 1. Also $\mathrm{k}(-2) = -2 - \sqrt{3} \approx -3.732$, which is just above $2x = -4$, as the asymptote

analysis predicts.

- $\mathrm{kh}(1) = \mathrm{k}(1) = 1$, and $\mathrm{kh}(2) = \mathrm{k}(3.732\ldots) = 3.732 - \sqrt{3.732^2 - 1} \approx 0.136$, inside $(0, 1]$.
- At $x = 1000$, $\mathrm{h}(x) - 2x \approx -0.0005$ and $\mathrm{k}(x) \approx 0.0005$, matching $-1/(2x)$ and $1/(2x)$.

Common slips

- Writing $\sqrt{x^2} = x$. It is $|x|$, which is why da, fg and gf are not the identity.
- Giving the domain of gf as $\mathbb{R}$ because the final formula $|x|$ makes sense everywhere. The domain of a composite is fixed by the first function applied.
- Missing the second branch of k for $x \leq -1$, or its asymptote $y = 2x$.

Beyond STEP

At university a function is a triple: a domain, a codomain and a rule. Two functions with the same rule but different domains, like fg and gf here, are genuinely different, and this precision matters as soon as you discuss inverses, injectivity and surjectivity. The function h is $e^{\operatorname{arcosh} x}$, and $\ln \mathrm{h}(x)$ is the standard formula for $\operatorname{arcosh} x$. Abbott's *Understanding Analysis* treats functions, continuity and monotonicity with this kind of care.

WORKED SOLUTION

STEP I 2013, Question 9

Mechanics • Projectiles and Collisions

STEP I 2013, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

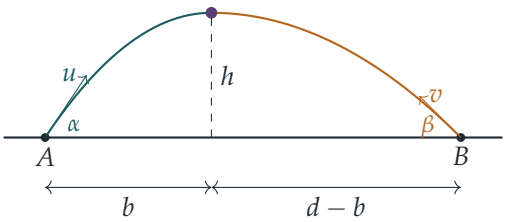
step.maths.org/questions/13-s1-q9

The key idea

At the top of its flight a projectile has no vertical velocity, so after a collision there the time to fall back down equals the time it took to rise. To land back where it started, each particle must therefore retrace its horizontal journey in the same time, which forces its horizontal velocity simply to be reversed. Put that into conservation of momentum, and use the fact that the two particles reach the top at the same instant, and everything follows.

Solution

We model A and B as particles moving freely under gravity, with no air resistance, and the impulse in the collision acts along the line of centres. Since the collision is direct and both particles are moving horizontally at that moment, this impulse is horizontal. Take the x-axis along the plane from A's starting point towards B's, so B starts at $x = d$.



Meeting at the tops

A has vertical velocity $u\sin\alpha - gt$, so it reaches its greatest height at time $t = u\sin\alpha/g$. Likewise B is highest at $t = v\sin\beta/g$. They were projected simultaneously and collide when each is at its top, so these times are equal:

$$T = \frac{u\sin\alpha}{g} = \frac{v\sin\beta}{g}, \qquad \text{so} \qquad u\sin\alpha = v\sin\beta. \tag{1}$$

Just before the collision, A has velocity $u\cos\alpha$ in the positive x-direction and B has velocity $v\cos\beta$ in the negative x-direction; neither has any vertical velocity.

After the collision

Ask yourself How can we say anything about the velocities after the collision when we know nothing about the coefficient of restitution?

The return condition does the work. The impulse is horizontal, so A still has zero vertical velocity just after the collision, at height h above the plane. Falling from rest (vertically) through h takes exactly as long as rising through h did, namely T. During that time A must move horizontally from $x = b$ back to $x = 0$, where $b = u\cos\alpha\, T$ is how far it travelled on the way up. So its new horizontal velocity is

$$-\frac{b}{T} = -u\cos\alpha :$$

A's horizontal velocity is exactly reversed, and in the same way so is B's. (In fact each particle retraces its own path backwards.)

Conservation of momentum in the x-direction now gives

$$mu\cos\alpha - Mv\cos\beta = -mu\cos\alpha + Mv\cos\beta, \qquad \text{so} \qquad mu\cos\alpha = Mv\cos\beta. \qquad (2)$$

In other words, the total momentum is zero both before and after. Dividing (2) by (1), which is allowed since α and β are strictly between 0 and $\frac{1}{2}\pi$ so both sides of (1) are non-zero,

$$m\cot\alpha = M\cot\beta,$$

as required.

Where the collision happens

In time T, A travels a horizontal distance $b = u\cos\alpha\, T$ and B travels $v\cos\beta\, T$. Together they cover the gap, so

$$u\cos\alpha\, T + v\cos\beta\, T = d.$$

By (2), $v\cos\beta = \dfrac{m}{M}\,u\cos\alpha$, so

$$b\left(1 + \frac{m}{M}\right) = d \qquad \Longrightarrow \qquad b = \frac{Md}{m+M},$$

as required. (The heavier B is, the slower it moves horizontally, and the further from A's start the meeting point lies.)

The height of the collision

Using $v^2 = u^2 + 2as$ vertically for A, the greatest height is

$$h = \frac{u^2\sin^2\alpha}{2g}, \qquad \text{while} \qquad b = u\cos\alpha\cdot\frac{u\sin\alpha}{g} = \frac{u^2\sin\alpha\cos\alpha}{g}.$$

Dividing,

$$\frac{h}{b} = \frac{\sin\alpha}{2\cos\alpha} = \tfrac{1}{2}\tan\alpha, \qquad \text{so} \qquad h = \tfrac{1}{2}\,b\tan\alpha.$$

Checks

- Symmetry: with $m = M$ we get $\alpha = \beta$ and $b = \frac{1}{2}d$, so the particles meet in the middle, as they must.
- The height formula says the line from A's start to the top of its path has gradient $\frac{1}{2}\tan\alpha$, half the launch gradient: the familiar fact that a parabola's tangent at the start meets the axis of symmetry at twice the height of the vertex. The same calculation for B gives $h = \frac{1}{2}(d-b)\tan\beta$, and this agrees with $\frac{1}{2}b\tan\alpha$ precisely because $m\cot\alpha = M\cot\beta$.
- Numerically, take $m = 1$, $M = 2$, $\alpha = 45^\circ$, $u = 10\,\mathrm{m\,s^{-1}}$, $g = 9.8\,\mathrm{m\,s^{-2}}$. Then $\cot\beta = \frac{1}{2}$, $v = \sqrt{50}/\sin\beta \approx 7.91\,\mathrm{m\,s^{-1}}$, and $d \approx 7.65\,\mathrm{m}$, giving $b = \frac{2}{3}d \approx 5.10\,\mathrm{m}$ and $h = \frac{1}{2}b \approx 2.55\,\mathrm{m}$. Stepping both motions forward in time by computer, with the horizontal velocities reversed at the collision, confirms that the particles meet at their tops at this point and both land back at their starting points.

Common slips

- Trying to use a coefficient of restitution: none is given, and none is needed, since the return condition fixes the velocities after impact.
- Forgetting that the particles must reach their tops at the same time, which gives (1); without it the cotangents cannot be produced.
- Sign errors in the momentum equation: B is moving in the negative direction before the collision.

Beyond STEP

The return condition amounts to time reversal: Newton's equations under gravity look the same if we run the clock backwards, so reversing a particle's velocity sends it back along its own path. Here the total momentum is zero, which means we are working in the centre-of-mass frame, and in that frame an elastic collision just reverses both velocities, exactly what happens here (so this collision is perfectly elastic). Centre-of-mass frames and conservation laws are explored beautifully in Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP I 2013, Question 10

Mechanics • Restitution and Friction

STEP I 2013, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s1-q10

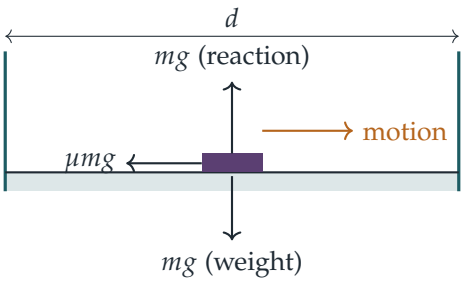
The key idea

Between collisions the only horizontal force is friction, so the puck has constant deceleration μg and the constant-acceleration formula $v^2 = u^2 + 2as$ links the speed leaving one barrier to the speed arriving at the next; Newton's law of restitution then multiplies that arrival speed by r. The result is a linear recurrence for v_i^2, and a linear recurrence $w_{i+1} = aw_i + b$ is solved by shifting to its fixed point, after which it is just a geometric sequence.

Solution

The recurrence

Model the puck as a particle moving in a straight line perpendicular to the barriers. While it is on the ice between collisions, the normal reaction is mg (no vertical acceleration), and since the puck is sliding, friction takes its limiting value μmg, opposing the motion.



By Newton's second law the deceleration is μg, constant, so for a traverse of length d starting with speed v_i (just after the ith collision) the speed u on arriving at the other barrier satisfies

$$u^2 = v_i^2 - 2\mu g d.$$

The barrier is fixed, so Newton's law of restitution says the puck leaves with speed r times its approach speed: $v_{i+1} = ru$. Squaring,

$$v_{i+1}^2 = r^2u^2 = r^2v_i^2 - 2r^2\mu gd, \qquad \text{so} \qquad v_{i+1}^2 - r^2v_i^2 = -2r^2\mu gd,$$

as required. The same argument applies to the first traverse if we write $v_0 = v$ for the starting speed, so the recurrence holds for all $i \geq 0$.

The number of traverses

Divide through by $2\mu gd$ and put $w_i = \dfrac{v_i^2}{2\mu gd}$, so that $w_0 = k$ and

$$w_{i+1} = r^2(w_i - 1).$$

"Comes to rest against a barrier after n traverses" means that the puck arrives at a barrier with zero speed at the end of the nth traverse. Then it leaves with zero speed, so $v_n = 0$, that is, $w_n = 0$ (and $w_i > 0$ for $i < n$).

Ask yourself How do we solve $w_{i+1} = r^2(w_i - 1)$?

Look for a constant solution $w_i = c$: we need $c = r^2(c-1)$, so $c = -\dfrac{r^2}{1-r^2}$, which makes sense because $r \neq 1$. Subtracting $c = r^2(c-1)$ from the recurrence gives

$$w_{i+1} - c = r^2(w_i - c),$$

so $w_i - c$ is geometric with ratio r^2:

$$w_i = c + r^{2i}(k - c).$$

Setting $w_n = 0$ gives $r^{2n}(k-c) = -c$, that is,

$$r^{2n}\left(k + \frac{r^2}{1-r^2}\right) = \frac{r^2}{1-r^2} \quad \Longrightarrow \quad r^{2n-2}\left(k(1-r^2) + r^2\right) = 1.$$

Multiplying through by r^{-2n} tidies this to

$$r^{-2n} = 1 + k\left(r^{-2} - 1\right), \qquad \text{so} \qquad \boxed{n = \frac{\ln\left(1 + k(r^{-2} - 1)\right)}{2\ln(1/r)}}$$

(for $0 < r < 1$, $\ln(1/r) > 0$, and the argument of the logarithm exceeds 1 because $k > 0$ and $r^{-2} > 1$, so $n > 0$ as it should be).

The case $r = \mathrm{e}^{-1}$

Now $r^{-2} = \mathrm{e}^2$ and $\ln(1/r) = 1$, so the formula becomes

$$n = \tfrac{1}{2}\ln\left(1 + k(\mathrm{e}^2 - 1)\right),$$

which is the required result.

The case $r = 1$

The recurrence becomes $w_{i+1} = w_i - 1$: each traverse uses up exactly $2\mu g d$ of v^2, because there is no loss at the barriers. So $w_i = k - i$, and $w_n = 0$ gives

$$n = k = \frac{v^2}{2\mu g d}.$$

This is just energy: the initial kinetic energy $\frac{1}{2}mv^2$ is all dissipated by friction over a total distance nd, so $\frac{1}{2}mv^2 = \mu mg\, nd$.

Checks

- Run the recurrence backwards with $r = \mathrm{e}^{-1}$ from $w_3 = 0$: $w_2 = 1$, $w_1 = \mathrm{e}^2 + 1$, $w_0 = \mathrm{e}^4 + \mathrm{e}^2 + 1 \approx 62.99$. The formula gives $\frac{1}{2}\ln\left(1 + (\mathrm{e}^4 + \mathrm{e}^2 + 1)(\mathrm{e}^2 - 1)\right) = \frac{1}{2}\ln \mathrm{e}^6 = 3$, as it should.
- As $r \to 1$, $r^{-2} - 1 \approx 2(1 - r)$ and $\ln(1/r) \approx 1 - r$, so the general formula tends to $\dfrac{2k(1-r)}{2(1-r)} = k$, matching the $r = 1$ answer.
- k is dimensionless (both v^2 and gd are speeds squared), as n must be.
- Stepping the motion numerically for random μ, d, r and v confirms the recurrence to high accuracy.

Common slips

- Applying restitution to v_i^2 with factor r rather than r^2: the law of restitution is about speeds, so squared speeds pick up r^2, and so does the friction term.
- Forgetting that n counts traverses, not collisions with nonzero rebound: the condition is $v_n = 0$, with $v_0 = v$.

Beyond STEP

The recurrence $w_{i+1} = aw_i + b$ is the simplest linear difference equation, and the trick of subtracting the fixed point is exactly the method of "complementary function plus particular solution" you will meet for linear differential equations. Bouncing problems like this (a ball bouncing on a floor is the classic) are also nice examples of energy dissipation and of infinitely many events in finite time when there is no friction. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats collisions and friction with more depth.

WORKED SOLUTION

STEP I 2013, Question 11

Mechanics • Friction and Equilibrium

STEP I 2013, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s1-q11

The key idea

The two strings make angles $\frac{1}{2}\pi - \alpha$ and α with the horizontal on opposite sides, so they are *perpendicular*. Two perpendicular forces $T\cos\beta$ and $T\sin\beta$ combine into a single force of size T, turned through β from the first one. Once the two strings are replaced by one pull of T at angle $\alpha + \beta$ to the horizontal, both parts become short exercises: limiting friction in (i), and in (ii) the realisation that the block is lifted clean off the surface.

Solution

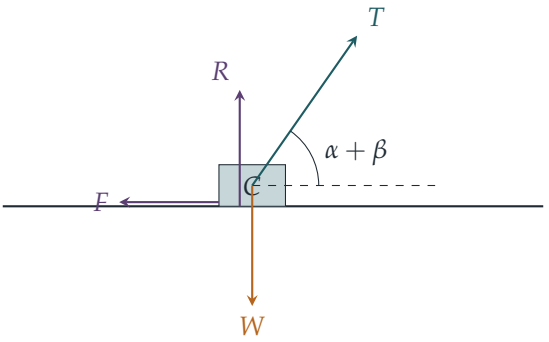
We model C as a particle and the strings as light. Take the x-axis horizontal towards B and the y-axis vertically up.

The resultant of the tensions

The string BC points at angle α above the positive x-axis, so its unit vector is $(\cos\alpha, \sin\alpha)$. The string AC makes angle $\frac{1}{2}\pi - \alpha$ with the negative x-axis, so it points at angle $\pi - (\frac{1}{2}\pi - \alpha) = \frac{1}{2}\pi + \alpha$ to the positive x-axis, with unit vector $(-\sin\alpha, \cos\alpha)$. These are perpendicular. The total pull of the strings is therefore

$$\begin{aligned} T\cos\beta\,(\cos\alpha, \sin\alpha) + T\sin\beta\,(-\sin\alpha, \cos\alpha) &= T\big(\cos\alpha\cos\beta - \sin\alpha\sin\beta, \\ &\qquad \sin\alpha\cos\beta + \cos\alpha\sin\beta\big) \\ &= T\big(\cos(\alpha+\beta),\ \sin(\alpha+\beta)\big). \end{aligned}$$

So the strings act exactly like a single force of magnitude T at angle $\alpha + \beta$ above the horizontal, pulling towards the B side. Since $0 < \alpha + \beta < \frac{1}{2}\pi$, both components are positive.



Part (i)

Let R be the normal reaction and F the friction force, which (if the block is at rest) acts horizontally away from B. For equilibrium,

$$\begin{aligned} \text{vertically:} \quad & R + T\sin(\alpha+\beta) = W, \qquad && \text{so } R = W - T\sin(\alpha+\beta), \\ \text{horizontally:} \quad & F = T\cos(\alpha+\beta). \end{aligned}$$

The condition $W > T\sin(\alpha+\beta)$ makes $R > 0$, so the block really is pressed onto the surface and friction is available. The block stays at rest provided the friction needed does not exceed its limiting value, $F \le \mu R$:

$$T\cos(\alpha+\beta) \le \mu\big(W - T\sin(\alpha+\beta)\big).$$

Now put $\mu = \tan\lambda = \sin\lambda/\cos\lambda$ and multiply through by $\cos\lambda$, which is positive because λ is acute:

$$T\cos(\alpha+\beta)\cos\lambda \le W\sin\lambda - T\sin(\alpha+\beta)\sin\lambda.$$

Rearranging and using $\cos(P-Q) = \cos P\cos Q + \sin P\sin Q$,

$$W\sin\lambda \ge T\big(\cos(\alpha+\beta)\cos\lambda + \sin(\alpha+\beta)\sin\lambda\big) = T\cos(\alpha+\beta-\lambda),$$

which is the required condition. (Every step is reversible, so in fact the block stays at rest *if and only if* this holds.)

? **Ask yourself** Why do we need the condition $W > T\sin(\alpha+\beta)$ at all?

Without it, the vertical equation would give $R \le 0$. A surface can push but not pull, so a negative R is impossible: the block would be lifted off (or only just touching), and there would be no friction to hold it. That is exactly the situation of part (ii).

Part (ii)

Now $W = T\tan\phi$ with $\alpha + \beta = 2\phi$, so $0 < \phi < \frac{1}{4}\pi$. Compare the weight with the upward pull of the strings:

$$\begin{aligned} T\sin(\alpha+\beta) - W &= T\big(2\sin\phi\cos\phi - \tan\phi\big) \\ &= T\tan\phi\,\big(2\cos^2\phi - 1\big) \\ &= T\tan\phi\cos 2\phi > 0, \end{aligned}$$

since $0 < \phi < \frac{1}{4}\pi$ makes both $\tan\phi$ and $\cos 2\phi$ positive. So the strings pull upwards harder than gravity pulls down. If the block stayed on the surface we would need $R = W - T\sin(\alpha+\beta) < 0$, which is impossible. Hence the block leaves the surface at once: $R = 0$, and with no contact there is no friction either.

The only forces are then the pull T at angle 2ϕ and the weight. The resultant force is

$$\big(T\cos 2\phi,\ T\sin 2\phi - T\tan\phi\big) = \big(T\cos 2\phi,\ T\tan\phi\cos 2\phi\big) = \frac{T\cos 2\phi}{\cos\phi}\,(\cos\phi,\ \sin\phi),$$

using the computation above for the vertical component. Its magnitude $T\cos 2\phi/\cos\phi$ is positive, so it is not zero. The block starts from rest, so its initial acceleration, and hence the direction in which it starts to move, is along the resultant force. That direction is $(\cos\phi, \sin\phi)$, at angle ϕ above the horizontal (towards B), as required.

Checks

- Take $T = 10\,\text{N}$, $\alpha = 20^\circ$, $\beta = 40^\circ$, so $\phi = 30^\circ$ and $W = 10\tan 30^\circ \approx 5.77\,\text{N}$. The upward pull is $10\sin 60^\circ \approx 8.66\,\text{N} > W$, and the resultant is $(10\cos 60^\circ,\ 8.66 - 5.77) = (5,\ 2.89)$, whose angle is $\arctan(2.89/5) = 30^\circ = \phi$.
- In (i), $\lambda = \frac{1}{2}\pi$ (infinitely rough) gives $W \geq T\sin(\alpha+\beta)$, which already holds, so the block never slides; $\lambda \to 0$ (smooth) gives $0 \geq T\cos(\alpha+\beta)$, which fails, so a smooth surface cannot hold it. Both are what we expect.
- The condition in (i) was also confirmed by computer against the direct test $F \leq \mu R$ on many random choices of α, β, T, W, μ.

Common slips

- Resolving each string separately and getting lost in the algebra, instead of noticing that the strings are perpendicular and combining them first.
- Writing $R = W - T\sin(\alpha+\beta)$ in part (ii) and carrying on with friction, without noticing that R would be negative.
- Forgetting to justify that a particle starting from rest moves initially along the resultant force.

Beyond STEP

The idea of λ, the *angle of friction*, is that the normal reaction and limiting friction combine into a single total reaction making angle λ with the normal; equilibrium is possible exactly when the total reaction can lie inside this "cone of friction". Part (i) is that statement in disguise, and the same picture handles friction in three dimensions and in rigid-body problems. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats friction, constraints and contact forces carefully.

WORKED SOLUTION

STEP I 2013, Question 12

Probability and Statistics • Sampling Without Replacement

STEP I 2013, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s1-q12

The key idea

Choosing k tablets "at random" means every set of k tablets left in the container is equally likely, so each day's probability is a ratio of counts: favourable selections over $\binom{\text{tablets left}}{k}$. For the last-day question, the trick is symmetry: the tablets left for the final day are just as random as the ones picked on the first. In (iii), the days chain together nicely, because after a good day the container looks exactly like a shorter holiday.

Solution

Throughout, think of the tablets as distinguishable (say, numbered), even though they look the same. This doesn't change any probability, and it makes all $\binom{N}{k}$ choices of k tablets from N equally likely.

Part (i)

With $k = 3$ and $n = 3$ there are 9 tablets, 3 of each type. The number of ways to choose 3 of them is $\binom{9}{3} = 84$. To get one of each type, we choose one of the 3 tablets of type 1, one of the 3 of type 2 and one of the 3 of type 3, so there are $3^3 = 27$ ways. Hence

$$\mathrm{P}(\text{one of each on day 1}) = \frac{27}{84} = \frac{9}{28},$$

as required.

Ask yourself Why don't I need to work through what happens on days 1 and 2 to find the last-day probability?

Over the holiday, the nine tablets are split at random into three groups of three: the day-1 group, the day-2 group and the day-3 group (what's left). Choosing the groups in the order 1, 2, 3 or in the order 3, 2, 1 gives the same distribution: each way of splitting the tablets into three labelled groups is equally likely. In particular, the set left for day 3

is a uniformly random set of three tablets, exactly like the set picked on day 1. So the probability that I am left with one tablet of each type on the last day is also

$$\frac{9}{28}.$$

Part (ii)

Now there are $3n$ tablets, n of each type. There are $\binom{3n}{3}$ equally likely choices, and n^3 of them contain one of each type. So

$$\begin{aligned}\mathrm{P}(\text{one of each on day 1}) &= \frac{n^3}{\binom{3n}{3}} = \frac{6n^3}{3n(3n-1)(3n-2)} \\ &= \frac{2n^2}{(3n-1)(3n-2)}.\end{aligned}$$

For $n = 3$ this is $\frac{18}{8\cdot 7} = \frac{9}{28}$, matching (i).

Part (iii)

Now $k = 2$ and there are $2n$ tablets, n of each of the two types. For $m = n, n-1, \ldots, 1$, let G_m be the event that I select one of each type on the day when m tablets of each type are in the container at the start of the day (so $m = n$ on day 1).

If the container holds m of each type, there are $\binom{2m}{2} = m(2m-1)$ choices of two tablets, and $m \cdot m$ of them are one of each type. So

$$\mathrm{P}(\text{one of each} \mid m \text{ of each left}) = \frac{m^2}{m(2m-1)} = \frac{m}{2m-1}.$$

If every day so far has been good, then each day has removed exactly one of each type, so the next day starts with $m-1$ of each. Multiplying the conditional probabilities day by day,

$$\mathrm{P}(\text{one of each type every day}) = \prod_{m=1}^{n} \frac{m}{2m-1} = \frac{n!}{1 \times 3 \times 5 \times \cdots \times (2n-1)}.$$

To tidy the product of odd numbers, multiply top and bottom by the even numbers $2 \times 4 \times \cdots \times (2n) = 2^n\, n!$:

$$1 \times 3 \times \cdots \times (2n-1) = \frac{(2n)!}{2^n\, n!}.$$

Hence

$$\boxed{\mathrm{P}(\text{one of each type every day}) = \frac{2^n\,(n!)^2}{(2n)!}}$$

(As a check, $n = 1$ gives 1, which is right: with one tablet of each type I must take both.)

Now use Stirling's approximation for $n!$ and for $(2n)!$:

$$(n!)^2 \approx 2n\pi \left(\frac{n}{\mathrm{e}}\right)^{2n},$$
$$(2n)! \approx \sqrt{4n\pi}\left(\frac{2n}{\mathrm{e}}\right)^{2n} = 2\sqrt{n\pi}\,2^{2n}\left(\frac{n}{\mathrm{e}}\right)^{2n}.$$

The factors $(n/\mathrm{e})^{2n}$ cancel, leaving

$$\frac{2^n\,(n!)^2}{(2n)!} \approx \frac{2^n \cdot 2n\pi}{2\sqrt{n\pi}\,2^{2n}} = 2^{-n}\,\frac{n\pi}{\sqrt{n\pi}} = 2^{-n}\sqrt{n\pi},$$

which is the required approximation.

Checks

- Exhaustive counting by computer of all ways of drawing the tablets confirms $\frac{9}{28}$ for both day 1 and day 3 in (i), the formula in (ii) for $n \le 7$, and the formula in (iii) for $n \le 4$.
- For $n = 10$ the exact probability in (iii) is $\frac{2^{10}(10!)^2}{20!} \approx 0.005542$, while $2^{-10}\sqrt{10\pi} \approx 0.005474$: about 1.3% low. The ratio tends to 1 as n grows (it is about 1.0025 at $n = 50$), as you'd expect from Stirling.
- Alternative count for (iii): split the $2n$ tablets into an ordered list of n pairs in $(2n)!/2^n$ equally likely ways; the good splits pair the type-1 tablets with the type-2 tablets day by day in $n! \times n!$ ways. The ratio is again $2^n(n!)^2/(2n)!$.

Common slips

- Drawing one tablet at a time and writing $\frac{3}{9} \times \frac{3}{8} \times \frac{3}{7}$, which is the chance of types 1, 2, 3 *in that order*. The types can come in 3! orders, giving $6 \times \frac{27}{504} = \frac{9}{28}$. (Or: the first tablet can be anything, so $1 \times \frac{6}{8} \times \frac{3}{7} = \frac{9}{28}$.)
- Trying to condition on everything that happens on day 1 for the last-day probability; it works, but is long, and symmetry gives the answer at once.
- In Stirling for $(2n)!$, replacing n by $2n$ everywhere, including inside the square root: $\sqrt{2(2n)\pi} = 2\sqrt{n\pi}$, not $\sqrt{2n\pi}$.

Beyond STEP

The answer to (iii) is $2^n/\binom{2n}{n}$, and the estimate $\binom{2n}{n} \approx 4^n/\sqrt{n\pi}$ appears all over probability: it gives the chance that a simple random walk is back at its starting point after $2n$ steps, and it drives the proof that such a walk in one or two dimensions returns home with probability 1. The symmetry argument in (i) is the idea of *exchangeability*: the order in which random draws are made doesn't change their joint distribution. Both ideas

are developed in Grimmett and Welsh, *Probability: An Introduction*, and in Blitzstein and Hwang, *Introduction to Probability*.

WORKED SOLUTION

STEP I 2013, Question 13

Probability and Statistics • Counting and Expectation

STEP I 2013, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s1-q13

The key idea

The numbers $1, \ldots, 52$ split into 26 disjoint "partner pairs" $\{k, 53-k\}$, and a discarded pair is just a partner pair that we have chosen both members of. So to count selections with exactly k complete pairs we choose which pairs are complete, then fill the remaining places with numbers that come from *different* pairs, with a free choice of which partner each time. For the expectation, linearity over the 26 pairs gives a quick confirmation.

Solution

Counting the selections

Write the integers as 26 pairs $\{1,52\}, \{2,51\}, \ldots, \{26,27\}$. Two integers from 1 to 52 sum to 53 exactly when they form one of these pairs, and each integer lies in exactly one pair. So X is the number of pairs both of whose members are among the seven chosen. Since $2X \leq 7$, X takes the values $0, 1, 2, 3$.

All $\binom{52}{7}$ selections are equally likely, so we count the selections with $X = k$.

- Choose the k complete pairs: $\binom{26}{k}$ ways. These use $2k$ of the seven numbers.
- The other $7-2k$ numbers must come from $7-2k$ *different* pairs among the remaining $26-k$ (two from the same pair would make another complete pair): $\binom{26-k}{7-2k}$ ways.
- From each of those pairs we take one of its two members: 2^{7-2k} ways.

Each selection with $X = k$ arises exactly once this way, so

$$\mathrm{P}(X = k) = \frac{\binom{26}{k}\binom{26-k}{7-2k}\,2^{7-2k}}{\binom{52}{7}}, \qquad k = 0, 1, 2, 3.$$

The numbers

$$\begin{aligned} X=0: &\quad \tbinom{26}{7}2^7 = 657\,800\times 128 = 84\,198\,400,\\ X=1: &\quad 26\tbinom{25}{5}2^5 = 26\times 53\,130\times 32 = 44\,204\,160,\\ X=2: &\quad \tbinom{26}{2}\tbinom{24}{3}2^3 = 325\times 2024\times 8 = 5\,262\,400,\\ X=3: &\quad \tbinom{26}{3}\tbinom{23}{1}2 = 2600\times 23\times 2 = 119\,600, \end{aligned}$$

and $\binom{52}{7} = 133\,784\,560$. These four counts do add up to $\binom{52}{7}$, a good check that nothing has been missed or double counted.

? **Ask yourself** How do we simplify fractions with nine-digit numbers in them?

The note in the question is the hint: $5593 = 7\times 17\times 47$ should be the denominator. Indeed $\binom{52}{7} = 23\,920\times 5593$, and every one of the four counts is a multiple of 23 920:

$$84\,198\,400 = 3520\times 23\,920, \qquad 44\,204\,160 = 1848\times 23\,920,$$
$$5\,262\,400 = 220\times 23\,920, \qquad 119\,600 = 5\times 23\,920.$$

So the distribution of X is

k	0	1	2	3
$\mathrm{P}(X=k)$	$\frac{3520}{5593}$	$\frac{1848}{5593}$	$\frac{220}{5593}$	$\frac{5}{5593}$

(the second is $\frac{264}{799}$ in lowest terms), and $3520+1848+220+5 = 5593$, as it must.

The expectation

$$\mathrm{E}(X) = \frac{0\times 3520 + 1\times 1848 + 2\times 220 + 3\times 5}{5593} = \frac{2303}{5593}.$$

Now $2303 = 7^2\times 47$ and $5593 = 7\times 17\times 47$, so

$$\mathrm{E}(X) = \frac{7^2\times 47}{7\times 17\times 47} = \frac{7}{17},$$

as required.

A quicker route to the expectation

For $j = 1,\ldots,26$ let $I_j = 1$ if both members of the jth pair are chosen and 0 otherwise, so $X = I_1 + \cdots + I_{26}$. Choosing both members of a given pair leaves 5 more from the other 50, so

$$\mathrm{P}(I_j = 1) = \frac{\binom{50}{5}}{\binom{52}{7}} = \frac{7\times 6}{52\times 51} = \frac{7}{442}.$$

By linearity of expectation (which needs no independence; the I_j are in fact dependent),

$$\mathrm{E}(X) = 26\times\frac{7}{442} = \frac{7}{17}.$$

Checks

- The four probabilities sum to 1, and the two methods give the same $\mathrm{E}(X) = \frac{7}{17} \approx 0.412$.
- A computer simulation of 300 000 random selections gave relative frequencies of about 0.630, 0.329, 0.039, 0.001 for $X = 0, 1, 2, 3$, against the exact values 0.6294, 0.3304, 0.0393, 0.0009.
- Sense check: $\mathrm{P}(X = 3)$ is tiny, as three complete pairs in only seven numbers ought to be.

Common slips

- Counting the selections with $X = 0$ as $\binom{52}{7}$ minus something rough, or as $\binom{26}{7}$ alone, forgetting the factor 2^7 for which member of each pair is taken.
- For $X = 1$, choosing the remaining five numbers as $\binom{50}{5}$: that allows a second complete pair and overcounts.

Beyond STEP

Writing a count as a sum of indicator variables and using linearity of expectation is one of the most useful tricks in first-year probability: it gives expectations even when the distribution is hopeless to find. Here the full distribution was within reach, but for, say, the expected number of fixed points of a random permutation, indicators are the natural way in. Blitzstein and Hwang, *Introduction to Probability*, uses this idea again and again.

WORKED SOLUTION

STEP II 2013, Question 1

Pure • Chords of the Logarithm

STEP II 2013, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s2-q1

This question has a full guided treatment in *Beyond STEP*, Section 3.5, π^e *or* e^π*?*: https://beyondstep.keithmcnulty.org/ch03/pi-e-or-e-pi.html

The key idea

Every inequality here is a comparison of p^q with q^p. Taking logarithms and dividing by pq makes it a comparison of $\frac{\ln p}{p}$ with $\frac{\ln q}{q}$, and $\frac{\ln x}{x}$ is the gradient of the line from the origin to the point $(x, \ln x)$ on the curve. So everything is about lines and chords meeting $y = \ln x$. The facts we need are that the curve is concave (it bends downwards) and that the tangent from the origin touches it at $x = \mathrm{e}$.

Solution

Part (i)

Suppose $y = mx$ touches $y = \ln x$ at $x = x_0$. The gradients agree and the curves meet, so

$$m = \frac{1}{x_0}, \qquad mx_0 = \ln x_0.$$

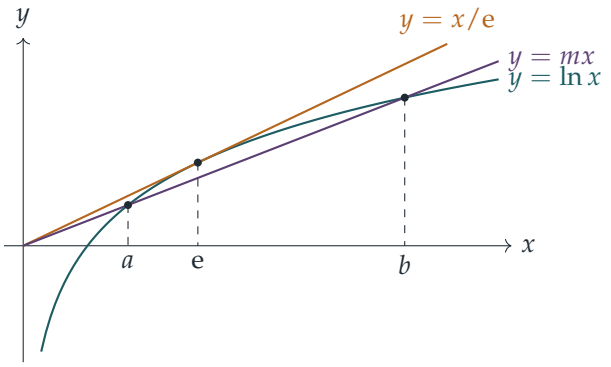
The first equation gives $mx_0 = 1$, so $\ln x_0 = 1$, $x_0 = \mathrm{e}$ and

$$m = \frac{1}{\mathrm{e}}.$$

Now suppose $y = mx$ meets the curve at $x = a$ and $x = b$. Then $\ln a = ma$ and $\ln b = mb$, so (as $a, b > 0$)

$$\frac{\ln a}{a} = m = \frac{\ln b}{b} \implies b \ln a = a \ln b \implies \ln\left(a^b\right) = \ln(b^a),$$

and since ln is one-to-one, $a^b = b^a$, as required.



For the sketch: if $m \leq 0$ the line meets the curve at most once, since $\ln x - mx$ is then strictly increasing. If $m \geq \frac{1}{\mathrm{e}}$ the line is the tangent or lies above it for $x > 0$, and the concave curve lies below its tangent $y = x/\mathrm{e}$, touching only at e, so again there are not two meeting points. So $0 < m < \frac{1}{\mathrm{e}}$, and the line $y = mx$ lies below the tangent, as in the figure. At $x = \mathrm{e}$ the curve is at height 1 and the line at height $m\mathrm{e} < 1$, so the curve is above the line there. Because the curve is concave and the line is straight, $\ln x - mx$ is concave. The set where it is positive is therefore a single interval, and its ends are the two meeting points a and b. That interval contains e, so $a < \mathrm{e} < b$.

Part (ii)

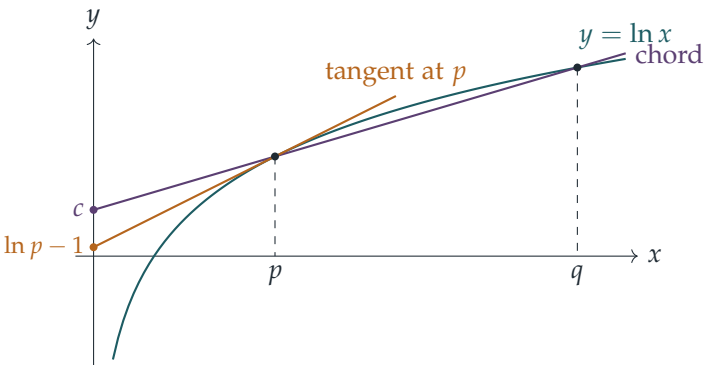
Here $\ln p = mp + c$ and $\ln q = mq + c$, with $0 < p < q$. Dividing by p and by q,

$$\frac{\ln p}{p} = m + \frac{c}{p}, \qquad \frac{\ln q}{q} = m + \frac{c}{q}.$$

Since $c > 0$ and $\frac{1}{p} > \frac{1}{q}$, we have $\frac{c}{p} > \frac{c}{q}$, so $\frac{\ln p}{p} > \frac{\ln q}{q}$. Multiplying by $pq > 0$ gives $q \ln p > p \ln q$, that is $\ln(p^q) > \ln(q^p)$, and so $p^q > q^p$.

In the picture, the lines from the origin to $(p, \ln p)$ and to $(q, \ln q)$ have gradients $\frac{\ln p}{p}$ and $\frac{\ln q}{q}$. If the chord crosses the y-axis above the origin, the nearer point subtends the steeper line from O. (The sign of c decides which way round the inequality goes. We use this again in part (iv).)

Part (iii)



Let the chord through $P = (p, \ln p)$ and $Q = (q, \ln q)$ have gradient s and y-intercept c, so $c = \ln p - sp$. Because the curve is concave, the chord from P to a point further right is less steep than the tangent at P, so

$$s < \frac{1}{p}.$$

(Or, by the mean value theorem, $s = \frac{1}{\xi}$ for some $\xi \in (p, q)$, and $\frac{1}{\xi} < \frac{1}{p}$.) The tangent at P is $y = \ln p + \frac{1}{p}(x - p)$, which meets the y-axis at $\ln p - 1$. The chord and the tangent both pass through P, and the chord is less steep. So moving left from P to $x = 0$, the chord stays above the tangent:

$$c = \ln p - sp > \ln p - 1 \geq 0,$$

using $p \geq \mathrm{e}$ at the last step. So the chord meets the y-axis at a positive value of y.

? **Ask yourself** How does this settle π^{e} against e^{π}?

Take $p = \mathrm{e}$ and $q = \pi$, which satisfy $\mathrm{e} \leq p < q$. The chord has $c > 0$, so part (ii) applies and gives $p^q > q^p$:

$$\mathrm{e}^{\pi} > \pi^{\mathrm{e}}.$$

Part (iv)

Now the chord through $(p, \ln p)$ and $(q, \ln q)$ has gradient $\frac{1}{\mathrm{e}}$, so it is $y = \frac{x}{\mathrm{e}} + c$ with

$$c = \ln p - \frac{p}{\mathrm{e}}.$$

The line $y = \frac{x}{\mathrm{e}}$ is the tangent at $x = \mathrm{e}$ from part (i), and the concave curve lies on or below it: $\ln x \leq \frac{x}{\mathrm{e}}$, with equality only at $x = \mathrm{e}$. So $c \leq 0$, with $c = 0$ only if $p = \mathrm{e}$. But $c = 0$ would make the chord the tangent itself, and the tangent meets the curve only at e, not at two points $p < q$. Hence

$$c < 0.$$

The chord is parallel to the tangent and lies below it. Now repeat the argument of part (ii) with the sign reversed:

$$\frac{\ln p}{p} = \frac{1}{\mathrm{e}} + \frac{c}{p}, \qquad \frac{\ln q}{q} = \frac{1}{\mathrm{e}} + \frac{c}{q}.$$

Since $c < 0$ and $\frac{1}{p} > \frac{1}{q}$, we get $\frac{c}{p} < \frac{c}{q}$, so $\frac{\ln p}{p} < \frac{\ln q}{q}$. Multiplying by $pq > 0$ gives $q \ln p < p \ln q$, and therefore

$$q^p > p^q.$$

(As in part (i), concavity also forces $p < \mathrm{e} < q$ here, but the argument does not need it.)

Checks

- $\mathrm{e}^\pi \approx 23.14$ and $\pi^\mathrm{e} \approx 22.46$, consistent with part (iii).
- Part (i): the pair $a = 2$, $b = 4$ has $2^4 = 4^2$, and indeed $\frac{\ln 2}{2} = \frac{\ln 4}{4}$ and $2 < \mathrm{e} < 4$.
- Part (iv): with $p = 1$, the chord gradient $\frac{\ln q}{q-1} = \frac{1}{\mathrm{e}}$ gives $q \approx 5.76$. Then $q^p \approx 5.76 > 1 = p^q$.
- Parts (ii) and (iii) were also confirmed by a computer check on many random chords.

Common slips

- Getting $q \ln p > p \ln q$ the wrong way round. Write out $\frac{\ln p}{p} > \frac{\ln q}{q}$ explicitly before multiplying by pq.
- In part (iii), stopping at "the curve is concave" without explaining why the intercept is positive. The tangent at p meets the axis at $\ln p - 1 \geq 0$, and the chord crosses above it.
- In part (iv), forgetting to rule out $c = 0$, which is needed for the strict inequality.

Beyond STEP

The function $\frac{\ln x}{x}$ has its maximum at e, and this explains the famous pair $2^4 = 4^2$. The positive solutions of $x^y = y^x$ with $x \neq y$ form a curve, which can be parametrised as $x = t^{1/(t-1)}$, $y = t^{t/(t-1)}$. More generally, "a chord of a concave function lies below the curve and above the tangents" is the basic inequality of convexity. It underlies Jensen's inequality and the AM–GM inequality. Bartle and Sherbert, *Introduction to Real Analysis*, covers the mean value theorem and convex functions carefully.

WORKED SOLUTION

STEP II 2013, Question 2

Pure • Reduction Formulae

STEP II 2013, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s2-q2

The key idea

The integrand $x^n(1-x)^n$ is symmetric under the reflection $x \mapsto 1-x$, and that symmetry lets us swap the roles of x and $1-x$ for free. Integration by parts then trades a power of x for a power of $1-x$, and the identity $x+(1-x)=1$ lets us split a product back into integrals we already know. Together these give a reduction formula, which unwinds to factorials for whole-number n and also works for half-integers.

Solution

Part (i)

Write $J_n = \int_0^1 x^{n-1}(1-x)^n \,\mathrm{d}x$ for $n \geq 1$. Substitute $u = 1-x$, so $\mathrm{d}x = -\mathrm{d}u$ and the limits $0, 1$ become $1, 0$:

$$J_n = \int_1^0 (1-u)^{n-1}u^n\,(-\mathrm{d}u) = \int_0^1 u^n(1-u)^{n-1}\,\mathrm{d}u,$$

which is the required equality (the name of the dummy variable doesn't matter).

Adding the two equal forms of J_n,

$$\begin{aligned} 2J_n &= \int_0^1 \Big[x^{n-1}(1-x)^n + x^n(1-x)^{n-1}\Big]\mathrm{d}x \\ &= \int_0^1 x^{n-1}(1-x)^{n-1}\big[(1-x)+x\big]\mathrm{d}x = I_{n-1}. \end{aligned}$$

Next integrate I_n by parts, with $u = x^n$ and $\mathrm{d}v = (1-x)^n\,\mathrm{d}x$, so $v = -\dfrac{(1-x)^{n+1}}{n+1}$:

$$I_n = \Big[-\frac{x^n(1-x)^{n+1}}{n+1}\Big]_0^1 + \frac{n}{n+1}\int_0^1 x^{n-1}(1-x)^{n+1}\,\mathrm{d}x.$$

The boundary term vanishes at both ends: at $x=1$ because of $(1-x)^{n+1}$, and at $x=0$

because of x^n with $n \geq 1$. This gives the stated formula

$$I_n = \frac{n}{n+1} \int_0^1 x^{n-1}(1-x)^{n+1}\,\mathrm{d}x.$$

Ask yourself How do we get from this integral back to I_n and I_{n-1}?

Split off one factor of $1-x$ and write it as $1-x$ literally:

$$x^{n-1}(1-x)^{n+1} = x^{n-1}(1-x)^n\,(1-x) = x^{n-1}(1-x)^n - x^n(1-x)^n.$$

Integrating, the right-hand side gives $J_n - I_n = \frac{1}{2}I_{n-1} - I_n$, using the result above. So

$$I_n = \frac{n}{n+1}\left(\tfrac{1}{2}I_{n-1} - I_n\right) \quad \Longrightarrow \quad I_n \cdot \frac{2n+1}{n+1} = \frac{n}{2(n+1)}I_{n-1},$$

and multiplying by $\dfrac{n+1}{2n+1}$,

$$I_n = \frac{n}{2(2n+1)}\,I_{n-1},$$

as required. Notice that nothing in this argument used n being an integer: only $n \geq 1$ was needed (for the boundary term and so that every integrand is continuous on $[0,1]$). We shall use that in part (iii).

Part (ii)

We prove $I_n = \dfrac{(n!)^2}{(2n+1)!}$ by induction, starting from $n=0$: $I_0 = \int_0^1 1\,\mathrm{d}x = 1 = \dfrac{(0!)^2}{1!}$. Suppose the formula holds for $n-1$, where $n \geq 1$. Then by part (i)

$$\begin{aligned} I_n = \frac{n}{2(2n+1)} \cdot \frac{\big((n-1)!\big)^2}{(2n-1)!} &= \frac{n^2\big((n-1)!\big)^2}{2n(2n+1)(2n-1)!} \\ &= \frac{(n!)^2}{(2n+1)!}, \end{aligned}$$

where we multiplied top and bottom by n and used $2n(2n+1)(2n-1)! = (2n+1)!$. This completes the induction, so the formula holds for every positive integer n (and for $n=0$).

Part (iii)

With $x = \sin^2\theta$ we have $\mathrm{d}x = 2\sin\theta\cos\theta\,\mathrm{d}\theta$, and θ runs from 0 to $\frac{\pi}{2}$. On this interval $\sin\theta$ and $\cos\theta$ are non-negative, so

$$\big(x(1-x)\big)^{1/2} = \big(\sin^2\theta\cos^2\theta\big)^{1/2} = \sin\theta\cos\theta.$$

Ask yourself Why do we need $\sin\theta\cos\theta \geq 0$ here?

Because $\sqrt{t^2} = |t|$, not t; on $[0, \frac{\pi}{2}]$ the modulus can simply be dropped. Hence

$$I_{\frac{1}{2}} = \int_0^{\pi/2} \sin\theta\cos\theta \cdot 2\sin\theta\cos\theta \,\mathrm{d}\theta = \int_0^{\pi/2} \tfrac{1}{2}\sin^2 2\theta \,\mathrm{d}\theta$$
$$= \int_0^{\pi/2} \tfrac{1}{4}(1-\cos 4\theta)\mathrm{d}\theta = \tfrac{1}{4}\Big[\theta - \tfrac{1}{4}\sin 4\theta\Big]_0^{\pi/2} = \frac{\pi}{8}.$$

For $I_{\frac{3}{2}}$, use the reduction formula from part (i) with $n = \frac{3}{2}$, which is allowed since $\frac{3}{2} \geq 1$ and the derivation never needed n to be an integer:

$$I_{\frac{3}{2}} = \frac{3/2}{2\cdot 4} I_{\frac{1}{2}} = \frac{3}{16}\cdot\frac{\pi}{8} = \frac{3\pi}{128}.$$

As a cross-check, the same substitution gives directly

$$I_{\frac{3}{2}} = \int_0^{\pi/2} 2\sin^4\theta\cos^4\theta \,\mathrm{d}\theta = \frac{1}{8}\int_0^{\pi/2} \sin^4 2\theta \,\mathrm{d}\theta = \frac{1}{8}\cdot\frac{3\pi}{16} = \frac{3\pi}{128},$$

using $\sin^4\phi = \frac{3}{8} - \frac{1}{2}\cos 2\phi + \frac{1}{8}\cos 4\phi$.

Checks

- $I_1 = \int_0^1 (x - x^2)\,\mathrm{d}x = \frac{1}{6}$, and the formula gives $\frac{1}{3!} = \frac{1}{6}$. Likewise $I_2 = \int_0^1 (x^2 - 2x^3 + x^4)\,\mathrm{d}x = \frac{1}{3} - \frac{1}{2} + \frac{1}{5} = \frac{1}{30} = \frac{4}{5!}$.
- $I_{\frac{1}{2}}$ is the area under the semicircle $y = \sqrt{\frac{1}{4} - (x - \frac{1}{2})^2}$ of radius $\frac{1}{2}$, which is $\frac{1}{2}\pi\left(\frac{1}{2}\right)^2 = \frac{\pi}{8}$.
- Numerically, $I_{\frac{3}{2}} \approx 0.073631$, matching $3\pi/128$; the reduction formula was also confirmed by computer for non-integer n such as $n = 2.7$.

Common slips

- Forgetting to check that the boundary term in the integration by parts vanishes, especially at $x = 0$.
- Writing $\sqrt{\sin^2\theta\cos^2\theta} = \sin\theta\cos\theta$ without noting the sign on $[0, \frac{\pi}{2}]$.
- Computing $I_{\frac{3}{2}}$ from the factorial formula of part (ii), which is only proved for integers; the reduction formula is the safe route.

Beyond STEP

I_n is a special case of the beta function $B(p,q) = \int_0^1 x^{p-1}(1-x)^{q-1}\,\mathrm{d}x$, here $I_n = B(n+1, n+1)$, and part (ii) is the integer case of $B(p,q) = \Gamma(p)\Gamma(q)/\Gamma(p+q)$. Putting

$n = \frac{1}{2}$ into that identity recovers $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, which is the deeper reason a π appears in part (iii). Integrals like these lie behind the beta distribution in statistics and Wallis's product for π; Siklos, *Advanced Problems in Mathematics*, has more reduction-formula problems in this spirit.

WORKED SOLUTION

STEP II 2013, Question 3

Pure • Roots of Cubics

STEP II 2013, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s2-q3

The key idea

Everything follows from the shape of a cubic with positive leading coefficient and three distinct real roots: it rises from $-\infty$, crosses the axis upwards, comes back down, then crosses upwards again. The value $f(0) = c$ tells us where 0 sits relative to the roots, and the turning points (the roots of a quadratic, $f'(x) = 0$) can be pinned down by their sum and product. The last part is a reminder that conditions on the turning points alone cannot guarantee that the curve actually reaches the axis three times.

Solution

Throughout, let

$$f(x) = x^3 + 3ax^2 + 3bx + c, \qquad f'(x) = 3\big(x^2 + 2ax + b\big).$$

When $f(x) = 0$ has three distinct real roots $\alpha < \beta < \gamma$, we can write $f(x) = (x-\alpha)(x-\beta)(x-\gamma)$, so, reading off the signs of the three factors,

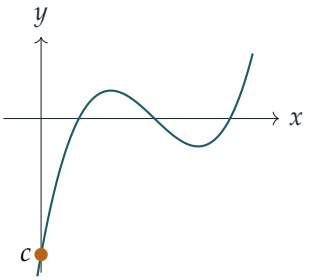
$$f(x) < 0 \text{ on } (-\infty, \alpha) \cup (\beta, \gamma), \qquad f(x) > 0 \text{ on } (\alpha, \beta) \cup (\gamma, \infty).$$

Also, by Rolle's theorem f' has a zero in (α, β) and another in (β, γ). Since f' is a quadratic it has no other zeros, so the turning points are at $p < q$ with

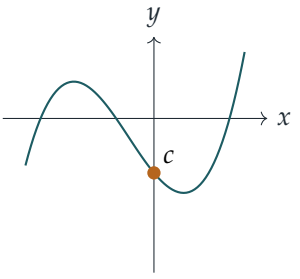
$$\alpha < p < \beta < q < \gamma.$$

Part (i)

The curve $y = f(x)$ meets the y-axis at $(0, c)$, which is below the x-axis since $c < 0$. From the sign pattern above, $f(0) < 0$ means that either $0 < \alpha$ or $\beta < 0 < \gamma$. These are the two pictures below.



all three roots positive



exactly one root positive

If $0 < \alpha$, all three roots are positive. If $\beta < 0 < \gamma$, then $\alpha < \beta < 0$, so exactly one root (γ) is positive. No root is zero, because $f(0) = c \neq 0$. So either exactly one root is positive or all three are, as required.

Part (ii)

Now $0 < \alpha < \beta < \gamma$.

The sign of c. Since $0 < \alpha$, the point 0 lies in $(-\infty, \alpha)$, where $f < 0$. So $c = f(0) < 0$.

The turning points. The equation $x^2 + 2ax + b = 0$ has the two distinct real roots p and q, so its discriminant is positive:

$$4a^2 - 4b > 0 \quad \Longrightarrow \quad a^2 > b.$$

Both turning points are positive, because $p > \alpha > 0$ and $q > p$. For the quadratic $x^2 + 2ax + b$, the sum and product of the roots are

$$p + q = -2a, \qquad pq = b.$$

Since $p, q > 0$, we get $-2a > 0$, so $a < 0$, and $b = pq > 0$. Putting these together,

$$a^2 > b > 0, \qquad a < 0, \qquad c < 0,$$

which is $(*)$.

(You can also get $a < 0$, $b > 0$ and $c < 0$ straight from the roots: $\alpha + \beta + \gamma = -3a$, $\alpha\beta + \beta\gamma + \gamma\alpha = 3b$ and $\alpha\beta\gamma = -c$ are all positive. But $a^2 > b$ really does need the turning points.)

Part (iii)

Now $c = f(0) > 0$, so 0 lies in (α, β) or in (γ, ∞), the two intervals where $f > 0$. That gives two candidate pictures: either $\alpha < 0 < \beta < \gamma$ (one negative root, two positive), or $\alpha < \beta < \gamma < 0$ (all three negative).

? **Ask yourself** How can the condition $ab < 0$ rule one of these out?

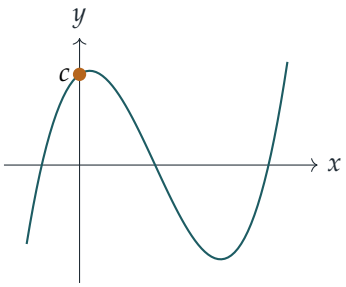
Look at the turning points again. If all three roots were negative, then $p < q < \gamma < 0$, so both turning points would be negative. Then

$$-2a = p + q < 0 \quad \text{and} \quad b = pq > 0,$$

so $a > 0$ and $b > 0$, giving $ab > 0$. This contradicts $ab < 0$. Hence the second picture is impossible, and

$$\alpha < 0 < \beta < \gamma :$$

one root is negative and the other two are positive.



This case really can happen. For example, the roots -1, 2 and 5 give $x^3 - 6x^2 + 3x + 10$, so $a = -2$, $b = 1$ and $c = 10$, with $ab = -2 < 0$ and $c > 0$; this is the curve sketched above.

Part (iv)

Ask yourself What extra condition, beyond $(*)$, does a cubic need to have three real roots?

Condition $(*)$ only controls the turning points and the sign of $f(0)$. For three real roots the local maximum must also be above the axis. So we should choose a and b to give two positive turning points, and then make c negative enough to pull the local maximum below the axis.

Take $a = -1$ and $b = \frac{3}{4}$, so that

$$f'(x) = 3\left(x^2 - 2x + \tfrac{3}{4}\right) = 3\left(x - \tfrac{1}{2}\right)\left(x - \tfrac{3}{2}\right).$$

The local maximum is at $x = \frac{1}{2}$, where

$$f\left(\tfrac{1}{2}\right) = \tfrac{1}{8} - \tfrac{3}{4} + \tfrac{9}{8} + c = \tfrac{1}{2} + c.$$

Choose $c = -1$. The equation is then

$$x^3 - 3x^2 + \tfrac{9}{4}x - 1 = 0 \qquad \left(a = -1,\ b = \tfrac{3}{4},\ c = -1\right),$$

and $a^2 = 1 > \frac{3}{4} = b > 0$, $a < 0$, $c < 0$, so $(*)$ holds.

Now count the roots. f is increasing on $(-\infty, \frac{1}{2}]$, so there $f(x) \le f(\frac{1}{2}) = -\frac{1}{2} < 0$. On $[\frac{1}{2}, \frac{3}{2}]$, f is decreasing, so it stays below $-\frac{1}{2}$. On $[\frac{3}{2}, \infty)$, f increases from $f(\frac{3}{2}) = -1$ to $+\infty$, so it crosses the axis exactly once. Hence there is exactly one real root (it lies between 2 and 3, since $f(2) = -\frac{1}{2}$ and $f(3) = \frac{23}{4}$).

Checks

- The cubic $(x-1)(x-2)(x-3) = x^3 - 6x^2 + 11x - 6$ has $a = -2$, $b = \frac{11}{3}$, $c = -6$, and indeed $a^2 = 4 > \frac{11}{3} > 0$, as part (ii) predicts.
- Reflecting the curve, $x \mapsto -x$, turns roots α, β, γ into $-\alpha, -\beta, -\gamma$ and changes the signs of a and c but not b. So ab and c both change sign, and the mirror image of part (i) says that when $c > 0$ either exactly one root is negative or all three are: exactly the two pictures we started from in part (iii).
- In part (iv) the only real root is $x \approx 2.178$. A computer check of thousands of random cubics with three distinct real roots also agreed with parts (i)–(iii).

Common slips

- Trying to prove $a^2 > b$ from the signs of the roots alone. It comes from the turning points being real and distinct.
- Assuming that $(*)$ is *equivalent* to three positive roots. Part (iv) shows it is only necessary, not sufficient.
- In part (iv), choosing c so that the local *minimum* is negative but the local maximum is still positive. That gives three roots, not one.

Beyond STEP

The sign arguments here are the start of a general theory: Descartes' rule of signs bounds the number of positive roots by the number of sign changes in the coefficients, and Sturm's theorem counts real roots in an interval exactly. For a cubic, the condition missed in part (iv) is the sign of the discriminant, which is (up to a positive factor) the product $-f(p)f(q)$ of the values at the turning points. Allenby's *Rings, Fields and Groups* discusses polynomials and their roots at first-year level.

WORKED SOLUTION

STEP II 2013, Question 4

Pure • Loci of Chord Midpoints

STEP II 2013, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s2-q4

The key idea

The midpoint of a chord can be found without ever finding the chord's ends: the x-coordinates of P and Q are the roots of a quadratic, and their average is minus half the ratio of the first two coefficients. After that, each locus is a matter of eliminating a parameter. Geometry explains every answer: the line from the centre of a circle to the midpoint of a chord is perpendicular to the chord.

Solution

The coordinates of M

The line is $y = b(x - a)$. Substituting into $x^2 + y^2 = 1$ gives

$$(1 + b^2)x^2 - 2ab^2x + a^2b^2 - 1 = 0.$$

Its roots are the x-coordinates x_P, x_Q of P and Q, so by the sum of the roots

$$x_M = \tfrac{1}{2}(x_P + x_Q) = \frac{ab^2}{1 + b^2}.$$

Since M lies on the line, $y_M = b(x_M - a) = b \cdot \dfrac{ab^2 - a - ab^2}{1 + b^2}$, so

$$M = \left(\frac{ab^2}{1 + b^2},\ -\frac{ab}{1 + b^2} \right).$$

For P and Q to exist and be distinct the discriminant must be positive:

$$4a^2b^4 - 4(1 + b^2)(a^2b^2 - 1) = 4\big(1 + b^2 - a^2b^2\big) > 0 \iff a^2b^2 < 1 + b^2.$$

(This says the distance $|ab|/\sqrt{1 + b^2}$ from O to the line is less than 1.) We need this condition in each part.

Part (i)

With $b > 0$ fixed,

$$-by_M = -b \cdot \left(-\frac{ab}{1+b^2}\right) = \frac{ab^2}{1+b^2} = x_M,$$

so $x = -by$ at every point of the locus, as required. This is the line through O with gradient $-1/b$, perpendicular to the chord, just as the key idea predicts.

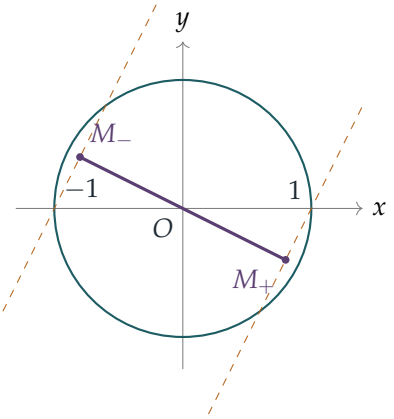
If $-1 \le a \le 1$ then $a^2b^2 \le b^2 < 1 + b^2$, so the line always meets the circle in two points and M is defined for every such a. Both coordinates of M are a times a constant, so as a runs over $[-1, 1]$ the point M moves steadily along the line $x = -by$ from the point given by $a = -1$ to the point given by $a = 1$. The locus is therefore the line segment joining

$$M_\pm = \pm\left(\frac{b^2}{1+b^2}, -\frac{b}{1+b^2}\right),$$

whose length is

$$2\,\frac{\sqrt{b^4+b^2}}{1+b^2} = \frac{2b\sqrt{1+b^2}}{1+b^2} = \frac{2b}{(1+b^2)^{1/2}},$$

using $b > 0$ to take the square root of b^2. This is the required length. The segment is centred at O; its end M_+ is the midpoint of the chord through $(1, 0)$, and M_- that of the chord through $(-1, 0)$. The sketch shows the case $b = 2$, where the ends are $\pm(\frac{4}{5}, -\frac{2}{5})$.



The dashed lines are the extreme chords, through $(\pm 1, 0)$ with gradient 2.

Part (ii)(a)

Now a is fixed with $0 < a < 1$. Then $a^2b^2 < b^2 < 1 + b^2$, so every line through $A = (a, 0)$ cuts the circle twice. From the formula for M,

$$x^2 + y^2 = \frac{a^2b^4 + a^2b^2}{(1+b^2)^2} = \frac{a^2b^2}{1+b^2} = a \cdot \frac{ab^2}{1+b^2} = ax.$$

So M lies on

$$x^2 + y^2 - ax = 0, \quad \text{that is} \quad \left(x - \tfrac{a}{2}\right)^2 + y^2 = \left(\tfrac{a}{2}\right)^2,$$

the circle with centre $(\frac{a}{2}, 0)$ and radius $\frac{a}{2}$, which has OA as a diameter.

Ask yourself Is the whole of this circle traced out?

Not quite. Since $x = ab^2/(1+b^2) = a - a/(1+b^2)$, we always have $0 \le x < a$, so the point $A = (a, 0)$ itself is never reached; it would need the vertical line through A, which has no gradient. Every other point is reached: putting $b = \tan\theta$ with $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$,

$$M = \big(a\sin^2\theta,\ -a\sin\theta\cos\theta\big) = \big(\tfrac{a}{2} - \tfrac{a}{2}\cos 2\theta,\ -\tfrac{a}{2}\sin 2\theta\big),$$

and as 2θ runs over $(-\pi, \pi)$ this covers the circle once, except for the point with $2\theta = \pm\pi$, which is A. So the locus is the circle on OA as diameter with the single point A removed. Geometrically: OM is perpendicular to the chord AM, so $\angle OMA$ is a right angle and M lies on the circle with diameter OA (with $M = O$ when $b = 0$). This circle lies inside the unit circle.

Part (ii)(b)

Now $a = 1/b$ with $0 < b \le 1$. The condition $a^2b^2 < 1 + b^2$ becomes $1 < 1 + b^2$, which holds. The line is $y = b(x - 1/b) = bx - 1$, so it always passes through $B = (0, -1)$, which is on the unit circle; thus B is always one of P, Q. The formula gives

$$M = \left(\frac{b}{1+b^2},\ -\frac{1}{1+b^2}\right), \qquad x^2 + y^2 = \frac{b^2+1}{(1+b^2)^2} = \frac{1}{1+b^2} = -y.$$

So M lies on

$$x^2 + y^2 + y = 0, \quad \text{that is} \quad x^2 + \big(y + \tfrac{1}{2}\big)^2 = \big(\tfrac{1}{2}\big)^2,$$

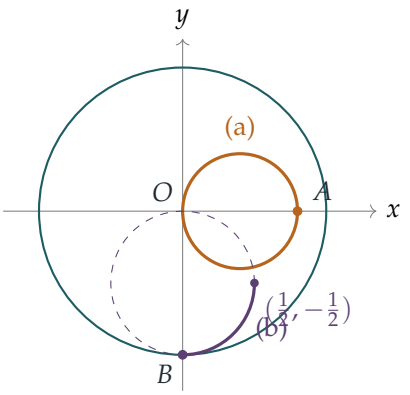
the circle with centre $(0, -\frac{1}{2})$ and radius $\frac{1}{2}$, which has OB as diameter (for the same right-angle reason as in (a)).

This time only part of the circle is traced. With $b = \tan\theta$, $0 < \theta \le \frac{\pi}{4}$,

$$M = \big(\tfrac{1}{2}\sin 2\theta,\ -\tfrac{1}{2} - \tfrac{1}{2}\cos 2\theta\big),$$

and 2θ runs over $(0, \frac{\pi}{2}]$. So the locus is a quarter of the circle: the arc from $B = (0, -1)$, which is not included (it would need $b = 0$), anticlockwise to $(\frac{1}{2}, -\frac{1}{2})$, which is included (from $b = 1$, $a = 1$). On this arc $0 < x \le \frac{1}{2}$ and $-1 < y \le -\frac{1}{2}$.

The sketch shows both loci with the unit circle, taking $a = 0.8$ in (a). The open circle marks the missing point A of (a) and the missing end B of (b).



In (b) the solid arc is the locus; the dashed remainder of the small circle is not reached.

Checks

- With $a = 0$ the line passes through O, so the chord is a diameter and M should be O: the formula gives $(0, 0)$.
- In (i) with $b = 1$: the chord through $(1, 0)$ with gradient 1 joins $(1, 0)$ and $(0, -1)$, with midpoint $(\frac{1}{2}, -\frac{1}{2})$, matching M_+. The segment then has length $\sqrt{2} = 2/\sqrt{2}$, as the formula says.
- The same chord is the $b = 1$ end of the arc in (ii)(b), and $(\frac{1}{2}, -\frac{1}{2})$ satisfies $x^2 + y^2 + y = \frac{1}{4} + \frac{1}{4} - \frac{1}{2} = 0$.
- In (ii)(a), $b = 1$ gives $M = (\frac{a}{2}, -\frac{a}{2})$, the lowest point of the circle on OA, as it should be for a chord at 45°.

Common slips

- Finding P and Q explicitly with the quadratic formula. It works, but the sum of the roots gives M in one line.
- Claiming the whole circle in (ii)(a) or (ii)(b). Always check which values the parameter actually takes and which points they give.
- Forgetting to check that the line actually cuts the circle; for (i) and both cases of (ii) it always does, but this needs saying.

Beyond STEP

The fact that midpoints of parallel chords lie on a line through the centre holds for any conic: for an ellipse that line is called a *conjugate diameter*, and it is a first glimpse of the projective idea of poles and polars. The circles in part (ii) are the classical theorem that an angle in a semicircle is a right angle, run backwards. For more on loci and

parametrisation as a problem-solving tool, Siklos, *Advanced Problems in Mathematics*, is a good next step.

WORKED SOLUTION

STEP II 2013, Question 5

Pure • Symmetry and Curve Sketching

STEP II 2013, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s2-q5

The key idea

If a function is unchanged by a substitution such as $x \mapsto 1-x$ or $x \mapsto 1/x$, then differentiating the identity with the chain rule ties the gradient at x to the gradient at the image point. At a point the substitution leaves fixed, the gradient must equal its own negative, so it is zero. For the equations in part (iii), the symmetry suggests writing everything in terms of $u = x^2 - x$, which is itself unchanged by $x \mapsto 1-x$; a cubic in u is then much easier to handle than a sextic in x.

Solution

Part (i)

Differentiating $\mathrm{f}(x) = \mathrm{f}(1-x)$ with respect to x, using the chain rule on the right,

$$\mathrm{f}'(x) = -\mathrm{f}'(1-x) \qquad \text{for all } x.$$

Putting $x = \frac{1}{2}$ gives $\mathrm{f}'(\frac{1}{2}) = -\mathrm{f}'(\frac{1}{2})$, so $\mathrm{f}'(\frac{1}{2}) = 0$.

Now suppose also that $\mathrm{f}(x) = \mathrm{f}(1/x)$ for $x \neq 0$. Differentiating, since $\frac{\mathrm{d}}{\mathrm{d}x}(\frac{1}{x}) = -\frac{1}{x^2}$,

$$\mathrm{f}'(x) = -\frac{1}{x^2}\,\mathrm{f}'\left(\frac{1}{x}\right) \qquad (x \neq 0).$$

At $x = -1$ (which is its own reciprocal) this reads $\mathrm{f}'(-1) = -\mathrm{f}'(-1)$, so $\mathrm{f}'(-1) = 0$.

At $x = 2$ it gives $\mathrm{f}'(2) = -\frac{1}{4}\mathrm{f}'(\frac{1}{2}) = 0$, by the first result. (Equally, the first identity with $x = 2$ gives $\mathrm{f}'(2) = -\mathrm{f}'(-1) = 0$.)

Part (ii)

Replace x by $1/x$. The numerator becomes

$$\left(\frac{1}{x^2} - \frac{1}{x} + 1\right)^3 = \frac{(1 - x + x^2)^3}{x^6},$$

and the denominator becomes

$$\left(\frac{1}{x^2}-\frac{1}{x}\right)^2=\frac{(1-x)^2}{x^4}.$$

Dividing, and using $x^2(1-x)^2=(x^2-x)^2$,

$$\mathrm{f}\left(\frac{1}{x}\right)=\frac{(x^2-x+1)^3}{x^6}\cdot\frac{x^4}{(1-x)^2}=\frac{(x^2-x+1)^3}{x^2(x-1)^2}=\mathrm{f}(x).$$

Replace x by $1-x$: we have $(1-x)^2-(1-x)=(1-x)(-x)=x^2-x$, so x^2-x is unchanged, and hence so is x^2-x+1. Therefore $\mathrm{f}(1-x)=\mathrm{f}(x)$.

The sketch. By part (i), $x=-1$, $\frac{1}{2}$ and 2 are stationary points; since there are exactly three, these are all of them. Their values are

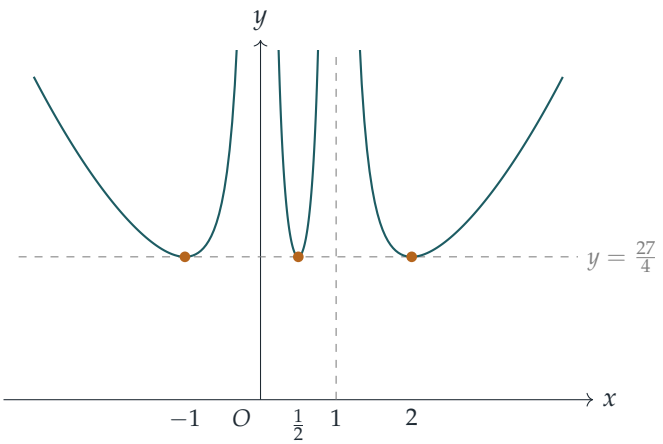
$$\mathrm{f}(-1)=\frac{3^3}{2^2}=\frac{27}{4},\qquad \mathrm{f}(\tfrac{1}{2})=\frac{(3/4)^3}{(1/4)^2}=\frac{27}{4},\qquad \mathrm{f}(2)=\frac{3^3}{2^2}=\frac{27}{4}.$$

All three are the same, as they must be: $\mathrm{f}(2)=\mathrm{f}(\frac{1}{2})$ and $\mathrm{f}(2)=\mathrm{f}(1-2)=\mathrm{f}(-1)$.

Other features:

- $x^2-x+1=(x-\frac{1}{2})^2+\frac{3}{4}>0$, so $\mathrm{f}(x)>0$ everywhere; the curve never meets the x-axis.
- As $x\to 0$ or $x\to 1$ the numerator tends to 1 and the denominator to 0^+, so $\mathrm{f}(x)\to+\infty$ from both sides: $x=0$ and $x=1$ are vertical asymptotes.
- For large $|x|$, $\mathrm{f}(x)\approx x^6/x^4=x^2$, so $\mathrm{f}(x)\to+\infty$ as $x\to\pm\infty$, roughly like a parabola.

So on each of the three intervals $(-\infty,0)$, $(0,1)$, $(1,\infty)$ the function is continuous and tends to $+\infty$ at both ends. Each interval must therefore contain a minimum, which is a stationary point; each contains exactly one of -1, $\frac{1}{2}$, 2, so that point is the minimum and there is no other turning point. Each branch is U-shaped with lowest value $\frac{27}{4}$.



Part (iii)

The equation $f(x) = \frac{27}{4}$**.** From the sketch, $\frac{27}{4}$ is the minimum value on every branch and is attained only at the three stationary points. So the roots are

$$x = -1, \qquad x = \tfrac{1}{2}, \qquad x = 2,$$

and $f(x) > \frac{27}{4}$ for every other x in the domain, that is for

$$x < -1, \quad -1 < x < 0, \quad 0 < x < \tfrac{1}{2}, \quad \tfrac{1}{2} < x < 1, \quad 1 < x < 2, \quad x > 2.$$

We can confirm this algebraically. Put $u = x^2 - x$ (so $u \neq 0$ in the domain); then $f(x) = \dfrac{(u+1)^3}{u^2}$ and

$$f(x) - \frac{27}{4} = \frac{4u^3 - 15u^2 + 12u + 4}{4u^2} = \frac{(u-2)^2(4u+1)}{4u^2}.$$

Since $4u + 1 = 4x^2 - 4x + 1 = (2x-1)^2 \geq 0$, this is never negative, and it is zero exactly when $u = 2$ (that is, $x^2 - x - 2 = 0$, so $x = 2$ or -1) or $x = \frac{1}{2}$.

Ask yourself How can we guess a root of $f(x) = \frac{343}{36}$ without solving a degree-six equation?

The equation $f(x) = \frac{343}{36}$**.** Note $\frac{343}{36} = \frac{7^3}{6^2}$, so we look for x with $x^2 - x + 1 = 7$ and $x^2 - x = 6$. Both hold for $x = 3$, so $f(3) = \frac{343}{36}$. The symmetries now hand us more roots for free: applying $x \mapsto 1 - x$ and $x \mapsto 1/x$ repeatedly to 3 gives

$$3, \quad 1 - 3 = -2, \quad \tfrac{1}{3}, \quad 1 - \tfrac{1}{3} = \tfrac{2}{3}, \quad \tfrac{3}{2}, \quad 1 - \tfrac{3}{2} = -\tfrac{1}{2}.$$

These are six distinct numbers, and clearing denominators turns $f(x) = \frac{343}{36}$ into a polynomial equation of degree six, so these are all the roots:

$$x = -2, \ -\tfrac{1}{2}, \ \tfrac{1}{3}, \ \tfrac{2}{3}, \ \tfrac{3}{2}, \ 3.$$

(In terms of u, the equation is $36u^3 - 235u^2 + 108u + 36 = 0$, which factorises as $(u-6)(9u+2)(4u-3) = 0$, giving the pairs $\{3, -2\}$, $\{\frac{1}{3}, \frac{2}{3}\}$ and $\{\frac{3}{2}, -\frac{1}{2}\}$.)

Since $\frac{343}{36} > \frac{27}{4}$, the horizontal line $y = \frac{343}{36}$ cuts each U-shaped branch twice, once on each side of its minimum, and the branch lies above the line outside these two crossings. The branch on $(-\infty, 0)$ is cut at -2 and $-\frac{1}{2}$, the one on $(0, 1)$ at $\frac{1}{3}$ and $\frac{2}{3}$, and the one on $(1, \infty)$ at $\frac{3}{2}$ and 3. Hence $f(x) > \frac{343}{36}$ exactly when

$$x < -2, \quad -\tfrac{1}{2} < x < 0, \quad 0 < x < \tfrac{1}{3}, \quad \tfrac{2}{3} < x < 1, \quad 1 < x < \tfrac{3}{2}, \quad x > 3.$$

Checks

- Try a value: $f(-\frac{1}{2}) = \dfrac{(7/4)^3}{(3/4)^2} = \dfrac{343/64}{9/16} = \dfrac{343}{36}$, as claimed.

- A point inside a claimed interval: $f(4) = \frac{13^3}{12^2} \approx 15.3 > \frac{343}{36} \approx 9.53$, while $f(2) = 6.75$ lies below, consistent with $(\frac{3}{2}, 3)$ being excluded.
- The six roots come in pairs summing to 1 and in reciprocal pairs, exactly as the two symmetries require; a computer check confirmed the inequalities at 10^5 random points.

Common slips

- Forgetting the factor $-1/x^2$ from the chain rule when differentiating $f(1/x)$. It doesn't matter at $x = -1$, but it is essential to see why the gradient at 2 vanishes.
- Writing the ranges as single intervals such as $x < -2$ or $x > 3$, and forgetting that the asymptotes at $x = 0$ and $x = 1$ split the solution set; $x = 0$ and $x = 1$ must be excluded.
- Treating the stationary points as maxima. Every branch tends to $+\infty$ at both ends, so they are minima.

Beyond STEP

The six maps generated by $x \mapsto 1 - x$ and $x \mapsto 1/x$ form a group (isomorphic to the symmetries of a triangle), and f is, up to a constant factor, the classical j-invariant of the cross-ratio: the function that is unchanged by all six. Functions invariant under a group of substitutions are a central theme of Galois theory and of the theory of elliptic curves. Allenby's *Rings, Fields and Groups* is a good place to meet groups of transformations like this one.

WORKED SOLUTION

STEP II 2013, Question 6

Pure • Convergence of Sequences

STEP II 2013, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s2-q6

This question has a full guided treatment in *Beyond STEP*, Section 2.2, *The Golden Ratio as a Limit*: https://beyondstep.keithmcnulty.org/ch02/golden-limit.html

The key idea

The terms $1, 2, \frac{3}{2}, \frac{5}{3}, \frac{8}{5}, \ldots$ jump alternately above and below their eventual limit, so the whole sequence is not monotonic and the theorem cannot be applied to it directly. Taking every second term, however, gives two monotonic sequences. Part (i) is the tool that shows this: it says that the step $u_{n+2} - u_n$ has the same sign as the previous step $u_n - u_{n-2}$, because the denominator is positive.

Solution

First note that every u_n is positive: $u_1 = 1 > 0$, and if $u_n > 0$ then $u_{n+1} = 1 + 1/u_n > 1 > 0$. So we never divide by zero, and every factor such as $1 + u_n$ is positive.

Part (i)

Using $(*)$ twice,

$$u_{n+2} - u_n = \frac{1}{u_{n+1}} - \frac{1}{u_{n-1}} = \frac{u_{n-1} - u_{n+1}}{u_{n+1}u_{n-1}}, \qquad u_{n+1} - u_{n-1} = \frac{1}{u_n} - \frac{1}{u_{n-2}} = \frac{u_{n-2} - u_n}{u_n u_{n-2}}.$$

(The second needs u_{n-2} to exist, which is why $n \geq 3$.) Substituting the second into the first, the two minus signs cancel:

$$u_{n+2} - u_n = \frac{u_n - u_{n-2}}{(u_n u_{n+1})(u_{n-2}u_{n-1})}.$$

Finally, multiplying $(*)$ by u_n gives $u_n u_{n+1} = u_n + 1$, and likewise $u_{n-2}u_{n-1} = u_{n-2} + 1$. Hence

$$u_{n+2} - u_n = \frac{u_n - u_{n-2}}{(1 + u_n)(1 + u_{n-2})},$$

as required.

Part (ii)

We use induction. The claim $1 \leq u_1 \leq 2$ is true since $u_1 = 1$. Suppose $1 \leq u_n \leq 2$. Taking reciprocals of positive numbers reverses the inequalities, so $\frac{1}{2} \leq 1/u_n \leq 1$, and therefore

$$\tfrac{3}{2} \leq u_{n+1} = 1 + \frac{1}{u_n} \leq 2,$$

which certainly gives $1 \leq u_{n+1} \leq 2$. By induction, $1 \leq u_n \leq 2$ for all n.

Part (iii)

The odd terms. We have $u_1 = 1$ and $u_3 = \frac{3}{2}$, so $u_3 - u_1 > 0$. For odd $n \geq 3$, part (i) says $u_{n+2} - u_n$ is $u_n - u_{n-2}$ multiplied by the positive number $1/\big((1+u_n)(1+u_{n-2})\big)$, so it has the same sign. By induction every step $u_{n+2} - u_n$ (n odd) is positive: the sequence $u_1, u_3, u_5, \ldots$ is increasing. By (ii) it is bounded above by 2, so by the theorem it tends to a limit L, with $1 \leq L \leq 2$ since all its terms lie in $[1, 2]$.

The even terms. Here $u_2 = 2$ and $u_4 = \frac{5}{3}$, so $u_4 - u_2 < 0$, and the same argument shows that $u_2, u_4, u_6, \ldots$ is decreasing. It is bounded below by 1.

> **Ask yourself** The theorem is about increasing sequences that are bounded above. How can we use it for a decreasing sequence?

Apply it to the negatives: $-u_2, -u_4, -u_6, \ldots$ is increasing and bounded above by -1, so it tends to a limit, and hence so does $u_2, u_4, \ldots$; call its limit M. Again $1 \leq M \leq 2$.

Finding the limits. Two steps of $(*)$ give a recurrence linking terms of the same parity:

$$u_{n+2} = 1 + \frac{1}{1 + \dfrac{1}{u_n}} = 1 + \frac{u_n}{u_n + 1} = \frac{2u_n + 1}{u_n + 1}.$$

> **Ask yourself** Why may we simply replace u_{n+2} and u_n by L?

Along the odd n, both u_n and u_{n+2} tend to L. The right-hand side is a continuous function of u_n (its denominator is at least 2), so it tends to $(2L+1)/(L+1)$. Limits are unique, so

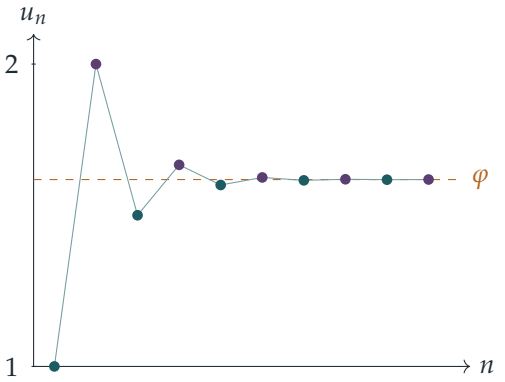
$$L = \frac{2L+1}{L+1} \quad \Longrightarrow \quad L^2 + L = 2L + 1 \quad \Longrightarrow \quad L^2 - L - 1 = 0,$$

so $L = \frac{1}{2}(1 \pm \sqrt{5})$. As $L \geq 1$, we must have $L = \frac{1}{2}(1 + \sqrt{5})$. Exactly the same equation holds for M, with $M \geq 1$, so

$$\lim_{k \to \infty} u_{2k-1} = \lim_{k \to \infty} u_{2k} = \tfrac{1}{2}\big(1 + \sqrt{5}\big).$$

The two limits might have been different (in which case the whole sequence would oscillate), but the equation has only one root that is at least 1.

The whole sequence. Write $\varphi = \frac{1}{2}(1+\sqrt{5})$ and let $\varepsilon > 0$. There is a K_1 such that $|u_n - \varphi| < \varepsilon$ for all odd $n > K_1$, and a K_2 such that the same holds for all even $n > K_2$. Then $|u_n - \varphi| < \varepsilon$ for every $n > \max(K_1, K_2)$, so $u_n \to \varphi$.



The odd terms (teal) climb towards φ and the even terms (plum) descend towards it.

Starting from $u_1 = 3$

The conclusion does not change. Now $u_1 = 3$, $u_2 = \frac{4}{3}$, $u_3 = \frac{7}{4}$, $u_4 = \frac{11}{7}$. The argument needs only small adjustments.

- Part (i) used only $(*)$ and the positivity of the terms, which still holds, so the identity is still true.
- Part (ii) fails at $n = 1$, but $1 \le u_2 = \frac{4}{3} \le 2$, and the inductive step is unchanged, so $1 \le u_n \le 2$ for all $n \ge 2$. Also $u_1 = 3 \ge 1$, so every term is at least 1.
- This time $u_3 - u_1 = -\frac{5}{4} < 0$ and $u_4 - u_2 = \frac{5}{21} > 0$. So the odd terms are decreasing and bounded below by 1, and the even terms are increasing and bounded above by 2: the roles are swapped, but both still converge.
- The recurrence $u_{n+2} = (2u_n + 1)/(u_n + 1)$ still holds and both limits are at least 1, so both are again φ.

So the sequence still tends to $\frac{1}{2}(1+\sqrt{5})$.

Checks

With $u_1 = 1$ the terms are $1, 2, \frac{3}{2}, \frac{5}{3}, \frac{8}{5}, \frac{13}{8}, \ldots$, ratios of consecutive Fibonacci numbers, and $u_{10} = \frac{89}{55} \approx 1.61818$, already very close to $\varphi \approx 1.61803$. With $u_1 = 3$ the terms are $3, \frac{4}{3}, \frac{7}{4}, \frac{11}{7}, \frac{18}{11}, \ldots$ (ratios of Lucas numbers), and these also approach 1.61803. You can also check the identity in (i) with $n = 3$: $u_5 - u_3 = \frac{1}{10}$, and $\dfrac{u_3 - u_1}{(1+u_3)(1+u_1)} = \dfrac{1/2}{5} = \frac{1}{10}$.

Common slips

- Assuming the whole sequence converges and solving $L = 1 + 1/L$ straight away. That gives the right number, but only after convergence has been proved, which is the point of the question.
- Forgetting the root $\frac{1}{2}(1 - \sqrt{5})$: it must be ruled out, here by $L \geq 1$.
- Applying the theorem to the decreasing even subsequence without saying how (for example, via $-u_{2k}$).

Beyond STEP

This is a model for a standard first-year analysis argument: the monotone convergence theorem, subsequences, and the fact that if the odd and even subsequences converge to the same limit then so does the whole sequence. In fact u_n is the nth convergent of the continued fraction $1 + \cfrac{1}{1 + \cfrac{1}{1 + \cdots}}$, and the map $x \mapsto 1 + 1/x$ is a contraction near φ, which is why any positive start leads to φ. Abbott's *Understanding Analysis* treats monotone sequences and subsequences carefully, and Davenport's *The Higher Arithmetic* covers continued fractions.

WORKED SOLUTION

STEP II 2013, Question 7

Pure • Pell Equation Solutions

STEP II 2013, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s2-q7

The key idea

Equations of the form $x^2 - 2y^2 = 1$ are Pell equations: once you have one solution, a fixed linear map produces infinitely many more. Parts (i) and (ii) tie these solutions to triangular numbers that are also squares. In part (iii) the trick is to move a^2 across and factorise $c^4 - a^2$ as a difference of two squares. Since b is prime, b^3 can only be split into two factors in very few ways. Part (ii) then provides a non-prime b for the last part.

Solution

Part (i)

The obvious solution is $x = 1$, $y = 0$, since $1^2 - 2 \cdot 0^2 = 1$.

Now suppose $p^2 - 2q^2 = 1$. Then

$$\begin{aligned}(3p + 4q)^2 - 2(2p + 3q)^2 &= (9p^2 + 24pq + 16q^2) - 2(4p^2 + 12pq + 9q^2) \\ &= p^2 - 2q^2 = 1,\end{aligned}$$

since the pq terms cancel. So $x = 3p + 4q$, $y = 2p + 3q$ is also a solution, as required.

Applying this to $(p, q) = (1, 0)$ gives $(3, 2)$, and applying it again to $(3, 2)$ gives $(3 \cdot 3 + 4 \cdot 2,\ 2 \cdot 3 + 3 \cdot 2) = (17, 12)$. Check: $9 - 8 = 1$ and $289 - 288 = 1$. So two solutions with x a positive odd integer and y a positive even integer are

$$(x, y) = (3, 2) \quad \text{and} \quad (x, y) = (17, 12).$$

(In fact the map always keeps x odd, as $3p + 4q \equiv p \pmod 2$, and keeps y even once it is even, as $2p + 3q \equiv q \pmod 2$. So every later solution in the chain also has the right parities: $(99, 70)$, $(577, 408)$, and so on.)

Part (ii)

Let x be odd and y even, and write $x = 2m+1$ and $y = 2n$ with m, n integers. Then $(*)$ becomes

$$(2m+1)^2 - 2(2n)^2 = 1 \iff 4m^2 + 4m = 8n^2 \iff n^2 = \tfrac{1}{2}m(m+1),$$

dividing by 8 in the last step. This is the required form. (Every step is reversible, so the two equations are equivalent for such x and y.)

So n^2 is the triangular number $1 + 2 + \cdots + m$. From part (i), $(3, 2)$ gives $m = 1$, $n = 1$ (so $1 = 1$), and $(17, 12)$ gives $m = 8$, $n = 6$ (so $36 = \frac{1}{2} \cdot 8 \cdot 9$). We use the second of these in part (iii).

Part (iii)

Rearranging and factorising as a difference of two squares,

$$b^3 = c^4 - a^2 = (c^2 - a)(c^2 + a).$$

Since $b^3 > 0$ and $c^2 + a > 0$, the factor $c^2 - a$ is a positive integer too. Also $a > 0$, so $c^2 - a < c^2 + a$.

Ask yourself How many ways are there to write b^3 as a product of two positive integers?

Because b is prime, the only positive divisors of b^3 are 1, b, b^2 and b^3. So the factorisations of b^3 into a smaller factor times a larger one are $1 \times b^3$ and $b \times b^2$. There are no others, and the two factors can never be equal, since b^3 is not a perfect square.

Case 1: $c^2 - a = 1$ and $c^2 + a = b^3$. Adding and subtracting,

$$c^2 = \tfrac{1}{2}(b^3 + 1), \qquad a = \tfrac{1}{2}(b^3 - 1).$$

Case 2: $c^2 - a = b$ and $c^2 + a = b^2$. Adding and subtracting,

$$c^2 = \tfrac{1}{2}b(b+1), \qquad a = \tfrac{1}{2}b(b-1).$$

(The question only asks for these formulae, but it is worth knowing which cases really happen. Case 2 never does. For $b = 2$ it would need $c^2 = 3$. If b is an odd prime, then b and $\frac{1}{2}(b+1)$ are coprime, because $1 \le \frac{1}{2}(b+1) < b$, and their product is the square c^2, so each must be a square. But a prime is never a square. Case 1 does happen: $b = 23$ gives $\frac{1}{2}(23^3 + 1) = 6084 = 78^2$, so $a = 6083$, $b = 23$, $c = 78$ is a solution.)

The final part

Case 2 fails for primes, but its formulae still give solutions if we drop the requirement that b is prime. Whenever $\frac{1}{2}b(b+1)$ is a perfect square c^2, put $a = \frac{1}{2}b(b-1)$. Then

$$a^2 + b^3 = (c^2 - b)^2 + b^3 = c^4 - 2bc^2 + b^2 + b^3 = c^4 - b^2(b+1) + b^2 + b^3 = c^4,$$

using $2c^2 = b(b+1)$. Part (ii) tells us exactly when $\frac{1}{2}b(b+1)$ is a square: $b = m$ with $n^2 = \frac{1}{2}m(m+1)$. The solution $m = 1$ gives $a = 0$, which is not allowed, but $m = 8$, $n = 6$ works. Take $b = 8$, $c = 6$, $a = \frac{1}{2} \cdot 8 \cdot 7 = 28$:

$$28^2 + 8^3 = 784 + 512 = 1296 = 6^4.$$

So $a = 28$, $b = 8$, $c = 6$ is a solution with $b = 2^3$ not prime.

Checks

- The next solution in the chain, $(99, 70)$, satisfies $9801 - 2 \cdot 4900 = 1$, and it gives $m = 49$, $n = 35$ with $\frac{1}{2} \cdot 49 \cdot 50 = 1225 = 35^2$.
- $6^4 = 1296$ and $28^2 + 8^3 = 784 + 512 = 1296$.
- A computer search over all $c < 400$ found $a = 6083$, $b = 23$, $c = 78$ as the only solution with b prime. It is a Case 1 solution: $78^4 - 6083^2 = (78^2 - 6083)(78^2 + 6083) = 1 \times 12167 = 23^3$.

Common slips

- Forgetting that $c^2 - a$ must be the *smaller* factor, which leads to extra "cases" with negative a.
- Missing the factorisation $1 \times b^3$ and finding only the case $b \times b^2$, or the other way round.
- In the final part, using $m = 1$ from $(3, 2)$, which gives $a = 0$; the question needs a to be positive.

Beyond STEP

The equation $x^2 - 2y^2 = 1$ is Pell's equation. Its solutions are exactly the powers $(3 + 2\sqrt{2})^k = x + y\sqrt{2}$, and the map in part (i) is just multiplication by $3 + 2\sqrt{2}$. This is the start of the theory of units in rings such as $\mathbb{Z}[\sqrt{2}]$ and of continued fractions. Equations like $a^2 + b^3 = c^4$ belong to the study of generalised Fermat equations. For a beautiful introduction to all of this, read Davenport, *The Higher Arithmetic*.

WORKED SOLUTION

STEP II 2013, Question 8

Pure • Calculus and Inequalities

STEP II 2013, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s2-q8

The key idea

Because f is decreasing, the region is a "staircase" that only gets lower as we move right, so the best rectangle sits in the bottom-left corner, on the line $y = \mathrm{f}(t)$ and against the y-axis, with its top right corner on the curve. That reduces part (i) to maximising a function of one variable. Part (ii) compares two areas, the whole region and the rectangle inside it, and part (iii) simply makes everything explicit for $\mathrm{f}(x) = 1/(1+x)$.

Solution

Part (i)

The region is $\{(x,y) : 0 \le x \le t,\ \mathrm{f}(t) \le y \le \mathrm{f}(x)\}$: the curve lies above the line $y = \mathrm{f}(t)$ for $x < t$ since f is decreasing.

Take any rectangle $[a,b] \times [c,d]$ inside the region. Its bottom edge must have $c \ge \mathrm{f}(t)$, its left edge $a \ge 0$, and its top right corner (b,d) must lie in the region, so $b \le t$ and $d \le \mathrm{f}(b)$. Hence its area is at most

$$b\big(\mathrm{f}(b) - \mathrm{f}(t)\big),$$

and this is achieved by the rectangle $[0,b] \times [\mathrm{f}(t), \mathrm{f}(b)]$, which does fit: for $0 \le x \le b$ we have $\mathrm{f}(x) \ge \mathrm{f}(b)$, so the rectangle lies under the curve. So

$$A_0(t) = \max_{0 \le x \le t} R(x), \qquad R(x) = x\big(\mathrm{f}(x) - \mathrm{f}(t)\big).$$

Now $R(0) = R(t) = 0$ while $R(x) > 0$ for $0 < x < t$, so for $t > 0$ the maximum is attained at an interior point $x_0 \in (0,t)$, where $R'(x_0) = 0$:

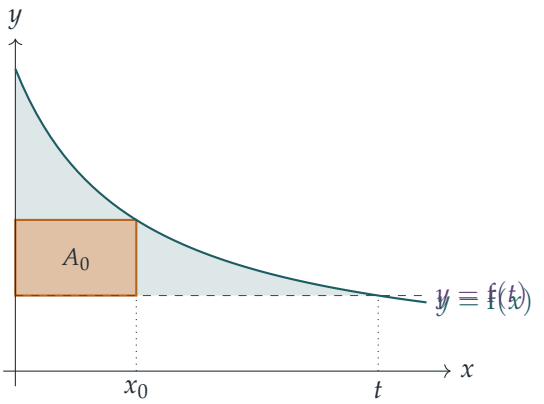
$$R'(x) = \mathrm{f}(x) + x\mathrm{f}'(x) - \mathrm{f}(t) = 0 \quad \Longrightarrow \quad x_0\mathrm{f}'(x_0) + \mathrm{f}(x_0) = \mathrm{f}(t).$$

Thus $A_0(t) = x_0\big(\mathrm{f}(x_0) - \mathrm{f}(t)\big)$ with x_0 satisfying this condition, as required. (For $t = 0$ the region has no width, $A_0(0) = 0$, and $x_0 = 0$ satisfies the condition trivially.)

Part (ii)

Write $t\,g(t) = \int_0^t f(x)\,dx$. Differentiating both sides, using the product rule on the left and the fundamental theorem of calculus on the right,

$$g(t) + t\,g'(t) = f(t) \quad \Longrightarrow \quad t\,g'(t) = f(t) - g(t).$$



The integral $\int_0^t \big(f(x) - f(t)\big)\,dx$ is exactly the area of the shaded region, between the curve and the line $y = f(t)$ for $0 \le x \le t$. The optimal rectangle from part (i) lies inside this region, and it does not fill it: for $x_0 < x < t$ the rectangle contributes nothing, but the region has height $f(x) - f(t) > 0$ there, since f is strictly decreasing. So that sliver has positive area, and

$$\int_0^t \big(f(x) - f(t)\big)\,dx > A_0(t).$$

? **Ask yourself** How do we turn the left-hand side into something involving g'?

Split the integral and use the definition of g, then the result just proved:

$$\begin{aligned}\int_0^t \big(f(x) - f(t)\big)\,dx &= t\,g(t) - t\,f(t)\\ &= -t\big(f(t) - g(t)\big)\\ &= -t\cdot t\,g'(t) = -t^2 g'(t).\end{aligned}$$

Hence $-t^2 g'(t) > A_0(t)$.

Part (iii)

Here $f(x) = \dfrac{1}{1+x}$ is positive and strictly decreasing for $x \ge 0$, so everything above applies.

The left-hand side. From the calculation at the end of part (ii),

$$-t^2 g'(t) = \int_0^t \frac{dx}{1+x} - \frac{t}{1+t} = \ln(1+t) - \frac{t}{1+t}.$$

The rectangle. With $\mathrm{f}'(x) = -\dfrac{1}{(1+x)^2}$,

$$x\mathrm{f}'(x) + \mathrm{f}(x) = \frac{-x + (1+x)}{(1+x)^2} = \frac{1}{(1+x)^2},$$

so the condition for x_0 is $(1+x_0)^2 = 1+t$. Write $s = \sqrt{1+t}$, so $s > 1$ and $x_0 = s-1$ (the positive root; note $0 < x_0 < t$). Then $1+t = s^2$ and

$$A_0(t) = (s-1)\left(\frac{1}{s} - \frac{1}{s^2}\right) = \frac{(s-1)^2}{s^2} = \left(1 - \frac{1}{s}\right)^2.$$

Ask yourself The two sides look messy: what is the natural variable?

Everything is cleaner in terms of s. We have $\ln(1+t) = 2\ln s$ and $\dfrac{t}{1+t} = 1 - \dfrac{1}{s^2}$, so $-t^2\mathrm{g}'(t) > A_0(t)$ becomes

$$\begin{aligned} & 2\ln s - 1 + \frac{1}{s^2} > 1 - \frac{2}{s} + \frac{1}{s^2} \\ \iff \quad & 2\ln s > 2 - \frac{2}{s} \\ \iff \quad & \ln s > 1 - \frac{1}{s}. \end{aligned}$$

Since $s = \sqrt{1+t}$ and $\ln s = \ln\sqrt{1+t}$, this is

$$\ln\sqrt{1+t} > 1 - \frac{1}{\sqrt{1+t}} \qquad (t > 0),$$

as required.

Checks

- With $t = 3$: $s = 2$, $x_0 = 1$, $A_0 = \frac{1}{4}$, and the region has area $\ln 4 - \frac{3}{4} \approx 0.636 > 0.25$. The final inequality reads $\ln 2 \approx 0.693 > 0.5$.
- A direct search over rectangles $x\big(\mathrm{f}(x) - \mathrm{f}(t)\big)$ for many values of t confirmed, by computer, that the maximum is $(1 - 1/\sqrt{1+t})^2$, attained at $x = \sqrt{1+t} - 1$.
- As $t \to 0^+$, both sides of the final inequality tend to 0; expanding, $\ln s \approx (s-1) - \frac{1}{2}(s-1)^2$ and $1 - \frac{1}{s} \approx (s-1) - (s-1)^2$, so the left side wins by about $\frac{1}{2}(s-1)^2$, as it should.

Common slips

- Assuming without comment that the best rectangle has its corner on the curve: it needs the decreasing property, as in part (i).

- Forgetting to say why the area inequality is *strict*: the rectangle misses the part of the region with $x_0 < x < t$.
- Taking the negative root $1 + x_0 = -\sqrt{1+t}$, which gives $x_0 < 0$.

Beyond STEP

The final inequality, $\ln s > 1 - 1/s$ for $s > 1$, is one half of the classic bounds $1 - \frac{1}{s} < \ln s < s - 1$, which come from comparing $\int_1^s \mathrm{d}u/u$ with rectangles; this question reaches it by a cleverer geometric route. Comparing integrals with areas of rectangles is the heart of the Riemann integral, which first-year analysis courses build carefully from upper and lower sums. Abbott's *Understanding Analysis* gives a very readable account.

WORKED SOLUTION

STEP II 2013, Question 9

Mechanics • Friction in Equilibrium

STEP II 2013, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s2-q9

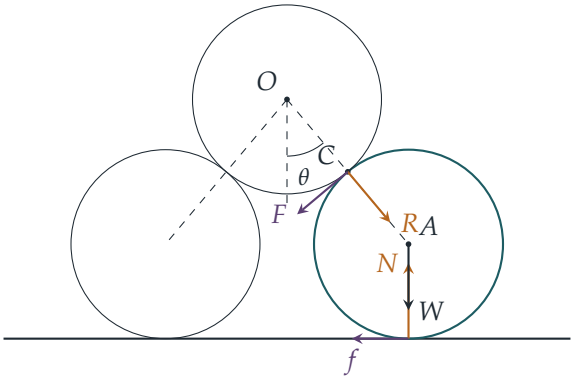
The key idea

Symmetry does most of the work. Treating all three discs as one system gives the ground reactions straight away. Then take moments about the centre of a bottom disc: the friction at the ground and the friction from the top disc both act along tangents, so they have the same lever arm (the radius) and must be equal in size. Once you know that, resolving forces on the bottom disc and on the top disc gives two equations for the two unknowns.

Solution

We model the discs as identical uniform rigid discs of radius r, each of weight W, all in one vertical plane. Let O be the centre of the top disc and A the centre of the right-hand bottom disc, so $OA = 2r$ and OA makes angle θ with the downward vertical. The contact point C between these two discs is the midpoint of OA.

By the left–right symmetry, the forces on the left-hand bottom disc mirror those on the right-hand one, so we only need to work with the right-hand disc. At C the right-hand disc feels a normal reaction R along OA (pushing it away from O) and a friction force F along the common tangent. At the ground it feels a normal reaction N and a friction force f.



The diagram shows the forces on the right-hand bottom disc. By Newton's third law the top disc feels R and F in the opposite directions at C, so friction pushes the top disc upwards and outwards along the tangent.

Part (i)

Take the three discs together as one system. The forces between discs are internal and cancel, so the only external forces are the three weights and the forces from the ground. The two ground reactions are equal by symmetry, and the frictional forces at the ground are horizontal, so resolving vertically gives

$$2N = 3W \quad \Longrightarrow \quad N = \tfrac{3}{2}W,$$

as required.

Part (ii)

> **Ask yourself** There seem to be three unknowns, R, F and f, on the bottom disc. Where does a third equation come from?

From moments. Take moments about A for the right-hand bottom disc. The weight W, the normal reaction N and the normal reaction R all pass through A, so they have no moment. The friction f acts along the tangent at the lowest point and F acts along the tangent at C, so each has lever arm r. They must turn the disc in opposite senses, so

$$fr = Fr \quad \Longrightarrow \quad f = F.$$

(This also fixes the directions in the diagram. The disc is pushed outwards, so f points inwards, and then F on the bottom disc must point down and inwards along the tangent to balance its moment.)

The unit vector from O to A is $(\sin\theta, -\cos\theta)$, and the tangent direction of F on the bottom disc is $(-\cos\theta, -\sin\theta)$. Resolving horizontally for the bottom disc:

$$R\sin\theta - F\cos\theta - f = 0 \quad \Longrightarrow \quad R\sin\theta = F(1+\cos\theta),$$

using $f = F$.

For the top disc, the two contacts give equal upward components, each $R\cos\theta + F\sin\theta$. Resolving vertically:

$$2(R\cos\theta + F\sin\theta) = W.$$

Substituting $F = \dfrac{R\sin\theta}{1+\cos\theta}$:

$$W = 2R\left(\cos\theta + \frac{\sin^2\theta}{1+\cos\theta}\right) = 2R\cdot\frac{\cos\theta+\cos^2\theta+\sin^2\theta}{1+\cos\theta} = 2R\cdot\frac{1+\cos\theta}{1+\cos\theta} = 2R.$$

So the normal reaction between two discs in contact is

$$R = \tfrac{1}{2}W,$$

and it does not depend on θ. Then

$$F = \frac{R\sin\theta}{1+\cos\theta} = \frac{W\sin\theta}{2(1+\cos\theta)},$$

which is the required frictional force.

As a consistency check, resolving vertically for the bottom disc gives $N = W + R\cos\theta + F\sin\theta = W + \frac{1}{2}W = \frac{3}{2}W$, agreeing with part (i).

Part (iii)

Using the half-angle formulae $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$ and $1+\cos\theta = 2\cos^2\frac{\theta}{2}$,

$$\frac{F}{R} = \frac{\sin\theta}{1+\cos\theta} = \tan\tfrac{\theta}{2}.$$

Equilibrium needs friction not to exceed its limiting value at every contact. Between the discs this means $F \le \mu R$, that is,

$$\tan\tfrac{\theta}{2} \le \mu.$$

(At the ground we need $f \le \mu N$, i.e. $\frac{1}{2}W\tan\frac{\theta}{2} \le \frac{3}{2}\mu W$. This is weaker, so the contact between the discs is the one that matters.)

> **?** **Ask yourself** What stops θ being as small as we like, which would make $\tan\frac{\theta}{2}$ tiny?

The bottom discs are not in contact. Their centres are a distance $2\cdot 2r\sin\theta = 4r\sin\theta$ apart, and this must exceed $2r$, so $\sin\theta > \frac{1}{2}$. Since $0 < \theta < \frac{\pi}{2}$, this means $\theta > \frac{\pi}{6}$, so $\frac{\theta}{2} > \frac{\pi}{12}$. As tan is increasing on $(0, \frac{\pi}{2})$,

$$\tan\tfrac{\theta}{2} > \tan\tfrac{\pi}{12} = \tan(45^\circ - 30^\circ) = \frac{1-\frac{1}{\sqrt{3}}}{1+\frac{1}{\sqrt{3}}} = \frac{\sqrt{3}-1}{\sqrt{3}+1} = \frac{(\sqrt{3}-1)^2}{2} = 2-\sqrt{3}.$$

So equilibrium requires $\mu \ge \tan\frac{\theta}{2} > 2-\sqrt{3}$. If $\mu < 2-\sqrt{3}$, no allowed value of θ satisfies this, and equilibrium is impossible.

Checks

- When $\theta \to 0$ the top disc sits directly on the others and $F \to 0$: no friction is needed, as you'd expect.
- For any sample angle, such as $\theta = 1$ radian, substituting $R = \frac{1}{2}W$, $F = f = \frac{1}{2}W\tan\frac{1}{2}$ and $N = \frac{3}{2}W$ makes both force components and the moment balance for each disc.
- $2 - \sqrt{3} \approx 0.2679$, which matches $\tan 15^\circ \approx 0.2679$. A numerical scan over θ just above 30° confirms equilibrium is impossible for $\mu = 0.2679$ and possible for $\mu = 0.268$.

Common slips

- Assuming the friction between the discs is zero or forgetting it altogether. Without it the bottom disc's moment equation forces $f = 0$, and nothing would stop the bottom discs sliding apart.
- Missing the condition that the bottom discs do not touch. It is the only thing that bounds θ from below, and so it is where $2 - \sqrt{3}$ comes from.
- Getting the direction of F wrong. Use the moment equation about A to fix it rather than guessing.

Beyond STEP

This is a statically determinate problem only because of symmetry and the moment equation; in many piles of rough bodies the friction forces cannot be found from equilibrium alone, and engineers need to model how the bodies deform. The neat result $F/R = \tan\frac{\theta}{2}$ is an example of the friction-angle idea: equilibrium is possible exactly when the total contact force lies within an angle $\arctan\mu$ of the normal. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats friction and rigid-body statics carefully at first-year level.

WORKED SOLUTION

STEP II 2013, Question 10

Mechanics • Projectile Motion

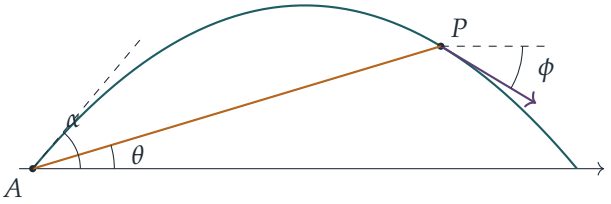
STEP II 2013, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s2-q10

The key idea

Write down the position and velocity at time t and you'll find that $\tan\theta$ and $\tan\phi$ are both *linear* in t. The identity in (i) drops straight out, and it turns both B and C into statements about the tangents of angles. Because both angles decrease steadily with time, "which comes first" is just a comparison of two numbers, and (i) lets us make that comparison at the single point B.



At a general point P, θ is the angle of elevation of P seen from A and ϕ is the angle the velocity makes with the horizontal (negative in the picture, since the particle is falling).

Solution

We model the particle as moving freely under gravity g with no air resistance. Let it be projected with speed u at time $t = 0$, with $0 < \alpha < 90°$. Taking A as origin, at time t

$$x = ut\cos\alpha, \qquad y = ut\sin\alpha - \tfrac{1}{2}gt^2, \qquad \dot{x} = u\cos\alpha, \qquad \dot{y} = u\sin\alpha - gt.$$

It lands when $y = 0$ again, at $t = 2u\sin\alpha/g$.

Part (i)

For $t > 0$ the angle of elevation of the particle from A satisfies

$$\tan\theta = \frac{y}{x} = \frac{ut\sin\alpha - \frac{1}{2}gt^2}{ut\cos\alpha} = \tan\alpha - \frac{gt}{2u\cos\alpha},$$

and the direction of motion satisfies

$$\tan\phi = \frac{\dot{y}}{\dot{x}} = \frac{u\sin\alpha - gt}{u\cos\alpha} = \tan\alpha - \frac{gt}{u\cos\alpha}.$$

Doubling the first and using the second,

$$2\tan\theta = 2\tan\alpha - \frac{gt}{u\cos\alpha} = \tan\alpha + \left(\tan\alpha - \frac{gt}{u\cos\alpha}\right) = \tan\alpha + \tan\phi,$$

as required.

It's worth recording what we'll use later: writing $k = \dfrac{g}{u\cos\alpha} > 0$,

$$\tan\theta = \tan\alpha - \tfrac{1}{2}kt, \qquad \tan\phi = \tan\alpha - kt, \tag{$*$}$$

so both $\tan\theta$ and $\tan\phi$ strictly decrease as time goes on. Over the flight θ falls from α to 0 and ϕ falls from α to $-\alpha$, so each of $\theta = \frac{1}{2}\alpha$ and $\phi = -\frac{1}{2}\alpha$ happens exactly once: B and C are well defined.

Part (ii)

Write $T = \tan\frac{1}{2}\alpha$; since $0 < \frac{1}{2}\alpha < 45^\circ$, we have $0 < T < 1$. At B, $\tan\theta = T$. At C, $\tan\phi = \tan(-\frac{1}{2}\alpha) = -T$. If B and C are the same point, both hold there, and (i) gives

$$2T = \tan\alpha - T, \qquad \text{that is} \qquad \tan\alpha = 3T.$$

By the double-angle formula $\tan\alpha = \dfrac{2T}{1-T^2}$, so

$$\frac{2T}{1-T^2} = 3T \implies 2 = 3(1-T^2) \implies T^2 = \tfrac{1}{3},$$

where we divided by T, which is allowed because $T > 0$. Taking the positive root, $\tan\frac{1}{2}\alpha = \frac{1}{\sqrt{3}}$, so $\frac{1}{2}\alpha = 30^\circ$ and

$$\alpha = 60^\circ.$$

Part (iii)

> **Ask yourself** How can we tell which point is reached first, without working out the two times?

By $(*)$, ϕ decreases throughout the flight. So C comes *after* B exactly when, at the moment the particle is at B, ϕ has not yet fallen to $-\frac{1}{2}\alpha$; that is, when $\tan\phi_B > -T$, where ϕ_B is the direction of motion at B.

At B, part (i) gives $\tan\phi_B = 2\tan\theta - \tan\alpha = 2T - \tan\alpha$. Hence

$$B \text{ is reached first} \iff 2T - \tan\alpha > -T \iff \tan\alpha < 3T.$$

Using $\tan\alpha = \dfrac{2T}{1-T^2}$ again, and dividing by $T > 0$ and multiplying by $1 - T^2 > 0$ (which keeps the inequality's direction),

$$\frac{2T}{1-T^2} < 3T \iff 2 < 3 - 3T^2 \iff T^2 < \tfrac{1}{3} \iff \tfrac{1}{2}\alpha < 30^\circ,$$

the last step because tan is increasing on $(0^\circ, 45^\circ)$. Every step is reversible, so

$$\alpha < 60^\circ \iff B \text{ is reached before } C.$$

So, given $\alpha < 60^\circ$, the particle reaches B first.

If you prefer the times explicitly: from $(*)$, $kt_B = 2(\tan\alpha - T)$ and $kt_C = \tan\alpha + T$, and $t_B < t_C$ is again equivalent to $\tan\alpha < 3T$.

Checks

- At $t = 0$ both formulas in $(*)$ give $\tan\alpha$, as they should: just after launch the particle is moving along the line of projection. At landing, $kt = 2\tan\alpha$, giving $\tan\theta = 0$ and $\tan\phi = -\tan\alpha$, the familiar symmetry of the trajectory.
- At $\alpha = 60^\circ$, $\tan\alpha = \sqrt{3} = 3\tan 30^\circ$, consistent with (ii).
- With $u = 10\,\mathrm{m\,s^{-1}}$, $g = 9.8\,\mathrm{m\,s^{-2}}$ and $\alpha = 45^\circ$, the formulas give $t_B \approx 0.845\,\mathrm{s}$ and $t_C \approx 1.020\,\mathrm{s}$, so B comes first; stepping the motion numerically by computer gives the same times, and also confirms that C comes first for $\alpha = 61^\circ$, 75° and 85°.

Common slips

- Treating θ (the direction of the particle *from* A) as if it were ϕ (the direction of its *velocity*). They agree only at the instant of launch.
- Getting the sign wrong at C: there $\tan\phi = -\tan\frac{1}{2}\alpha$, not $+\tan\frac{1}{2}\alpha$.
- In (iii), claiming the answer without saying why "$\tan\phi_B > -T$" means B comes first. It's the monotonic decrease of ϕ that justifies it.

Beyond STEP

The identity in (i) says that the chord from A to any point of a parabola has a slope equal to the average of the slopes at its two ends, a special property of quadratics (it's the mean value theorem with the midpoint pinned down exactly). In a first-year mechanics course you'd meet the same ideas when studying projectiles on inclined planes, where the angle of the line from the launch point plays exactly the role of θ here. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good place to see projectile problems treated with this kind of care.

WORKED SOLUTION

STEP II 2013, Question 11

Mechanics • Successive Collisions

STEP II 2013, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

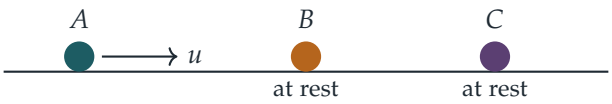
step.maths.org/questions/13-s2-q11

The key idea

Every collision here is between two equal masses, so each one is the same little calculation: conservation of momentum plus Newton's law of restitution. Do it once in general, then reuse it. Whether a further collision happens is just a question of whether the particle behind is moving faster than the particle in front, so the whole question comes down to comparing velocities.

Solution

Model the particles as identical particles of mass m on a smooth surface, so no horizontal forces act between collisions and each particle moves at constant velocity. Call them A, B, C in order along the line, with A the one projected towards B and C. Take the direction of A's initial motion as positive.



A general collision

Suppose a particle with velocity p catches up with an identical particle in front of it with velocity $q < p$, and afterwards they have velocities p' and q'. Momentum (cancelling m) and restitution give

$$p' + q' = p + q, \qquad q' - p' = e(p - q).$$

Adding and subtracting,

$$p' = \tfrac{1}{2}\big[(p+q) - e(p-q)\big], \qquad q' = \tfrac{1}{2}\big[(p+q) + e(p-q)\big]. \tag{$*$}$$

Since $e > 0$ and $p > q$, we always get $q' > p'$: after a collision the front particle moves faster than the rear one, so those two cannot meet again unless something else changes

one of their velocities.

Part (i)

First collision (A hits B): put $p = u$, $q = 0$ in $(*)$:

$$v_A = \tfrac{1}{2}u(1-e), \qquad v_B = \tfrac{1}{2}u(1+e).$$

Now $v_B > v_A$, so A cannot catch B; but B moves towards the stationary C, so the second collision is B with C.

Second collision (B hits C): put $p = \frac{1}{2}u(1+e)$, $q = 0$:

$$v_B = \tfrac{1}{2}\cdot\tfrac{1}{2}u(1+e)(1-e) = \tfrac{1}{4}u(1-e^2), \qquad v_C = \tfrac{1}{4}u(1+e)^2.$$

A is untouched, so it still has $v_A = \frac{1}{2}u(1-e)$. All three velocities are positive (as $0 < e < 1$), so these are the speeds $\frac{1}{2}u(1-e)$, $\frac{1}{4}u(1-e^2)$, $\frac{1}{4}u(1+e)^2$, as required.

A third collision. C is faster than B (by $(*)$), so B and C separate. The only possible next collision is A catching B, which happens exactly when $v_A > v_B$:

$$v_A - v_B = \tfrac{1}{4}u(1-e)\big[2-(1+e)\big] = \tfrac{1}{4}u(1-e)^2 > 0,$$

since $e \neq 1$. So A is moving faster than B and behind it, and a third collision (A with B) occurs for every e in $0 < e < 1$.

Part (ii)

Third collision (A hits B): here $p = \frac{1}{2}u(1-e)$ and $q = \frac{1}{4}u(1-e^2)$, so

$$p+q = \tfrac{1}{4}u(1-e)\big[2+(1+e)\big] = \tfrac{1}{4}u(1-e)(3+e), \qquad p-q = \tfrac{1}{4}u(1-e)^2.$$

By $(*)$, the new velocity of B is

$$\begin{aligned} v_B &= \tfrac{1}{2}\Big[\tfrac{1}{4}u(1-e)(3+e) + \tfrac{1}{4}eu(1-e)^2\Big] \\ &= \tfrac{1}{8}u(1-e)\big[3+e+e-e^2\big] \\ &= \tfrac{1}{8}u(1-e)(3+2e-e^2) = \tfrac{1}{8}u(1-e)(1+e)(3-e), \end{aligned}$$

and (for completeness) $v_A = \frac{1}{8}u(1-e)\big[3+e-e(1-e)\big] = \frac{1}{8}u(1-e)(3+e^2)$, which is less than v_B.

> **Ask yourself** After the third collision, which pairs could possibly collide again?

A and B have just collided, so by $(*)$ B is now faster than A and they are separating. C is still moving at $\frac{1}{4}u(1+e)^2$, and there is nothing beyond C. So a fourth collision happens if and only if B catches C, that is, if and only if $v_B > v_C$: dividing by $\frac{1}{8}u(1+e) > 0$,

$$\begin{aligned} \tfrac{1}{8}u(1-e)(1+e)(3-e) > \tfrac{1}{4}u(1+e)^2 &\iff (1-e)(3-e) > 2(1+e) \\ &\iff e^2 - 6e + 1 > 0. \end{aligned}$$

The quadratic $e^2 - 6e + 1$ has roots $e = 3 \pm 2\sqrt{2}$, and it is positive outside them. The larger root $3 + 2\sqrt{2}$ exceeds 1, while $3 - 2\sqrt{2} \approx 0.172$ lies in $(0, 1)$. So, for $0 < e < 1$,

$$\text{there is a fourth collision} \iff e < 3 - 2\sqrt{2}.$$

At $e = 3 - 2\sqrt{2}$ exactly, B and C move with equal velocity and never meet, so the inequality is strict.

Checks

- Momentum: after the second collision the total is $\frac{1}{4}u\left[2(1-e) + (1-e^2) + (1+e)^2\right] = \frac{1}{4}u \cdot 4 = u$, as it should be.
- Limiting cases: when $e \to 1$ the velocities after two collisions tend to 0, 0, u (the familiar "Newton's cradle" exchange) and $v_A - v_B \to 0$, so the third collision only just survives; the critical value $3 - 2\sqrt{2}$ lies well below 1, consistent with this.
- Numerically, with $e = 0.1$ the velocities after the third collision are $v_B \approx 0.3589u$ and $v_C = 0.3025u$, so B does catch C; with $e = 0.2$, $v_B = 0.336u < v_C = 0.36u$ and it does not. A computer simulation of the motion confirmed exactly three collisions for every $e > 0.1716$ and at least four below it.

Common slips

- Forgetting to justify *which* pair collides next. Each time, check that the two particles that have just collided are separating and that the remaining pair is closing.
- Getting the sign wrong in the restitution law: it is (speed of separation) $= e \times$ (speed of approach), so $q' - p' = e(p - q)$, with the front particle's velocity first on the left.
- Keeping the root $3 + 2\sqrt{2}$, which is outside the allowed range $0 < e < 1$.

Beyond STEP

The rule $(*)$ is a linear map taking the pair of velocities before a collision to the pair after, so a sequence of collisions is a product of matrices; asking how many collisions occur in total is a classic problem, with a beautiful geometric answer for elastic billiard-ball systems by "unfolding" the motion into a straight line in a wedge. For collisions, momentum and frames of reference treated with undergraduate care, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP II 2013, Question 12

Probability and Statistics • Poisson Distribution Moments

STEP II 2013, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s2-q12

The key idea

X keeps the odd values of U and Y keeps the even ones, so every expectation is a Poisson sum taken over only the odd or only the even terms. The trick for $\mathrm{E}(U)$ and $\mathrm{E}(U(U-1))$ still works: cancel k or $k(k-1)$ against $k!$ and pull out λ or λ^2. Each step down in k switches between odd and even, which is how α and β swap places. Note also that $\alpha+\beta=\mathrm{e}^{\lambda}$, so the factor $\mathrm{e}^{-\lambda}$ is $1/(\alpha+\beta)$.

Solution

Throughout, $\mathrm{P}(U=k)=\mathrm{e}^{-\lambda}\lambda^k/k!$ for $k=0,1,2,\ldots$, with $\lambda>0$. Adding the two series term by term gives

$$\alpha+\beta=\sum_{k\geq 0}\frac{\lambda^k}{k!}=\mathrm{e}^{\lambda},\qquad\text{so}\qquad \mathrm{e}^{-\lambda}=\frac{1}{\alpha+\beta}.$$

(In fact $\alpha=\cosh\lambda$ and $\beta=\sinh\lambda$.)

Part (i)

X is 0 unless U is odd, so only odd k contribute:

$$\mathrm{E}(X)=\sum_{k\ \text{odd}}k\,\frac{\mathrm{e}^{-\lambda}\lambda^k}{k!}=\mathrm{e}^{-\lambda}\lambda\sum_{k\ \text{odd}}\frac{\lambda^{k-1}}{(k-1)!}=\mathrm{e}^{-\lambda}\lambda\left(1+\frac{\lambda^2}{2!}+\frac{\lambda^4}{4!}+\cdots\right),$$

because $k-1$ runs over $0,2,4,\ldots$. So

$$\mathrm{E}(X)=\frac{\lambda\alpha}{\alpha+\beta}.$$

In the same way, Y picks up the even $k\geq 2$ (the $k=0$ term is zero anyway), and then $k-1$ runs over $1,3,5,\ldots$:

$$\mathrm{E}(Y)=\mathrm{e}^{-\lambda}\lambda\sum_{k\ \text{even},k\geq 2}\frac{\lambda^{k-1}}{(k-1)!}=\frac{\lambda\beta}{\alpha+\beta}.$$

As a check, $\mathrm{E}(X)+\mathrm{E}(Y)=\lambda=\mathrm{E}(U)$, as it must be, because $X+Y=U$.

Part (ii)

Ask yourself How do we sum $k^2\lambda^k/k!$ when k^2 doesn't cancel neatly against $k!$?

Write $k^2=k(k-1)+k$. The $k(k-1)$ part cancels two factors of $k!$ and the k part cancels one. For odd k, the $k(k-1)$ sum starts at $k=3$ (it vanishes at $k=1$), and $k-2$ runs over $1,3,5,\ldots$:

$$\sum_{k\text{ odd}} k(k-1)\frac{\lambda^k}{k!}=\lambda^2\sum_{k\text{ odd},k\geq 3}\frac{\lambda^{k-2}}{(k-2)!}=\lambda^2\beta,\qquad \sum_{k\text{ odd}} k\frac{\lambda^k}{k!}=\lambda\alpha$$

(the second sum is the one from part (i)). Hence

$$\mathrm{E}(X^2)=\mathrm{e}^{-\lambda}\left(\lambda\alpha+\lambda^2\beta\right)=\frac{\lambda\alpha+\lambda^2\beta}{\alpha+\beta},$$

and subtracting $\mathrm{E}(X)^2$ gives

$$\mathrm{Var}(X)=\frac{\lambda\alpha+\lambda^2\beta}{\alpha+\beta}-\frac{\lambda^2\alpha^2}{(\alpha+\beta)^2},$$

as required.

For Y the parities swap. For even k the $k(k-1)$ sum starts at $k=2$ with $k-2$ running over $0,2,4,\ldots$, which gives $\lambda^2\alpha$. The k sum gives $\lambda\beta$, as in part (i). So

$$\mathrm{Var}(Y)=\frac{\lambda\beta+\lambda^2\alpha}{\alpha+\beta}-\frac{\lambda^2\beta^2}{(\alpha+\beta)^2}.$$

The final question. Because $X+Y=U$, we have $\mathrm{Var}(X+Y)=\mathrm{Var}(U)=\lambda$. Adding the two variances,

$$\begin{aligned}\mathrm{Var}(X)+\mathrm{Var}(Y)&=\frac{\lambda(\alpha+\beta)+\lambda^2(\alpha+\beta)}{\alpha+\beta}-\frac{\lambda^2(\alpha^2+\beta^2)}{(\alpha+\beta)^2}\\&=\lambda+\lambda^2\frac{(\alpha+\beta)^2-\alpha^2-\beta^2}{(\alpha+\beta)^2}=\lambda+\frac{2\lambda^2\alpha\beta}{(\alpha+\beta)^2}.\end{aligned}$$

Ask yourself Why should we expect the two sides to differ?

In general $\mathrm{Var}(X+Y)=\mathrm{Var}(X)+\mathrm{Var}(Y)+2\,\mathrm{Cov}(X,Y)$. Here $XY=0$ always, since at most one of X, Y is non-zero. So $\mathrm{Cov}(X,Y)=-\mathrm{E}(X)\mathrm{E}(Y)=-\lambda^2\alpha\beta/(\alpha+\beta)^2$, which matches the calculation above. Equality would need $\lambda^2\alpha\beta=0$. For $\lambda>0$, though, $\alpha\geq 1$ and $\beta\geq\lambda>0$, so $\lambda^2\alpha\beta>0$. Therefore **there are no non-zero values of** λ for which $\mathrm{Var}(X)+\mathrm{Var}(Y)=\mathrm{Var}(X+Y)$: the left-hand side is always strictly larger. In fact $\alpha=\cosh\lambda$ and $\beta=\sinh\lambda$, so the excess is $\lambda^2\sinh 2\lambda/\mathrm{e}^{2\lambda}=\frac{1}{2}\lambda^2\left(1-\mathrm{e}^{-4\lambda}\right)$.

Checks

- As $\lambda \to 0$, U is almost always 0 or 1, so Y is almost always 0. Indeed $\mathrm{E}(Y) \approx \lambda^2 \to 0$ and $\mathrm{E}(X) \approx \lambda$.
- For large λ, $\alpha \approx \beta \approx \frac{1}{2}\mathrm{e}^{\lambda}$, so $\mathrm{E}(X) \approx \mathrm{E}(Y) \approx \frac{1}{2}\lambda$: odd and even values become equally likely.
- At $\lambda = 1$, summing the Poisson probabilities directly gives $\mathrm{Var}(X) \approx 0.6778$ and $\mathrm{Var}(Y) \approx 0.8131$, so their sum exceeds $\mathrm{Var}(U) = 1$ by about $0.4908 = \frac{1}{2}(1 - \mathrm{e}^{-4})$. Both formulas agree.

Common slips

- Getting the parity wrong after cancelling: if k is odd, then $k-1$ is even (giving α) and $k-2$ is odd (giving β).
- Forgetting that $\mathrm{e}^{-\lambda} = 1/(\alpha+\beta)$, which leaves answers that aren't in the requested form.
- Assuming the variances add. X and Y are far from independent: knowing $X \neq 0$ forces $Y = 0$.

Beyond STEP

Splitting a series into its even and odd parts is the decomposition $\mathrm{e}^{\lambda} = \cosh\lambda + \sinh\lambda$. With probability generating functions it becomes routine: the even part of $G(s)$ is $\frac{1}{2}\big(G(s) + G(-s)\big)$. The covariance identity used at the end is one of the first tools in a university probability course. Grimmett and Welsh, *Probability: An Introduction*, covers both generating functions and covariance.

WORKED SOLUTION

STEP II 2013, Question 13

Probability and Statistics • Runs in Coin Tossing

STEP II 2013, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s2-q13

The key idea

The event $S = n$ says exactly what the first $n+1$ tosses are: n equal tosses, then one that breaks the run. The same is true of $A = n$: n alternating tosses, then a repeat. So each probability is a sum of two products, one for each possible first toss. Comparing S with A then comes down to factorising a difference of four terms, and a factorisation is easier to see if we keep p and q as independent letters rather than replacing q by $1-p$.

Solution

Throughout, tosses are independent, each a head with probability p and a tail with probability q, and $p, q > 0$ with $p \neq q$.

The general formulae

$S = n$ happens exactly when tosses $1, \ldots, n$ all match the first and toss $n+1$ is different. If the first toss is a head this is $\mathrm{H}^n\mathrm{T}$, with probability $p^n q$; if it is a tail it is $\mathrm{T}^n\mathrm{H}$, with probability $q^n p$. These cases are disjoint, so

$$\mathrm{P}(S = n) = p^n q + q^n p = pq\left(p^{n-1} + q^{n-1}\right).$$

$A = n$ happens exactly when tosses $1, \ldots, n$ alternate and toss $n+1$ is the *same* as toss n, which ends the alternation.

- For $n = 2m$ (even): the run $\mathrm{HTHT}\cdots\mathrm{HT}$ has m heads and m tails and ends in T, so the next toss must be T, giving $p^m q^m \cdot q$. Starting with a tail gives $q^m p^m \cdot p$. Hence
$$\mathrm{P}(A = 2m) = p^m q^m (p+q).$$
- For $n = 2m+1$ (odd): the run $\mathrm{HTH}\cdots\mathrm{H}$ has $m+1$ heads and m tails and ends in H, so the next toss must be H, giving $p^{m+2}q^m$. Starting with a tail gives $q^{m+2}p^m$. Hence
$$\mathrm{P}(A = 2m+1) = p^m q^m\left(p^2 + q^2\right).$$

Part (i)

$A = 1$ means the second toss repeats the first: HH or TT. So $\mathrm{P}(A = 1) = p^2 + q^2$ (the case $m = 0$ above). $S = 1$ means the second toss differs from the first: HT or TH, so

$$\mathrm{P}(S = 1) = pq + qp = 2pq.$$

Then

$$\mathrm{P}(A = 1) - \mathrm{P}(S = 1) = p^2 - 2pq + q^2 = (p - q)^2 > 0,$$

strictly because $p \neq q$. So $\mathrm{P}(S = 1) < \mathrm{P}(A = 1)$.

Part (ii)

From the formulae,

$$\mathrm{P}(S = 2) = p^2q + q^2p = pq(p + q) = \mathrm{P}(A = 2),$$

as required. For length 3,

$$\mathrm{P}(S = 3) = p^3q + q^3p = pq\left(p^2 + q^2\right) = \mathrm{P}(A = 3),$$

so these two are equal as well.

Part (iii)

Even lengths. Take $n > 1$. Dividing both probabilities by $pq > 0$,

$$\frac{\mathrm{P}(S = 2n)}{pq} = p^{2n-1} + q^{2n-1}, \qquad \frac{\mathrm{P}(A = 2n)}{pq} = p^{n-1}q^{n-1}(p + q) = p^nq^{n-1} + p^{n-1}q^n.$$

? **Ask yourself** How can we tell the sign of $p^{2n-1} + q^{2n-1} - p^nq^{n-1} - p^{n-1}q^n$?

Pair each pure power with the cross term that shares its highest power of the same letter:

$$\begin{aligned} p^{2n-1} + q^{2n-1} - p^nq^{n-1} - p^{n-1}q^n &= p^n\left(p^{n-1} - q^{n-1}\right) - q^n\left(p^{n-1} - q^{n-1}\right) \\ &= \left(p^n - q^n\right)\left(p^{n-1} - q^{n-1}\right). \end{aligned}$$

Since $p, q > 0$ and $p \neq q$, for any $k \geq 1$ the number $p^k - q^k$ is non-zero and has the same sign as $p - q$ (as $x \mapsto x^k$ is strictly increasing on $x > 0$). Here $n \geq 2$, so both n and $n - 1$ are at least 1, and the two brackets are non-zero with the same sign. Their product is positive, and multiplying back by $pq > 0$,

$$\mathrm{P}(S = 2n) > \mathrm{P}(A = 2n) \qquad (n > 1).$$

(For $n = 1$ the second bracket is $p^0 - q^0 = 0$, which is the equality of part (ii).)

Odd lengths. In the same way, again with $n > 1$,

$$\frac{\mathrm{P}(S = 2n+1)}{pq} = p^{2n} + q^{2n},$$

$$\frac{\mathrm{P}(A = 2n+1)}{pq} = p^{n-1}q^{n-1}\left(p^2 + q^2\right) = p^{n+1}q^{n-1} + p^{n-1}q^{n+1},$$

and

$$\begin{aligned} p^{2n} + q^{2n} - p^{n+1}q^{n-1} - p^{n-1}q^{n+1} &= p^{n+1}\left(p^{n-1} - q^{n-1}\right) - q^{n+1}\left(p^{n-1} - q^{n-1}\right) \\ &= \left(p^{n+1} - q^{n+1}\right)\left(p^{n-1} - q^{n-1}\right). \end{aligned}$$

By the same argument both brackets are non-zero with the sign of $p - q$, so

$$\mathrm{P}(S = 2n+1) > \mathrm{P}(A = 2n+1) \qquad (n > 1).$$

Again the factor $p^{n-1} - q^{n-1}$ vanishes at $n = 1$, which matches $\mathrm{P}(S = 3) = \mathrm{P}(A = 3)$.

So, in summary: a straight run of length 1 is less likely than an alternating one, lengths 2 and 3 are equally likely, and every length from 4 upwards is more likely as a straight run.

Checks

- Take $p = \frac{1}{3}$, $q = \frac{2}{3}$. Then $\mathrm{P}(S = 4) = pq(p^3 + q^3) = \frac{2}{9} \cdot \frac{1}{3} = \frac{2}{27} \approx 0.074$ and $\mathrm{P}(A = 4) = p^2q^2 = \frac{4}{81} \approx 0.049$, so $S = 4$ is indeed more likely.
- Summing $\mathrm{P}(S = n)$ over all $n \geq 1$ gives $q \cdot \frac{p}{1-p} + p \cdot \frac{q}{1-q} = p + q = 1$, as it should.
- Each formula was confirmed by computer, enumerating all sequences of 14 tosses with exact fractions for several values of p, for lengths up to 13.

Common slips

- Forgetting the toss that *ends* the run. $S = n$ needs toss $n+1$ to differ; without it you are computing $\mathrm{P}(S \geq n)$.
- Ending an alternating run on the wrong letter: a run of odd length ends on the same letter it started with, and one of even length ends on the other letter.
- Substituting $q = 1 - p$ in part (iii): the expressions become polynomials in p whose factorisation is much harder to spot.

Beyond STEP

S is a geometric-type random variable: given the first toss, the run length is the waiting time for the first change. Runs like these underlie the runs test in statistics, used to

judge whether a sequence of outcomes looks random, and generating functions give a neat way to find the full distribution of longest runs. Grimmett and Welsh, *Probability: An Introduction*, covers generating functions and waiting times at the right level.

WORKED SOLUTION

STEP III 2013, Question 1

Pure • Half-Angle Substitution

STEP III 2013, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s3-q1

The key idea

The half-angle substitution $t = \tan\frac{1}{2}x$ turns any rational function of $\sin x$ into a rational function of t, so the first integral becomes $\int \mathrm{d}t/(\text{quadratic})$, which we finish by completing the square. Getting the answer into the required form then needs the $\tan(A-B)$ formula. For I_n, the hint is designed so that the awkward denominator cancels: $\sin^{n+1}x + 2\sin^n x = \sin^n x\,(2+\sin x)$. That gives a recurrence which climbs from I_0, the only integral that genuinely needs the first part.

Solution

The substitution

With $t = \tan\frac{1}{2}x$, the chain rule gives

$$\frac{\mathrm{d}t}{\mathrm{d}x} = \tfrac{1}{2}\sec^2\tfrac{1}{2}x = \tfrac{1}{2}\left(1+\tan^2\tfrac{1}{2}x\right) = \tfrac{1}{2}(1+t^2).$$

For the sine, use the double-angle formula and then write $\cos^2\frac{1}{2}x = 1/\sec^2\frac{1}{2}x$:

$$\sin x = 2\sin\tfrac{1}{2}x\cos\tfrac{1}{2}x = 2\tan\tfrac{1}{2}x\cos^2\tfrac{1}{2}x = \frac{2\tan\frac{1}{2}x}{\sec^2\frac{1}{2}x} = \frac{2t}{1+t^2}.$$

Both results are shown.

The first integral

As x runs from 0 to $\frac{1}{2}\pi$, t runs from $\tan 0 = 0$ to $\tan\frac{1}{4}\pi = 1$, and $\mathrm{d}x = \dfrac{2\,\mathrm{d}t}{1+t^2}$. Also

$$1 + a\sin x = \frac{1+t^2+2at}{1+t^2},$$

so the factors $1+t^2$ cancel and

$$\int_0^{\frac{1}{2}\pi} \frac{\mathrm{d}x}{1+a\sin x} = \int_0^1 \frac{2\,\mathrm{d}t}{t^2+2at+1} = \int_0^1 \frac{2\,\mathrm{d}t}{(t+a)^2+(1-a^2)}.$$

Since $0<a<1$, the constant $1-a^2$ is positive; write $r=\sqrt{1-a^2}$. Using $\int \mathrm{d}u/(u^2+r^2) = \frac{1}{r}\arctan\frac{u}{r}$,

$$\int_0^{\frac{1}{2}\pi} \frac{\mathrm{d}x}{1+a\sin x} = \frac{2}{r}\Big[\arctan\frac{t+a}{r}\Big]_0^1 = \frac{2}{r}(\alpha-\beta),$$

where

$$\tan\alpha = \frac{1+a}{r} = \frac{1+a}{\sqrt{1-a}\sqrt{1+a}} = \frac{\sqrt{1+a}}{\sqrt{1-a}}, \qquad \tan\beta = \frac{a}{r},$$

with $\alpha,\beta\in(0,\frac{1}{2}\pi)$.

Ask yourself This doesn't look like the answer we want. How do we turn a difference of two arctangents into a single one?

Take the tangent of the difference. We have

$$\tan\alpha-\tan\beta = \frac{(1+a)-a}{r} = \frac{1}{r}, \qquad 1+\tan\alpha\tan\beta = 1+\frac{a(1+a)}{1-a^2} = 1+\frac{a}{1-a} = \frac{1}{1-a},$$

so

$$\tan(\alpha-\beta) = \frac{\tan\alpha-\tan\beta}{1+\tan\alpha\tan\beta} = \frac{1-a}{r} = \frac{1-a}{\sqrt{1-a}\sqrt{1+a}} = \frac{\sqrt{1-a}}{\sqrt{1+a}}.$$

We need $\alpha-\beta$ to lie in $(-\frac{1}{2}\pi,\frac{1}{2}\pi)$, the range of arctan, before we can invert. It does: both angles are in $(0,\frac{1}{2}\pi)$, and $\tan\alpha>\tan\beta$ (their difference is $1/r>0$), so $0<\alpha-\beta<\frac{1}{2}\pi$. Hence $\alpha-\beta=\arctan\dfrac{\sqrt{1-a}}{\sqrt{1+a}}$ and

$$\int_0^{\frac{1}{2}\pi} \frac{\mathrm{d}x}{1+a\sin x} = \frac{2}{\sqrt{1-a^2}}\arctan\frac{\sqrt{1-a}}{\sqrt{1+a}},$$

as required.

Evaluating I_3

Following the hint,

$$I_{n+1}+2I_n = \int_0^{\frac{1}{2}\pi} \frac{\sin^n x\,(\sin x+2)}{2+\sin x}\,\mathrm{d}x = \int_0^{\frac{1}{2}\pi} \sin^n x\,\mathrm{d}x.$$

We need the right-hand side for $n=0,1,2$:

$$\int_0^{\frac{1}{2}\pi} 1\,\mathrm{d}x = \tfrac{1}{2}\pi, \qquad \int_0^{\frac{1}{2}\pi} \sin x\,\mathrm{d}x = 1, \qquad \int_0^{\frac{1}{2}\pi} \sin^2 x\,\mathrm{d}x = \int_0^{\frac{1}{2}\pi} \tfrac{1}{2}(1-\cos 2x)\,\mathrm{d}x = \tfrac{1}{4}\pi.$$

Ask yourself The recurrence links each I_n to the next, but where does it start?

We need I_0 separately, and this is where the first part comes in. Taking out a factor of 2 and using $a = \frac{1}{2}$,

$$I_0 = \int_0^{\frac{1}{2}\pi} \frac{\mathrm{d}x}{2+\sin x} = \frac{1}{2}\int_0^{\frac{1}{2}\pi} \frac{\mathrm{d}x}{1+\frac{1}{2}\sin x} = \frac{1}{2}\cdot\frac{2}{\sqrt{3/4}}\arctan\frac{\sqrt{1/2}}{\sqrt{3/2}}$$
$$= \frac{2}{\sqrt{3}}\arctan\frac{1}{\sqrt{3}} = \frac{2}{\sqrt{3}}\cdot\frac{\pi}{6} = \frac{\pi}{3\sqrt{3}}.$$

Now climb the recurrence $I_{n+1} = \int_0^{\frac{1}{2}\pi} \sin^n x\,\mathrm{d}x - 2I_n$:

$$I_1 = \tfrac{1}{2}\pi - 2I_0 = \frac{\pi}{2} - \frac{2\pi}{3\sqrt{3}},$$
$$I_2 = 1 - 2I_1 = 1 - \pi + \frac{4\pi}{3\sqrt{3}},$$
$$I_3 = \tfrac{1}{4}\pi - 2I_2 = \frac{\pi}{4} - 2 + 2\pi - \frac{8\pi}{3\sqrt{3}}.$$

Therefore

$$I_3 = \frac{9\pi}{4} - \frac{8\sqrt{3}\,\pi}{9} - 2.$$

Checks

- As $a \to 0$ the first result should give $\int_0^{\frac{1}{2}\pi} 1\,\mathrm{d}x = \frac{1}{2}\pi$; indeed it tends to $2\arctan 1 = \frac{1}{2}\pi$.
- Since $0 \le \sin^3 x/(2+\sin x) \le \frac{1}{3}$ on $[0, \frac{1}{2}\pi]$, we need $0 < I_3 < \frac{1}{6}\pi \approx 0.52$. Our answer is $I_3 \approx 0.2318$, and numerical integration of I_3 agrees to many decimal places. Similarly $I_0 = \pi/(3\sqrt{3}) \approx 0.6046$, which lies between $\frac{\pi}{6}$ and $\frac{\pi}{4}$ as it must.
- The values decrease, $I_0 > I_1 > I_2 > I_3 > 0$, as they should, because $0 \le \sin x \le 1$ on the interval.

Common slips

- Forgetting to change the limits to $t = 0$ and $t = 1$, or forgetting the factor $2/(1+t^2)$ from $\mathrm{d}x$.
- Using $\tan(\alpha - \beta)$ to conclude $\alpha - \beta = \arctan(\dots)$ without checking that $\alpha - \beta$ lies in $(-\frac{1}{2}\pi, \frac{1}{2}\pi)$.
- Applying the first result to I_0 with $a = 2$: it only holds for $0 < a < 1$, so the factor 2 must be taken out first.

Beyond STEP

The $t = \tan\frac{1}{2}x$ substitution (often called the Weierstrass substitution) shows that every rational function of $\sin x$ and $\cos x$ has an elementary antiderivative, since partial fractions handle any rational function of t. Geometrically it is the stereographic parametrisation of the unit circle, $(\cos x, \sin x) = \left(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2}\right)$, which also produces every Pythagorean triple from rational t. Integrals like $\int_0^{2\pi} \mathrm{d}x/(1 + a\sin x)$ are also a classic first application of contour integration in complex analysis; Needham's *Visual Complex Analysis* is a beautiful place to see why.

WORKED SOLUTION

STEP III 2013, Question 2

Pure • Maclaurin Series

STEP III 2013, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s3-q2

The key idea

Clearing the square root first, by writing $y\sqrt{1-x^2} = \arcsin x$, makes the first derivative painless. The higher-order relation then comes from differentiating that first-order equation repeatedly with Leibniz's rule. At $x = 0$ most of the terms vanish and we are left with a two-step recurrence for the derivatives, which builds the Maclaurin coefficients one at a time. The final sum looks like the series evaluated at a well-chosen value of x.

Solution

The first-order equation

Multiply up to get $y\sqrt{1-x^2} = \arcsin x$ and differentiate both sides with respect to x, using the product rule on the left:

$$\sqrt{1-x^2}\,\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{xy}{\sqrt{1-x^2}} = \frac{1}{\sqrt{1-x^2}}.$$

Multiplying through by $\sqrt{1-x^2}$ (non-zero for $|x| < 1$) gives

$$(1-x^2)\frac{\mathrm{d}y}{\mathrm{d}x} - xy - 1 = 0,$$

as required.

The higher-order relation

Write $y^{(k)}$ for $\dfrac{\mathrm{d}^k y}{\mathrm{d}x^k}$. Differentiate the first-order equation $n+1$ times. The constant 1 disappears, and Leibniz's rule handles the two products. Since $1-x^2$ has derivatives

$-2x$, -2 and then 0, and x has derivatives 1 and then 0,

$$\frac{\mathrm{d}^{n+1}}{\mathrm{d}x^{n+1}}\Big[(1-x^2)y'\Big] = (1-x^2)y^{(n+2)} + \binom{n+1}{1}(-2x)\,y^{(n+1)} + \binom{n+1}{2}(-2)\,y^{(n)}$$
$$= (1-x^2)y^{(n+2)} - 2(n+1)x\,y^{(n+1)} - n(n+1)\,y^{(n)},$$

$$\frac{\mathrm{d}^{n+1}}{\mathrm{d}x^{n+1}}\Big[xy\Big] = x\,y^{(n+1)} + (n+1)\,y^{(n)}.$$

Subtracting the second from the first and collecting terms,

$$(1-x^2)y^{(n+2)} - \big(2(n+1)+1\big)x\,y^{(n+1)} - \big(n(n+1)+(n+1)\big)y^{(n)} = 0,$$

that is,

$$(1-x^2)y^{(n+2)} - (2n+3)x\,y^{(n+1)} - (n+1)^2 y^{(n)} = 0.$$

Nothing here needed $n \geq 1$: differentiating once ($n = 0$) gives the same pattern, $(1-x^2)y'' - 3xy' - y = 0$, and we will use that case too.

The Maclaurin series

Put $x = 0$. The middle term carries a factor x, so it drops out and

$$y^{(n+2)}(0) = (n+1)^2\,y^{(n)}(0) \qquad (n \geq 0).$$

The recurrence moves in steps of two, so we need two starting values. From the definition, $y(0) = \arcsin 0 = 0$, and the first-order equation at $x = 0$ gives $y'(0) - 0 - 1 = 0$, so $y'(0) = 1$.

Ask yourself How do we get every derivative from just these two values?

The recurrence links $y^{(n)}(0)$ only to $y^{(n+2)}(0)$, so the even derivatives form one chain started by $y(0)$ and the odd derivatives form another started by $y'(0)$.

Even powers. Since $y(0) = 0$, the recurrence gives $y''(0) = 1^2 \cdot 0 = 0$, then $y^{(4)}(0) = 0$, and so on: $y^{(2k)}(0) = 0$ for every k. So the coefficient of every even power of x is zero. (This fits with y being an odd function, as a quotient of the odd function $\arcsin x$ by the even function $\sqrt{1-x^2}$.)

Odd powers. Using the recurrence with $n = 1, 3, \ldots, 2k-1$,

$$y^{(2k+1)}(0) = (2k)^2(2k-2)^2 \cdots 4^2 \cdot 2^2 \cdot y'(0) = \big(2 \cdot 4 \cdots 2k\big)^2 = \big(2^k\,k!\big)^2,$$

which also holds for $k = 0$. Dividing by $(2k+1)!$ gives the Maclaurin coefficients:

$$\frac{\arcsin x}{\sqrt{1-x^2}} = \sum_{k=0}^{\infty} \frac{2^{2k}(k!)^2}{(2k+1)!}\,x^{2k+1} = x + \frac{2^2}{3!}x^3 + \frac{2^2 \cdot 4^2}{5!}x^5 + \cdots,$$

so the general odd term is $\dfrac{2^{2k}(k!)^2}{(2k+1)!}x^{2k+1}$, and the general even term is $0 \cdot x^{2k}$.

The infinite sum

The nth term of the sum is $\dfrac{(n!)^2}{(2n+1)!}$, with $n = 0$ giving the leading 1. Comparing with the series, we want to remove the factor 2^{2n}, so choose x with $2^{2n}x^{2n+1}$ a constant multiple of 1: $x = \frac{1}{2}$ gives $2^{2n}\left(\frac{1}{2}\right)^{2n+1} = \frac{1}{2}$. Hence

$$\frac{\arcsin\frac{1}{2}}{\sqrt{1-\frac{1}{4}}} = \frac{1}{2}\sum_{n=0}^{\infty}\frac{(n!)^2}{(2n+1)!}.$$

The left-hand side is $\dfrac{\pi/6}{\sqrt{3}/2} = \dfrac{\pi}{3\sqrt{3}}$, so

$$1 + \frac{1}{3!} + \frac{2^2}{5!} + \frac{2^2\times 3^2}{7!} + \cdots = \frac{2\pi}{3\sqrt{3}} = \frac{2\sqrt{3}\,\pi}{9}.$$

Checks

- Multiplying the known series $\arcsin x = x + \frac{1}{6}x^3 + \cdots$ by $(1-x^2)^{-1/2} = 1 + \frac{1}{2}x^2 + \cdots$ gives $x + \left(\frac{1}{6}+\frac{1}{2}\right)x^3 + \cdots = x + \frac{2}{3}x^3 + \cdots$, and indeed $\frac{2^2}{3!} = \frac{2}{3}$.
- The first five terms of the sum are $1 + 0.16667 + 0.03333 + 0.00714 + 0.00159 \approx 1.2087$, already close to $\frac{2\pi}{3\sqrt{3}} \approx 1.20920$; the terms shrink by a factor of about 4 each time, so the remainder is small.

Common slips

- Forgetting the binomial coefficient $\binom{n+1}{2}$ on the $y^{(n)}$ term in Leibniz's rule, which spoils the $(n+1)^2$.
- Using the recurrence only for $n \geq 1$ and so never establishing $y''(0) = 0$; the case $n = 0$ (one differentiation) is needed for the even chain.
- Substituting $x = 1$ or $x = \frac{1}{4}$ in the final part: the factor 2^{2n} has to be cancelled by x^{2n}, which forces $x = \frac{1}{2}$.

Beyond STEP

Here we "ignored questions of convergence", but a first analysis course makes this precise: the series has radius of convergence 1 by the ratio test, and Taylor's theorem with remainder shows it really does equal the function inside that interval. The coefficients $\frac{2^{2k}(k!)^2}{(2k+1)!}$ are closely related to central binomial coefficients, and the same function generates the series for $(\arcsin x)^2$ after one integration. For the analysis behind power series, see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP III 2013, Question 3

Pure • Vectors and the Sphere

STEP III 2013, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s3-q3

The key idea

Everything flows from two facts: each $\mathbf{p}_i$ is a unit vector, and the four of them add to $\mathbf{0}$. Expanding a squared length as a dot product turns $(XP_i)^2$ into something *linear* in $\mathbf{x}\cdot\mathbf{p}_i$, and sums of linear terms collapse because $\sum \mathbf{p}_i = \mathbf{0}$. For the fourth powers a quadratic term survives, so we need actual coordinates, which is exactly what part (ii) provides.

Solution

The dot products $\mathbf{p}_i \cdot \mathbf{p}_j$

Since the tetrahedron is regular, all six edges have the same length. For $i \neq j$,

$$(P_iP_j)^2 = (\mathbf{p}_i - \mathbf{p}_j)\cdot(\mathbf{p}_i - \mathbf{p}_j) = |\mathbf{p}_i|^2 + |\mathbf{p}_j|^2 - 2\,\mathbf{p}_i\cdot\mathbf{p}_j = 2 - 2\,\mathbf{p}_i\cdot\mathbf{p}_j,$$

using $|\mathbf{p}_i| = 1$ (each vertex lies on the unit sphere). Equal edge lengths therefore force every $\mathbf{p}_i \cdot \mathbf{p}_j$ $(i \neq j)$ to take the same value; call it c.

Now take the dot product of $\mathbf{p}_1 + \mathbf{p}_2 + \mathbf{p}_3 + \mathbf{p}_4 = \mathbf{0}$ with $\mathbf{p}_1$:

$$\mathbf{p}_1\cdot\mathbf{p}_1 + \mathbf{p}_1\cdot\mathbf{p}_2 + \mathbf{p}_1\cdot\mathbf{p}_3 + \mathbf{p}_1\cdot\mathbf{p}_4 = 0 \quad\Longrightarrow\quad 1 + 3c = 0,$$

so $c = -\frac{1}{3}$. That is, $\mathbf{p}_i\cdot\mathbf{p}_j = -\frac{1}{3}$ for all $i \neq j$, as required.

Part (i)

Since $|\mathbf{x}| = 1$ too,

$$(XP_i)^2 = (\mathbf{p}_i - \mathbf{x})\cdot(\mathbf{p}_i - \mathbf{x}) = |\mathbf{p}_i|^2 - 2\,\mathbf{x}\cdot\mathbf{p}_i + |\mathbf{x}|^2 = 2 - 2\,\mathbf{x}\cdot\mathbf{p}_i.$$

Summing over i and using the distributive law for the dot product,

$$\sum_{i=1}^{4}(XP_i)^2 = 8 - 2\,\mathbf{x}\cdot\left(\mathbf{p}_1 + \mathbf{p}_2 + \mathbf{p}_3 + \mathbf{p}_4\right) = 8 - 2\,\mathbf{x}\cdot\mathbf{0} = 8,$$

as required.

Part (ii)

With $\mathbf{p}_1 = (0,0,1)$ and $\mathbf{p}_2 = (a,0,b)$, the condition $\mathbf{p}_1 \cdot \mathbf{p}_2 = -\frac{1}{3}$ gives $b = -\frac{1}{3}$ at once. Then $|\mathbf{p}_2| = 1$ gives

$$a^2 = 1 - b^2 = \tfrac{8}{9}, \qquad \text{so} \qquad a = \frac{2\sqrt{2}}{3},$$

taking the positive root since $a > 0$.

Let $\mathbf{p}_3 = (u,v,w)$. Three conditions pin it down:

$$\mathbf{p}_1 \cdot \mathbf{p}_3 = -\tfrac{1}{3} \implies w = -\tfrac{1}{3},$$

$$\mathbf{p}_2 \cdot \mathbf{p}_3 = -\tfrac{1}{3} \implies \tfrac{2\sqrt{2}}{3}u + \tfrac{1}{9} = -\tfrac{1}{3} \implies u = -\tfrac{4}{9} \cdot \tfrac{3}{2\sqrt{2}} = -\frac{\sqrt{2}}{3},$$

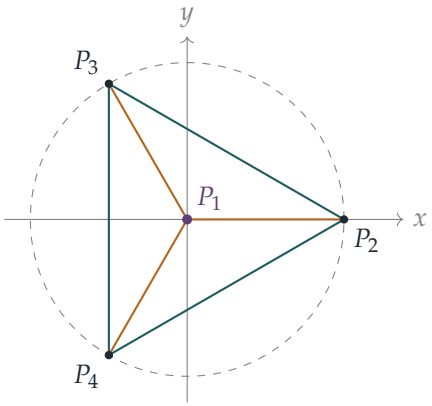
$$|\mathbf{p}_3| = 1 \implies v^2 = 1 - \tfrac{2}{9} - \tfrac{1}{9} = \tfrac{6}{9} \implies v = \pm\frac{\sqrt{6}}{3}.$$

Exactly the same equations hold for $\mathbf{p}_4$, so P_3 and P_4 are the two points this gives (they must be different points, so one takes each sign). Labelling them one way round,

$$P_3 = \Big(-\frac{\sqrt{2}}{3}, \frac{\sqrt{6}}{3}, -\frac{1}{3}\Big), \qquad P_4 = \Big(-\frac{\sqrt{2}}{3}, -\frac{\sqrt{6}}{3}, -\frac{1}{3}\Big).$$

(The labels of P_3 and P_4 can be swapped.) As a check, the four position vectors do add to $\mathbf{0}$, and $\mathbf{p}_3 \cdot \mathbf{p}_4 = \frac{2}{9} - \frac{6}{9} + \frac{1}{9} = -\frac{1}{3}$.

Seen from above, P_2, P_3, P_4 form an equilateral triangle in the plane $z = -\frac{1}{3}$, with P_1 directly above its centre:



Part (iii)

From part (i), $(XP_i)^2 = 2(1 - \mathbf{x} \cdot \mathbf{p}_i)$, so squaring,

$$(XP_i)^4 = 4(1 - \mathbf{x} \cdot \mathbf{p}_i)^2 \qquad \implies \qquad \sum_{i=1}^{4}(XP_i)^4 = 4\sum_{i=1}^{4}(1 - \mathbf{x} \cdot \mathbf{p}_i)^2,$$

as required. Expanding the bracket and using $\sum \mathbf{x}\cdot\mathbf{p}_i = 0$ once more,

$$\sum_{i=1}^{4}(XP_i)^4 = 4\Big(4 - 2\sum_{i=1}^{4}\mathbf{x}\cdot\mathbf{p}_i + \sum_{i=1}^{4}(\mathbf{x}\cdot\mathbf{p}_i)^2\Big) = 16 + 4\sum_{i=1}^{4}(\mathbf{x}\cdot\mathbf{p}_i)^2.$$

Ask yourself The linear terms vanished again. Why can't the same trick dispose of $\sum(\mathbf{x}\cdot\mathbf{p}_i)^2$?

A sum of squares is not linear in the $\mathbf{p}_i$, so knowing $\sum \mathbf{p}_i = \mathbf{0}$ says nothing directly about it. This is where the coordinates come in. With $\mathbf{x} = (x, y, z)$ and the vertices from part (ii),

$$\mathbf{x}\cdot\mathbf{p}_1 = z, \qquad \mathbf{x}\cdot\mathbf{p}_2 = \tfrac{1}{3}\big(2\sqrt{2}\,x - z\big),$$
$$\mathbf{x}\cdot\mathbf{p}_3 = \tfrac{1}{3}\big(-\sqrt{2}\,x + \sqrt{6}\,y - z\big), \qquad \mathbf{x}\cdot\mathbf{p}_4 = \tfrac{1}{3}\big(-\sqrt{2}\,x - \sqrt{6}\,y - z\big).$$

Square and add, collecting terms:

- x^2: $\tfrac{1}{9}(8+2+2) = \tfrac{4}{3}$; y^2: $\tfrac{1}{9}(6+6) = \tfrac{4}{3}$; z^2: $1 + \tfrac{1}{9}(1+1+1) = \tfrac{4}{3}$.
- xy and yz: these appear only in $\mathbf{x}\cdot\mathbf{p}_3$ and $\mathbf{x}\cdot\mathbf{p}_4$, which differ only in the sign of the y term, so they cancel.
- xz: $\tfrac{1}{9}\big(-4\sqrt{2} + 2\sqrt{2} + 2\sqrt{2}\big) = 0$.

Hence

$$\sum_{i=1}^{4}(\mathbf{x}\cdot\mathbf{p}_i)^2 = \tfrac{4}{3}\big(x^2 + y^2 + z^2\big) = \tfrac{4}{3},$$

because X lies on the unit sphere. Therefore

$$\sum_{i=1}^{4}(XP_i)^4 = 16 + 4\cdot\tfrac{4}{3} = \frac{64}{3},$$

which does not depend on the position of X, as required.

Ask yourself We used particular coordinates. Does the conclusion hold for every regular tetrahedron inscribed in the sphere?

Yes. Any such tetrahedron can be rotated about O so that P_1 goes to $(0, 0, 1)$ and P_2 into the half-plane $y = 0$, $x > 0$; rotations preserve lengths, and as X ranges over the sphere so does its rotated image. So the coordinates of part (ii) lose no generality.

Checks

- Take $X = P_1$. The distances are 0 and three edges, each of length $\sqrt{2 - 2(-\tfrac{1}{3})} = \sqrt{8/3}$. Then $\sum(XP_i)^2 = 3\cdot\tfrac{8}{3} = 8$ and $\sum(XP_i)^4 = 3\cdot\tfrac{64}{9} = \tfrac{64}{3}$, matching both results.

- Take $X = (0, 0, -1)$, the point opposite P_1. Then $XP_1 = 2$, and for the other three $\mathbf{x} \cdot \mathbf{p}_i = \frac{1}{3}$, so $(XP_i)^2 = \frac{4}{3}$. This gives $4 + 3 \cdot \frac{4}{3} = 8$ and $16 + 3 \cdot \frac{16}{9} = \frac{64}{3} \approx 21.33$.
- A computer check at thousands of random points on the sphere gave 8 and 21.333… every time.

Common slips

- Asserting that all the $\mathbf{p}_i \cdot \mathbf{p}_j$ are equal without saying why: it is the equal edge lengths, through $(P_iP_j)^2 = 2 - 2\,\mathbf{p}_i \cdot \mathbf{p}_j$.
- Forgetting that $|\mathbf{x}| = 1$, which is needed both for $(XP_i)^2 = 2 - 2\,\mathbf{x} \cdot \mathbf{p}_i$ and in the final step $x^2 + y^2 + z^2 = 1$.
- Arithmetic slips in the cross terms of part (iii): a sign error there leaves a spurious dependence on X.

Beyond STEP

The heart of part (iii) is the identity $\sum_i (\mathbf{x} \cdot \mathbf{p}_i)^2 = \frac{4}{3}|\mathbf{x}|^2$ for *every* vector $\mathbf{x}$. In matrix language, $\sum_i \mathbf{p}_i \mathbf{p}_i^{\mathsf{T}} = \frac{4}{3} I$: the four vertex vectors behave like an over-complete orthonormal basis, called a *tight frame*. This has to be true by symmetry, since the matrix commutes with every rotation of the tetrahedron and so must be a multiple of I, and its trace is 4. Point sets on a sphere whose low-degree averages match the sphere's own averages are called *spherical designs*; they are used in numerical integration and coding theory. For the linear algebra behind this (outer products, symmetric matrices and eigenvalues), see Strang, *Introduction to Linear Algebra*.

WORKED SOLUTION

STEP III 2013, Question 4

Pure • Roots of Unity Products

STEP III 2013, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s3-q4

This question has a full guided treatment in *Beyond STEP*, Section 5.3, *Factorising* $z^{2n} + 1$:
https://beyondstep.keithmcnulty.org/ch05/z2n-plus-one.html

The key idea

The roots of $z^{2n} = -1$ come in conjugate pairs $e^{\pm i\theta}$, and each pair multiplies to a real quadratic, which gives the factorisation. Then a single well-chosen value of z turns the factorisation into a statement about cosines: at $z = i$ the $z^2 + 1$ part of every quadratic vanishes, leaving just the $-2z\cos\theta$ term. In part (ii) that same substitution would make both sides zero, so we first cancel the one factor that vanishes at $z = i$.

Solution

The factorisation

Expanding,

$$\begin{aligned}(z - e^{i\theta})(z - e^{-i\theta}) &= z^2 - z\left(e^{i\theta} + e^{-i\theta}\right) + e^{i\theta}e^{-i\theta} \\ &= z^2 - 2z\cos\theta + 1,\end{aligned}$$

since $e^{i\theta} + e^{-i\theta} = 2\cos\theta$.

Now $e^{i\theta}$ satisfies $z^{2n} = -1 = e^{i\pi}$ exactly when $2n\theta$ is an odd multiple of π, that is, $\theta = \frac{(2k-1)\pi}{2n}$ for an integer k. The values in $(-\pi, \pi]$ are

$$\theta = \pm\frac{\pi}{2n},\ \pm\frac{3\pi}{2n},\ \ldots,\ \pm\frac{(2n-1)\pi}{2n},$$

because $\frac{(2n-1)\pi}{2n} < \pi$ while $\frac{(2n+1)\pi}{2n} > \pi$. These are $2n$ distinct roots, so they are all the $(2n)$th roots of -1:

$$e^{\pm i\theta_k}, \qquad \theta_k = \frac{(2k-1)\pi}{2n}, \quad k = 1, 2, \ldots, n.$$

Since $z^{2n}+1$ is monic of degree $2n$ with these $2n$ distinct roots,

$$z^{2n}+1=\prod_{k=1}^{n}(z-\mathrm{e}^{\mathrm{i}\theta_k})(z-\mathrm{e}^{-\mathrm{i}\theta_k})=\prod_{k=1}^{n}\left(z^2-2z\cos\theta_k+1\right),$$

using the first result on each conjugate pair. This is the required identity.

Part (i)

Put $z=\mathrm{i}$. Each factor becomes

$$\mathrm{i}^2-2\mathrm{i}\cos\theta_k+1=-2\mathrm{i}\cos\theta_k,$$

and the left-hand side is $\mathrm{i}^{2n}+1=(-1)^n+1=2$, as n is even. So

$$2=(-2\mathrm{i})^n\prod_{k=1}^{n}\cos\theta_k.$$

With n even, $(-2)^n=2^n$ and $\mathrm{i}^n=(\mathrm{i}^2)^{n/2}=(-1)^{\frac{1}{2}n}$, so $(-2\mathrm{i})^n=2^n(-1)^{\frac{1}{2}n}$. Therefore

$$\prod_{k=1}^{n}\cos\theta_k=\frac{2}{2^n(-1)^{\frac{1}{2}n}}=(-1)^{\frac{1}{2}n}\,2^{1-n},$$

using $1/(-1)^{\frac{1}{2}n}=(-1)^{\frac{1}{2}n}$. The product on the left is exactly $\cos\frac{\pi}{2n}\cos\frac{3\pi}{2n}\cdots\cos\frac{(2n-1)\pi}{2n}$, as required.

Part (ii)

? **Ask yourself** Why not just put $z=\mathrm{i}$ in the factorisation again?

When n is odd, $\mathrm{i}^{2n}+1=(-1)^n+1=0$, so the left-hand side vanishes. On the right, the middle value $k=m$, where $m=\frac{1}{2}(n+1)$, gives $\theta_m=\frac{n\pi}{2n}=\frac{\pi}{2}$, so that factor is z^2+1, which is also zero at $z=\mathrm{i}$. We get $0=0$ and learn nothing. The cure is to remove the factor z^2+1 from both sides first.

Using the given fact and the factorisation,

$$(1+z^2)\left(1-z^2+z^4-\cdots+z^{2n-2}\right)=(z^2+1)\prod_{k\neq m}\left(z^2-2z\cos\theta_k+1\right).$$

These are polynomials, and z^2+1 is not the zero polynomial, so we may cancel it (polynomial division is unique):

$$1-z^2+z^4-\cdots+z^{2n-2}=\prod_{k\neq m}\left(z^2-2z\cos\theta_k+1\right),$$

an identity of polynomials, valid for every z, including $z=\mathrm{i}$.

Now substitute $z = \mathrm{i}$. On the left, the term $(-1)^j z^{2j}$ becomes $(-1)^j(-1)^j = 1$, and there are n terms ($j = 0, 1, \ldots, n-1$), so the left-hand side is n. On the right there are $n-1$ factors, each equal to $-2\mathrm{i}\cos\theta_k$:

$$n = (-2\mathrm{i})^{n-1} \prod_{k \neq m} \cos\theta_k.$$

As $n-1$ is even, $(-2\mathrm{i})^{n-1} = 2^{n-1}(-1)^{\frac{1}{2}(n-1)}$, exactly as in part (i).

To turn the product into squares, pair k with $n+1-k$. Since

$$\theta_{n+1-k} = \frac{(2n+1-2k)\pi}{2n} = \pi - \theta_k,$$

we have $\cos\theta_{n+1-k} = -\cos\theta_k$. The values $k \neq m$ split into the $\frac{1}{2}(n-1)$ pairs $\{k, n+1-k\}$ with $1 \leq k \leq \frac{1}{2}(n-1)$, so

$$\prod_{k \neq m} \cos\theta_k = \prod_{k=1}^{(n-1)/2} \left(-\cos^2\theta_k\right) = (-1)^{\frac{1}{2}(n-1)}\, P, \qquad P = \prod_{k=1}^{(n-1)/2} \cos^2\theta_k.$$

Putting this together, the two signs $(-1)^{\frac{1}{2}(n-1)}$ multiply to $+1$:

$$n = 2^{n-1}(-1)^{\frac{1}{2}(n-1)}(-1)^{\frac{1}{2}(n-1)} P = 2^{n-1} P,$$

so $P = n\,2^{1-n}$. The last factor in P has $k = \frac{1}{2}(n-1)$, that is, $\theta_k = \frac{(n-2)\pi}{2n}$, so P is precisely

$$\cos^2\frac{\pi}{2n}\cos^2\frac{3\pi}{2n}\cdots\cos^2\frac{(n-2)\pi}{2n} = n\,2^{1-n},$$

as required. (For $n = 1$ the product is empty, equal to $1 = 1 \cdot 2^0$, which is consistent.)

Checks

- $n = 2$ in part (i): $\cos\frac{\pi}{4}\cos\frac{3\pi}{4} = -\frac{1}{2} = (-1)^1 2^{-1}$. For $n = 4$ the product is $\frac{1}{8}$, matching $(-1)^2 2^{-3}$.
- $n = 3$ in part (ii): $\cos^2\frac{\pi}{6} = \frac{3}{4} = 3 \cdot 2^{-2}$. For $n = 5$, $\cos^2\frac{\pi}{10}\cos^2\frac{3\pi}{10} = 0.3125 = \frac{5}{16}$.
- Both identities were also confirmed by computer for all n up to 39, and the factorisation was tested at random complex z.

Common slips

- Substituting $z = \mathrm{i}$ straight into the factorisation in part (ii), which only gives $0 = 0$.
- Losing the sign: in part (i) the answer really is negative when $n \equiv 2 \pmod 4$, because $\cos\theta_k < 0$ for the $\frac{1}{2}n$ angles beyond $\frac{\pi}{2}$.
- Forgetting that $\cos\theta_{n+1-k} = -\cos\theta_k$, and so getting the wrong sign when forming squares.

Beyond STEP

The same trick, factorising $z^N \pm 1$ over its roots of unity and then evaluating at a clever point, gives a whole family of product formulas, such as $\prod_{k=1}^{n-1} \sin \frac{k\pi}{n} = \frac{n}{2^{n-1}}$ (try $z = 1$ in $\frac{z^n-1}{z-1}$). Letting $n \to \infty$ in such finite products leads to Euler's infinite product $\sin \pi x = \pi x \prod_{k \geq 1} \left(1 - \frac{x^2}{k^2}\right)$, a first taste of how complex analysis builds functions from their zeros. Needham's *Visual Complex Analysis* is a beautiful place to see roots of unity and these ideas geometrically.

WORKED SOLUTION

STEP III 2013, Question 5

Pure • Rationality and Coprimality

STEP III 2013, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s3-q5

The key idea

Every step runs on one tool: if a number divides a product and is coprime to one factor, it divides the other factor. Put a rational number in lowest terms, clear denominators, and watch which side each factor has to land on. In part (ii) this traps the denominator b between two facts that can't both hold unless b is tiny.

We shall use one small fact repeatedly: *if p and q are coprime then so are p^s and q^t* for any positive integers s, t. Indeed, a prime dividing both p^s and q^t would divide p and q (a prime dividing a product divides one of the factors, which is the given result with a prime), contradicting coprimality.

Solution

Part (i)

From $q^nN = p^n$ we see that p^n divides q^nN. Since p and q are coprime, so are p^n and q^n, and the given result (with $a = p^n$, $b = q^n$, $c = N$) tells us that p^n divides N. So $N = kp^n$ for some positive integer k, as required.

Substituting back, $q^nkp^n = p^n$, so $q^nk = 1$. Both are positive integers, so $k = 1$ and $q^n = 1$, giving

$$q = 1 \quad (\text{and } N = p^n).$$

Now take any positive integers n and N and suppose $\sqrt[n]{N}$ is rational. It is positive, so we can write $\sqrt[n]{N} = p/q$ with p, q positive integers and the fraction in lowest terms, i.e. p and q coprime. Raising to the nth power, $q^nN = p^n$, and the work above gives $q = 1$. Hence $\sqrt[n]{N} = p$ is a positive integer. So $\sqrt[n]{N}$ is either a positive integer or irrational.

Part (ii)

We have $a^ad^b = b^ac^b$ with $\gcd(a, b) = \gcd(c, d) = 1$.

Showing $d^b = b^a$. The right-hand side shows that b^a divides $a^a d^b$. As a and b are coprime, b^a and a^a are coprime, so b^a divides d^b. Symmetrically, d^b divides $b^a c^b$, and d^b is coprime to c^b, so d^b divides b^a. Two positive integers that divide each other are equal:

$$d^b = b^a,$$

and then cancelling in the original equation gives $a^a = c^b$ as well.

If p is a prime factor of d, then p divides $d^b = b^a$, and a prime dividing a power of b divides b. So p is a prime factor of b.

Powers of p. Let p^m and p^n be the highest powers of p dividing d and b, with $m, n \geq 1$ by the previous paragraph. By unique factorisation, the highest power of p dividing d^b is p^{mb}, and that dividing b^a is p^{na}. Since $d^b = b^a$, these are equal, so $mb = na$ and

$$b = \frac{na}{m}.$$

Ask yourself How does $b = na/m$ give a bound on p^n?

Rewrite it as $mb = na$: so b divides na. But b is coprime to a, so by the given result b divides n, and in particular $b \leq n$. Also p^n divides b, so $p^n \leq b$. Together,

$$p^n \leq b \leq n.$$

But $p \geq 2$ and $n > 0$, so the given fact $y^x > x$ (with $y = p$, $x = n$) says $p^n > n$. This contradiction shows that our assumption was impossible: d has *no* prime factor. Hence $d = 1$, so $b^a = d^b = 1$, and therefore

$$b = 1.$$

The final part

Let r be a positive rational with r^r rational. Write $r = a/b$ in lowest terms (a, b positive and coprime), and $r^r = c/d$ in lowest terms (c, d positive, since $r^r > 0$, and coprime). Then

$$\left(\frac{a}{b}\right)^{a/b} = \frac{c}{d} \quad \Longrightarrow \quad \left(\frac{a}{b}\right)^a = \left(\frac{c}{d}\right)^b \quad \Longrightarrow \quad a^a d^b = b^a c^b,$$

raising both sides to the power b and clearing denominators. This is exactly the situation of part (ii), so $b = 1$ and $r = a$ is a positive integer.

Checks

- Integer r really do work: $r = 2$ gives $r^r = 4$; and the conclusion $c^b = a^a$ with $b = 1$ just says $c = a^a$, as it should.
- A non-example: $r = \frac{1}{2}$ gives $r^r = 1/\sqrt{2}$, irrational, consistent with the result. In part (i), $\sqrt[3]{27} = 3$ and $\sqrt[3]{24}$ is irrational: both alternatives occur.
- A computer search over all coprime $a, b \leq 24$, $c < 200$, $d < 60$ found only solutions of $a^a d^b = b^a c^b$ with $b = d = 1$ (namely $a = 1, 2, 3$ with $c = a^a$).

Common slips

- Forgetting to say *why* p^n and q^n are coprime: the given result is about a and b coprime, so you need the powers to be coprime.
- Concluding $p^n \leq n$ from $b = na/m$ without the step $b \mid n$: it is coprimality of a and b that turns $mb = na$ into $b \mid n$.
- Writing $(a/b)^{a/b} = c/d$ but not insisting that c/d is in lowest terms; without $\gcd(c, d) = 1$, the step $d^b \mid b^a$ fails.

Beyond STEP

The given "coprime divisor" result is Euclid's lemma, usually proved from Bézout's identity, and it is the engine behind unique factorisation into primes. The first part is the general version of the irrationality of $\sqrt{2}$; the second is a first taste of transcendence theory, where the Gelfond–Schneider theorem shows, for instance, that $\sqrt{2}^{\sqrt{2}}$ is not even algebraic. Davenport's *The Higher Arithmetic* develops these foundations beautifully.

WORKED SOLUTION

STEP III 2013, Question 6

Pure • Complex Modulus Inequalities

STEP III 2013, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s3-q6

This question has a full guided treatment in *Beyond STEP*, Section 5.8, *A Hidden Distance in the Disc*: https://beyondstep.keithmcnulty.org/ch05/poincare-disc.html

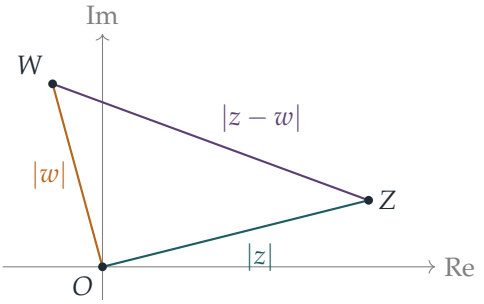
The key idea

Everything rests on writing a squared modulus as a number times its conjugate, $|u|^2 = uu^*$. Expanding $|z-w|^2$ and $|1-zw^*|^2$ this way produces the same cross terms $zw^* + z^*w$ in both, which is exactly why the single quantity E appears in each. Once both moduli are written as "something squared minus E", the final inequality is a comparison of two fractions of the form $\frac{A-E}{B-E}$ and $\frac{A}{B}$, and that comes down to the sign of $B - A$.

Solution

The triangle inequality

Mark the origin O and the points Z and W representing z and w in the Argand diagram. The distances OZ and OW are $|z|$ and $|w|$, and the distance WZ is $|z-w|$.



The straight segment WZ is the shortest path from W to Z, so it is no longer than the

path $W \to O \to Z$:

$$|z - w| = WZ \leq WO + OZ = |w| + |z|.$$

Equality holds exactly when O lies on the segment WZ, that is, when z and w point in opposite directions (or one of them is 0). If O, Z, W are collinear without O between them, the "triangle" is flat, but the shortest-path argument still applies.

Part (i)

Using $|u|^2 = uu^*$ and $(z-w)^* = z^* - w^*$,

$$\begin{aligned} |z-w|^2 &= (z-w)(z^*-w^*) \\ &= zz^* + ww^* - zw^* - z^*w \\ &= |z|^2 + |w|^2 - (zw^* + z^*w). \end{aligned}$$

Since $|zw| = |z||w|$, we have $(|z| + |w|)^2 = |z|^2 + |w|^2 + 2|zw|$, so

$$(|z| + |w|)^2 - E = |z|^2 + |w|^2 + 2|zw| - zw^* - z^*w - 2|zw| = |z-w|^2,$$

as required. Rearranging,

$$E = (|z| + |w|)^2 - |z-w|^2.$$

The right-hand side is a difference of squares of real numbers, so E is real. By the triangle inequality, $0 \leq |z-w| \leq |z| + |w|$, and squaring non-negative quantities preserves the inequality, so $E \geq 0$.

(Directly: z^*w is the conjugate of zw^*, so $E = 2\,\mathrm{Re}(zw^*) + 2|zw^*|$, which is real and at least 0 because $\mathrm{Re}\, u \geq -|u|$.)

Part (ii)

In the same way, using $(zw^*)^* = z^*w$,

$$\begin{aligned} |1 - zw^*|^2 &= (1 - zw^*)(1 - z^*w) \\ &= 1 - zw^* - z^*w + zz^*ww^* \\ &= 1 + |zw|^2 - (zw^* + z^*w) \\ &= \left(1 + 2|zw| + |zw|^2\right) - (zw^* + z^*w + 2|zw|) \\ &= (1 + |zw|)^2 - E, \end{aligned}$$

as required.

The final inequality

Write

$$A = (|z| + |w|)^2, \qquad B = (1 + |zw|)^2,$$

so that parts (i) and (ii) say $|z-w|^2 = A - E$ and $|1 - zw^*|^2 = B - E$.

First, the left-hand side makes sense: if $|z| > 1$ and $|w| > 1$ then $|zw^*| = |z||w| > 1$, so $zw^* \neq 1$ and $B - E = |1 - zw^*|^2 > 0$. Also $B > 0$.

Both sides of the required inequality are non-negative, so it is equivalent to the inequality between their squares,

$$\frac{A - E}{B - E} \leq \frac{A}{B}.$$

Multiplying through by the positive quantity $B(B - E)$, this is equivalent to

$$(A - E)B \leq A(B - E) \iff -EB \leq -EA \iff E(B - A) \geq 0.$$

Ask yourself What is the sign of $B - A$, and where does the condition on $|z|$ and $|w|$ come in?

Writing $a = |z|$ and $b = |w|$,

$$B - A = (1 + ab)^2 - (a + b)^2 = 1 + a^2b^2 - a^2 - b^2 = (1 - a^2)(1 - b^2).$$

If $a > 1$ and $b > 1$, both brackets are negative, so $B - A > 0$. Since $E \geq 0$ by part (i), $E(B - A) \geq 0$. Every step above was an equivalence, so

$$\frac{|z - w|}{|1 - zw^*|} \leq \frac{|z| + |w|}{1 + |zw|}.$$

Equality holds exactly when $E = 0$, that is, when z and w point in opposite directions.

When $|z| < 1$ and $|w| < 1$

Yes, the inequality still holds. Now $|zw^*| = ab < 1$, so again $zw^* \neq 1$ and $B - E > 0$, and the chain of equivalences goes through unchanged. This time both brackets in $(1 - a^2)(1 - b^2)$ are positive, so again $B - A > 0$, giving $E(B - A) \geq 0$ and hence the inequality.

What matters is only that $1 - a^2$ and $1 - b^2$ have the same sign. If one modulus is above 1 and the other below, $B - A < 0$ and the inequality can fail: for $z = 2$, $w = 0.4$ the left-hand side is $1.6/0.2 = 8$ while the right-hand side is $2.4/1.8 = \frac{4}{3}$.

Checks

- Take $z = 2$, $w = 2\mathrm{i}$. Then $|z - w| = 2\sqrt{2}$ and $1 - zw^* = 1 - 2(-2\mathrm{i}) = 1 + 4\mathrm{i}$, so the left-hand side is $2\sqrt{2}/\sqrt{17} \approx 0.686$, while the right-hand side is $4/5 = 0.8$. Scaling both by $\frac{1}{4}$ ($z = \frac{1}{2}$, $w = \frac{1}{2}\mathrm{i}$) gives the same two numbers, $0.686 \leq 0.8$, illustrating the $|z|, |w| < 1$ case.
- Equality case: $z = 2 + \mathrm{i}$, $w = -1.5z$ gives $E = 0$, and both sides equal 0.6577 to four decimal places.
- A computer check on several hundred thousand random pairs confirmed the two identities, that $E \geq 0$, and the inequality in both cases.

Common slips

- Writing $|z-w|^2 = |z|^2 + |w|^2 - 2zw^*$: the cross terms are $zw^* + z^*w$, which are conjugates of each other but not equal in general.
- Cross-multiplying without checking that $|1 - zw^*| \neq 0$; here $|zw| \neq 1$ guarantees it.
- Answering "no" to the last part because the derivation used $|z|, |w| > 1$: the factorisation $(1-a^2)(1-b^2)$ shows that only the agreement of signs matters.

Beyond STEP

Since $|1 - zw^*| = |1 - z^*w|$, the left-hand side is the modulus of $\dfrac{z-w}{1-z^*w}$. For fixed z with $|z| < 1$, the map $w \mapsto \dfrac{z-w}{1-z^*w}$ is a Möbius transformation taking the unit disc onto itself, and $\left|\frac{z-w}{1-z^*w}\right|$ is the pseudo-hyperbolic distance between z and w. The inequality in this question says that this distance is at most the "hyperbolic sum" of the distances from z and from w to 0: a strong form of the triangle inequality, and one of the foundations of hyperbolic geometry. Needham's *Visual Complex Analysis* explains Möbius transformations and the hyperbolic disc with beautiful pictures.

WORKED SOLUTION

STEP III 2013, Question 7

Pure • Energy Methods for ODEs

STEP III 2013, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s3-q7

This question has a full guided treatment in *Beyond STEP*, Section 8.2, *Bounding Without Solving*: https://beyondstep.keithmcnulty.org/ch08/bounding-without-solving.html

The key idea

Each equation looks like a particle on a spring: a "force" term depending only on the unknown, plus (in (ii) and (iii)) a friction term proportional to the derivative. Multiplying the equation by the derivative turns it into the derivative of an "energy". With no friction the energy is conserved; with friction that never pushes the wrong way, the energy can only decrease, so it never exceeds its starting value. Part (iii) asks us to build the energy ourselves and to show that the friction coefficient is never negative.

Solution

Part (i)

Write $y' = \frac{\mathrm{d}y}{\mathrm{d}x}$. By the chain rule,

$$\begin{aligned}\frac{\mathrm{dE}}{\mathrm{d}x} &= 2y'y'' + 2y^3y' \\ &= 2y'\left(y'' + y^3\right) = 0,\end{aligned}$$

using the differential equation. So E is constant, and its value is $\mathrm{E}(0) = 0^2 + \frac{1}{2}\cdot 1^4 = \frac{1}{2}$. Hence for every x,

$$\tfrac{1}{2}y^4 = \tfrac{1}{2} - (y')^2 \le \tfrac{1}{2},$$

since $(y')^2 \ge 0$. So $y^4 \le 1$, which gives $|y(x)| \le 1$ for all x.

Part (ii)

Differentiating in the same way, and then using the equation in the form $v'' + \sinh v = -xv'$,

$$\begin{aligned}\frac{\mathrm{dE}}{\mathrm{d}x} &= 2v'v'' + 2\sinh v\, v' \\ &= 2v'\big(v'' + \sinh v\big) \\ &= -2x\,(v')^2.\end{aligned}$$

For $x \geq 0$ both x and $(v')^2$ are non-negative, so $\frac{\mathrm{dE}}{\mathrm{d}x} \leq 0$, as required.

So E is non-increasing on $x \geq 0$ and $\mathrm{E}(x) \leq \mathrm{E}(0)$ there. At $x = 0$, $\cosh(\ln 3) = \frac{1}{2}\big(3 + \frac{1}{3}\big) = \frac{5}{3}$, so $\mathrm{E}(0) = 0 + 2 \cdot \frac{5}{3} = \frac{10}{3}$. Therefore, for $x \geq 0$,

$$2\cosh v(x) = \mathrm{E}(x) - (v')^2 \leq \mathrm{E}(x) \leq \tfrac{10}{3},$$

so $\cosh v(x) \leq \frac{5}{3}$.

Part (iii)

We look for an energy of the form $\mathrm{E} = (w')^2 + F(w)$, as in (ii). Then

$$\frac{\mathrm{dE}}{\mathrm{d}x} = 2w'w'' + F'(w)\,w' = w'\big(2w'' + F'(w)\big),$$

and we want the bracket to become a multiple of w' via the equation. That happens if $F'(w) = 2(w\cosh w + 2\sinh w)$.

? **Ask yourself** What function of w has derivative $w\cosh w + 2\sinh w$?

Integrating by parts, $\int w\cosh w\,\mathrm{d}w = w\sinh w - \cosh w$, and $\int 2\sinh w\,\mathrm{d}w = 2\cosh w$. Adding,

$$\frac{\mathrm{d}}{\mathrm{d}w}\big(w\sinh w + \cosh w\big) = \sinh w + w\cosh w + \sinh w = w\cosh w + 2\sinh w.$$

So we take

$$\mathrm{E}(x) = \left(\frac{\mathrm{d}w}{\mathrm{d}x}\right)^2 + 2\big(w\sinh w + \cosh w\big).$$

Writing $g(x) = 5\cosh x - 4\sinh x - 3$, the equation says $w'' + (w\cosh w + 2\sinh w) = -g(x)\,w'$, so

$$\frac{\mathrm{dE}}{\mathrm{d}x} = 2w'\big(w'' + w\cosh w + 2\sinh w\big) = -2g(x)\,(w')^2.$$

? **Ask yourself** Is $g(x)$ ever negative?

In exponentials,

$$\begin{aligned} g(x) &= \tfrac{5}{2}\left(\mathrm{e}^{x}+\mathrm{e}^{-x}\right)-2\left(\mathrm{e}^{x}-\mathrm{e}^{-x}\right)-3 \\ &= \tfrac{1}{2}\mathrm{e}^{x}-3+\tfrac{9}{2}\mathrm{e}^{-x} \\ &= \tfrac{1}{2}\mathrm{e}^{-x}\left(\mathrm{e}^{2x}-6\mathrm{e}^{x}+9\right)=\tfrac{1}{2}\mathrm{e}^{-x}\left(\mathrm{e}^{x}-3\right)^{2} \;\geq\; 0 \end{aligned}$$

for every x (it is zero only at $x=\ln 3$). Hence $\frac{\mathrm{dE}}{\mathrm{d}x}\leq 0$, and $\mathrm{E}(x)\leq \mathrm{E}(0)$ for $x\geq 0$. At $x=0$, $w=0$ and $w'=\frac{1}{\sqrt{2}}$, so $\mathrm{E}(0)=\frac{1}{2}+2(0+1)=\frac{5}{2}$. Therefore, for $x\geq 0$,

$$2\left(w\sinh w+\cosh w\right)\leq \mathrm{E}(x)\leq \tfrac{5}{2}, \qquad \text{so} \qquad w\sinh w+\cosh w\leq \tfrac{5}{4}.$$

Finally, $w\sinh w\geq 0$ for every real w, because w and $\sinh w$ always have the same sign. Dropping this term gives $\cosh w(x)\leq \frac{5}{4}$ for $x\geq 0$, as required.

Checks

- In (ii) the bound is sharp at the start: $\cosh v(0)=\cosh(\ln 3)=\frac{5}{3}$.
- In (iii), $g(0)=5-0-3=2$, which agrees with $\frac{1}{2}(1-3)^2=2$ from the factorised form.
- A numerical solution of each equation confirms the picture: in (i) E stays at 0.5; in (ii) E falls from $\frac{10}{3}$; in (iii) E falls from 2.5 and $\cosh w$ never rises above about 1.03, well inside the bound 1.25.

Common slips

- Forgetting the factor w' from the chain rule when differentiating $F(w)$ with respect to x.
- In (iii), stopping at $w\sinh w+\cosh w\leq\frac{5}{4}$ without explaining why $w\sinh w\geq 0$ lets us drop that term.
- Claiming $g(x)>0$: it vanishes at $x=\ln 3$, so only $\leq$ holds (which is all we need).

Beyond STEP

These energy functions are *Lyapunov functions*: quantities that never increase along solutions, used to prove that a system stays bounded or settles down without solving it. The same idea, conservation of energy plus dissipation by damping, is the basis of stability theory for nonlinear oscillators. Boyce and DiPrima, *Elementary Differential Equations*, covers Lyapunov's method in its chapter on nonlinear systems and stability.

WORKED SOLUTION

STEP III 2013, Question 8

Pure • Polar Equations of Conics

STEP III 2013, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s3-q8

The key idea

The relation $r = ks$ with $k < 1$ says that E is an ellipse with focus O and directrix D, and in polar coordinates centred on O it becomes a very simple equation for r in terms of θ. A chord through O is just two radii pointing in opposite directions, so $1/l$ comes out as a simple expression in $\cos^2\theta$. The n equally spaced lines then turn the sum into a sum of $\cos 2\theta$ terms, which is exactly what the opening sum disposes of.

Solution

The opening sum

The terms form a geometric progression with first term $\mathrm{e}^{2\mathrm{i}\alpha}$ and common ratio $w = \mathrm{e}^{2\pi\mathrm{i}/n}$. Since $n \geq 2$, the angle $2\pi/n$ lies in $(0, \pi]$, so $w \neq 1$ and we may use the usual formula:

$$\sum_{r=0}^{n-1} \mathrm{e}^{2\mathrm{i}(\alpha + r\pi/n)} = \mathrm{e}^{2\mathrm{i}\alpha}\,\frac{1-w^n}{1-w} = \mathrm{e}^{2\mathrm{i}\alpha}\,\frac{1-\mathrm{e}^{2\pi\mathrm{i}}}{1-w} = 0.$$

Taking real parts, $\displaystyle\sum_{r=0}^{n-1} \cos\left(2\alpha + 2r\pi/n\right) = 0$; this is the form we shall use.

The distance s

Measure θ from the perpendicular from O to D, as in the diagram. The foot of that perpendicular is at distance d from O, and P is at distance $r\cos\theta$ from O in the same direction, so for P on the same side of D as O,

$$s = d - r\cos\theta.$$

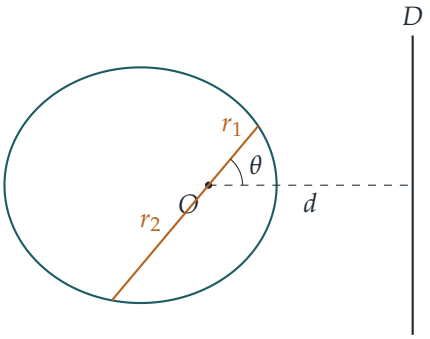
(This holds for every θ: when $\cos\theta < 0$, P is further from D than O is, and the formula correctly gives $s > d$.)

The polar equation of E

Every point of E is on the same side of D as O. Indeed, if P were on the other side we would have $s = r\cos\theta - d$, and $r = ks$ would give $r(1 - k\cos\theta) = -kd < 0$, which is impossible because $1 - k\cos\theta > 0$ for $0 < k < 1$. So we can use $s = d - r\cos\theta$, and $r = ks$ becomes

$$r = k(d - r\cos\theta) \quad \Longrightarrow \quad r = \frac{kd}{1 + k\cos\theta}.$$

The denominator is positive for every θ since $k < 1$, so each ray from O meets E exactly once.



The length of a chord through O

A line through O making angle θ with the perpendicular meets E at the points in directions θ and $\theta + \pi$, which lie on opposite sides of O. Since $\cos(\theta + \pi) = -\cos\theta$, the chord length is

$$l = \frac{kd}{1 + k\cos\theta} + \frac{kd}{1 - k\cos\theta} = \frac{2kd}{1 - k^2\cos^2\theta}, \qquad \text{so} \qquad \frac{1}{l} = \frac{1 - k^2\cos^2\theta}{2kd}.$$

Note that this depends only on the line, not on which of its two directions we call θ.

Ask yourself How do we sum $\cos^2\theta$ over the n lines?

Use the double-angle formula $\cos^2\theta = \frac{1}{2}(1 + \cos 2\theta)$: the $\cos 2\theta$ terms are exactly the real parts of the opening sum.

The sum over the n lines

Since adjacent lines are at angle π/n, the n lines make angles $\theta_j = \alpha + j\pi/n$ $(j = 1, \ldots, n)$ for some fixed α. These n values of $2\theta_j$ are $2\alpha + 2j\pi/n$; reducing $j = n$ to $j = 0$ (which changes $2\theta_j$ by 2π) they are the angles of the opening sum. Hence

$$\sum_{j=1}^{n} \cos^2\theta_j = \frac{n}{2} + \frac{1}{2}\sum_{j=1}^{n} \cos 2\theta_j = \frac{n}{2} + 0 = \frac{n}{2},$$

and so

$$\sum_{j=1}^{n} \frac{1}{l_j} = \frac{1}{2kd}\left(n - k^2 \cdot \frac{n}{2}\right) = \frac{(2-k^2)n}{4kd},$$

which is the required result. The condition $n \geq 2$ was used exactly once, to ensure $w \neq 1$ in the opening sum.

Checks

- Take $n = 2$, $k = \frac{1}{2}$, $d = 1$ with lines along and perpendicular to OD. Along OD the chord is the major axis, of length $\frac{1/2}{3/2} + \frac{1/2}{1/2} = \frac{4}{3}$; perpendicular to it the chord (the latus rectum) has length $2kd = 1$. So $\frac{3}{4} + 1 = \frac{7}{4}$, and the formula gives $\frac{(7/4)\cdot 2}{2} = \frac{7}{4}$.
- For $n = 1$ the result fails, as it should: a single line's $1/l$ depends on its direction. The case $n = 1$ is exactly where $w = 1$.
- A computer check, finding each chord directly from $r = ks$ for random d, k, n and α, agreed with the formula; for example $d = 1$, $k = \frac{1}{2}$, $n = 3$ gives $2.625 = \frac{21}{8}$ for any α.

Common slips

- Writing $s = d + r\cos\theta$: in the diagram θ is measured towards D, so moving out along OP brings you closer to D.
- Taking the second half of the chord as $kd/(1 + k\cos\theta)$ again, forgetting that the opposite direction has angle $\theta + \pi$.
- Summing the angles $\alpha + j\pi/n$ directly instead of $2\theta_j$: it is the doubled angles that go once round the circle evenly, which is why the opening sum has the factor 2.

Beyond STEP

The focus–directrix definition gives every conic the polar equation $r = \ell/(1 + e\cos\theta)$, with $e < 1$, $e = 1$ and $e > 1$ for ellipse, parabola and hyperbola; this is the form in which Kepler's orbits appear when one solves Newton's equations for planetary motion. The fact that $1/l_1 + 1/l_2$ is the same for any two perpendicular focal chords is a classical property, and the result here generalises it. Kleppner and Kolenkow, *An Introduction to Mechanics*, derives the orbit equation this way.

WORKED SOLUTION

STEP III 2013, Question 9

Mechanics • Buoyancy and SHM

STEP III 2013, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s3-q9

The key idea

The submerged part of the sphere is a spherical cap, and its volume comes from slicing it into discs, which is a volume of revolution. After that the question is Newton's second law with buoyancy as the only force besides weight. For small oscillations we put $x = x_0 + y$, linearise the cubic on the right about the equilibrium, and read off ω^2 from an equation of the form $\ddot{y} = -\omega^2 y$.

Solution

The displaced volume

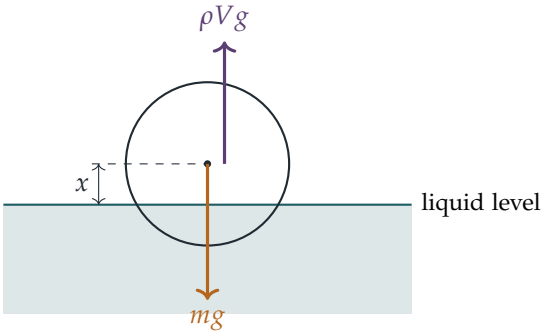
Take the centre of the sphere as origin with z measured upwards. The liquid surface is then the plane $z = -x$, and the submerged part is the part of the ball with $-R \le z \le -x$. This makes sense for $-R < x < R$, so it covers the case in the question. At height z the cross-section is a disc of radius $\sqrt{R^2 - z^2}$, so

$$\begin{aligned} V &= \int_{-R}^{-x} \pi \left(R^2 - z^2\right) \mathrm{d}z = \pi \left[R^2 z - \tfrac{1}{3} z^3\right]_{-R}^{-x} \\ &= \pi \left(-R^2 x + \tfrac{1}{3} x^3 + R^3 - \tfrac{1}{3} R^3\right) = \frac{\pi}{3} \left(2R^3 - 3R^2 x + x^3\right), \end{aligned}$$

which is the required volume.

The equation of motion

Model the sphere as a rigid body moving vertically. Because the tank is large, the liquid level does not move as the sphere bobs, so $\ddot{x}$ is the upward acceleration of the sphere. Its mass is $m = \frac{4}{3}\pi R^3 \rho_s$. Two forces act on it: its weight mg downwards, and the buoyancy force $\rho V g$ upwards.



Resolving vertically, with upwards positive,

$$m\ddot{x} = \rho V g - mg \quad \Longrightarrow \quad m\,(g + \ddot{x}) = \rho g V.$$

Substituting for m and V:

$$\tfrac{4}{3}\pi R^3 \rho_s\,(g + \ddot{x}) = \tfrac{\pi}{3}\left(2R^3 - 3R^2 x + x^3\right)\rho g,$$

and multiplying by $3/\pi$ gives

$$4R^3\rho_s\,(g + \ddot{x}) = \left(2R^3 - 3R^2 x + x^3\right)\rho g,$$

as required.

The equilibrium density

In equilibrium $\ddot{x} = 0$. Putting $x = \frac{1}{2}R$:

$$4R^3\rho_s g = \left(2R^3 - \tfrac{3}{2}R^3 + \tfrac{1}{8}R^3\right)\rho g = \tfrac{5}{8}R^3\rho g,$$

so

$$\rho_s = \frac{5}{32}\,\rho.$$

This is less than $\frac{1}{2}\rho$, as it should be: the centre is above the surface, so less than half the sphere is under water.

Small oscillations

Write $f(x) = 2R^3 - 3R^2x + x^3$, and put $x = \frac{1}{2}R + y$ with y small. Then $\ddot{x} = \ddot{y}$.

> **Ask yourself** How do we turn the cubic into something linear in y?

We expand f about the equilibrium and keep only first-order terms: $f(\frac{1}{2}R + y) = f(\frac{1}{2}R) + f'(\frac{1}{2}R)\,y + O(y^2)$. Here

$$f'(x) = -3R^2 + 3x^2, \qquad f'(\tfrac{1}{2}R) = -3R^2 + \tfrac{3}{4}R^2 = -\tfrac{9}{4}R^2,$$

and $f(\frac{1}{2}R) = \frac{5}{8}R^3$ from before. Using $\rho_s = \frac{5}{32}\rho$, the left-hand side of the equation of motion is $4R^3 \cdot \frac{5}{32}\rho\,(g + \ddot{y}) = \frac{5}{8}R^3\rho\,(g + \ddot{y})$. So, to first order,

$$\tfrac{5}{8}R^3\rho\,(g + \ddot{y}) = \left(\tfrac{5}{8}R^3 - \tfrac{9}{4}R^2 y\right)\rho g.$$

The $\frac{5}{8}R^3\rho g$ terms cancel (that is just the equilibrium condition), leaving

$$\tfrac{5}{8}R^3\ddot{y} = -\tfrac{9}{4}R^2 g\, y \quad \Longrightarrow \quad \ddot{y} = -\frac{18g}{5R}\, y.$$

This is simple harmonic motion with $\omega^2 = \dfrac{18g}{5R}$, so the period of small oscillations is

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{5R}{18g}}.$$

Checks

- Volume: at $x = R$ the formula gives 0 (just touching the surface), at $x = 0$ it gives $\frac{2}{3}\pi R^3$ (half the sphere), and at $x = -R$ it gives $\frac{4}{3}\pi R^3$ (fully submerged).
- Dimensions: $\omega^2 = 18g/(5R)$ has units s^{-2}, as it should.
- Numbers: with $R = 1\,\mathrm{m}$ and $g = 9.81\,\mathrm{m\,s^{-2}}$ the period is about 1.06 s. Stepping the full nonlinear equation forward by computer from a small displacement gives the same period to better than one part in 10^4.

Common slips

- Taking the height of the submerged cap as x rather than $R - x$: x is the height of the centre *above* the liquid.
- Forgetting that the restoring force comes from $f'(\frac{1}{2}R)$, which is negative. A sign error here gives $\ddot{y} = +\omega^2 y$, which isn't oscillation at all.
- Dropping the sphere's own mass on the left: $\ddot{x}$ multiplies ρ_s, not ρ. (Real floating bodies also drag some liquid with them, the "added mass", which the question tells us to ignore.)

Beyond STEP

The method used here, expanding the force about an equilibrium and keeping the linear term, is the standard way to find small oscillations of any system. Its sign gives stability, and its size gives the frequency. In a first-year course this becomes linearisation of $\ddot{x} = F(x)$, and then the study of normal modes when there are several degrees of freedom. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers small oscillations about equilibrium and fluid buoyancy at this level.

WORKED SOLUTION

STEP III 2013, Question 10

Mechanics • Rotational Impact

STEP III 2013, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

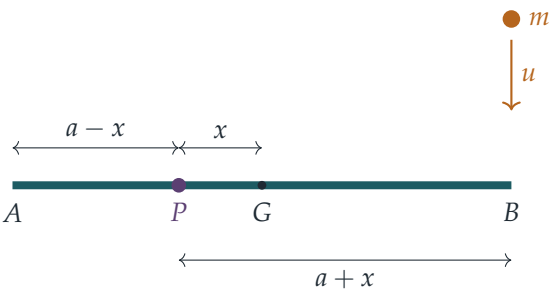
step.maths.org/questions/13-s3-q10

The key idea

The moment of inertia comes straight from the parallel axis theorem, once we notice how far the centre of the rod is from P. For the impact, the axis exerts an unknown impulse on the rod, so linear momentum is *not* conserved; but that impulse acts through P, so it has no moment about P. Conservation of angular momentum about P, together with Newton's experimental law at B, gives two equations for the two unknowns.

Solution

We model the rod as a uniform rigid rod, the particle as a particle, and the impact as instantaneous, so that the only horizontal impulses are those at B and at the axis (finite forces such as gravity give no impulse during the instant of impact).



The moment of inertia

Let G be the midpoint of the rod, its centre of mass. Since $AG = a$ and $AP = a - x$, the distance PG is x (with G on the B side of P when $x > 0$). The moment of inertia of a uniform rod about a perpendicular axis through its centre is $\frac{1}{3}Ma^2$ (for a rod of length $2a$). By the parallel axis theorem,

$$I_P = \tfrac{1}{3}Ma^2 + Mx^2 = \tfrac{1}{3}M(a^2 + 3x^2),$$

as required.

If you prefer to work from first principles: with line density $M/(2a)$ and s measured along the rod from P, the rod runs from $s = -(a-x)$ to $s = a+x$, and

$$I_P = \frac{M}{2a}\int_{-(a-x)}^{a+x} s^2\,\mathrm{d}s = \frac{M}{6a}\left[(a+x)^3 + (a-x)^3\right] = \frac{M}{6a}(2a^3 + 6ax^2),$$

which is the same thing.

The impact

The distance from P to B is $2a - (a-x) = a+x$. Let v be the particle's velocity immediately after impact (in the original direction of motion) and ω the rod's angular velocity, so that B then moves with speed $(a+x)\omega$ in the same direction.

Ask yourself Why can't we use conservation of linear momentum here?

Because the axis is fixed, it supplies an impulse on the rod during the collision, of unknown size. That impulse acts at P, so its moment about P is zero. The impulse between the particle and the rod is internal to the system of rod and particle. So the total angular momentum of the system about P is conserved:

$$mu(a+x) = mv(a+x) + I_P\,\omega. \tag{1}$$

Newton's experimental law at B (speed of separation is e times speed of approach) gives

$$(a+x)\omega - v = eu. \tag{2}$$

From (2), $v = (a+x)\omega - eu$. Substituting into (1),

$$\begin{aligned} mu(a+x) &= m(a+x)\left[(a+x)\omega - eu\right] + I_P\,\omega \\ mu(1+e)(a+x) &= \left[m(a+x)^2 + \tfrac{1}{3}M(a^2+3x^2)\right]\omega. \end{aligned}$$

Multiplying through by 3,

$$\omega = \frac{3mu(1+e)(a+x)}{M(a^2+3x^2) + 3m(a+x)^2},$$

which is the given result.

The case $m = 2M$

Put $m = 2M$ and cancel M:

$$\omega = \frac{6u(1+e)(a+x)}{a^2+3x^2+6(a+x)^2}.$$

The variable that appears naturally is the lever arm $s = a+x$, the distance PB, which lies in $[0, 2a]$. With $x = s-a$, the denominator is

$$a^2 + 3(s-a)^2 + 6s^2 = 9s^2 - 6as + 4a^2,$$

so

$$\omega = \frac{6u(1+e)\,s}{9s^2 - 6as + 4a^2}.$$

At $s = 0$ (the axis at B) we have $\omega = 0$, so the maximum occurs with $s > 0$, and there we can divide top and bottom by s:

$$\omega = \frac{6u(1+e)}{9s + \dfrac{4a^2}{s} - 6a}.$$

We want the denominator as small as possible. Since $s > 0$, the AM–GM inequality gives

$$9s + \frac{4a^2}{s} \;\geq\; 2\sqrt{9s \cdot \frac{4a^2}{s}} = 12a,$$

with equality exactly when $9s = 4a^2/s$, that is $s = \frac{2}{3}a$, which does lie in $(0, 2a]$. So the denominator is at least $12a - 6a = 6a$, with equality only at $s = \frac{2}{3}a$, and so

$$\omega \leq \frac{6u(1+e)}{6a} = \frac{u(1+e)}{a}.$$

The greatest angular velocity is therefore $u(1+e)/a$, attained when $a + x = \frac{2}{3}a$, that is

$$x = -\tfrac{1}{3}a.$$

The negative sign just means that P is on the B side of the centre: P is a distance $\frac{4}{3}a$ from A and $\frac{2}{3}a$ from B.

Checks

- At $x = a$ (the axis at A) with $e = 1$ the formula gives $\omega = 3mu/\big(a(M+3m)\big)$; letting $M \to 0$ gives $\omega = u/a$, so B moves at $2u$, which is the familiar result for a very light "target" struck elastically.
- In the optimal case with $a = 1$, $u = 1$, $e = \frac{1}{2}$, the rod turns at $\omega = 1.5$, so B moves at 1, and the particle's velocity after impact is $v = 1 - \frac{1}{2} = \frac{1}{2}$. Then the angular momentum about P is $2M \cdot \frac{1}{2} \cdot \frac{2}{3} + \frac{1}{3}M\big(1 + \frac{1}{3}\big) \cdot 1.5 = \frac{4}{3}M$, matching $2M \cdot 1 \cdot \frac{2}{3}$ before impact.
- A fine search by computer over $x \in [-a, a]$ confirms the maximum at $x = -a/3$.

Common slips

- Conserving linear momentum: the fixed axis gives an impulse, so only angular momentum about P is conserved.
- Taking the distance PB as $a - x$ or $2a - x$ rather than $a + x$.
- Rejecting $x = -\frac{1}{3}a$ because it is negative: x is a signed position, and any x in $[-a, a]$ puts P on the rod.

Beyond STEP

Choosing the point about which to take moments so that an unknown reaction drops out is one of the most useful tricks in rigid-body dynamics; it reappears in the "centre of percussion", the point of a bat or rod at which a blow produces no impulse at the pivot. First-year mechanics develops this into the general theory of angular momentum and the inertia tensor. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats impulsive rotation and the centre of percussion very clearly.

WORKED SOLUTION

STEP III 2013, Question 11

Mechanics • Elastic Springs and Energy

STEP III 2013, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s3-q11

The key idea

Everything is fixed by symmetry: the triangle stays horizontal, its centre stays directly below the suspension point, and it can only move up and down. So one particle and one spring tell the whole story. The geometry of the vertical plane through a spring gives its length in terms of θ. Resolving vertically gives the equilibrium, and conservation of energy (elastic, gravitational and kinetic) gives the speed.

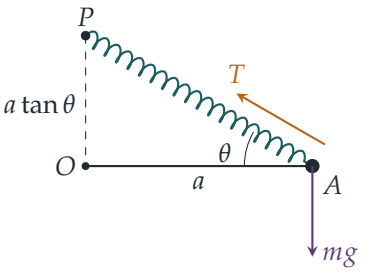
Solution

Model and geometry

Treat the masses as particles, the rods as light and rigid, and the springs as light and obeying Hooke's law. Let P be the point of suspension and O the centre of the triangle, directly below P. The distance from the centre of an equilateral triangle of side s to a vertex is $s/\sqrt{3}$, so each vertex is at horizontal distance

$$OA = \frac{\sqrt{3}a}{\sqrt{3}} = a$$

from O. Take a vertex A and look at the vertical plane through P, O and A:



The spring PA makes angle θ with the horizontal OA, so the spring has length $a/\cos\theta$ and A is a depth $a\tan\theta$ below P.

The tension

The extension of each spring is

$$x = \frac{a}{\cos\theta} - a = \frac{a(1-\cos\theta)}{\cos\theta},$$

so by Hooke's law, with modulus kmg and natural length a,

$$T = \frac{kmg}{a}\,x = \frac{kmg(1-\cos\theta)}{\cos\theta},$$

as required.

Equilibrium at $\theta = \frac{1}{6}\pi$

The forces on the particle at A are its weight, the spring tension along AP and the forces from the two rods. The rods lie in the horizontal plane, so their forces are horizontal.

? **Ask yourself** Why can we ignore the rod forces?

We resolve vertically, and the rod forces have no vertical component. (Horizontally the tension component $T\cos\theta$ is balanced by the rods, which is why the rods are needed, but we don't need that equation.) Resolving vertically at A:

$$T\sin\theta = mg \quad\Longrightarrow\quad k(1-\cos\theta)\tan\theta = 1.$$

At $\theta = \frac{1}{6}\pi$, $\cos\theta = \frac{\sqrt{3}}{2}$ and $\tan\theta = \frac{1}{\sqrt{3}}$, so

$$k\cdot\frac{2-\sqrt{3}}{2}\cdot\frac{1}{\sqrt{3}} = 1 \quad\Longrightarrow\quad k = \frac{2\sqrt{3}}{2-\sqrt{3}} = \frac{2\sqrt{3}(2+\sqrt{3})}{(2-\sqrt{3})(2+\sqrt{3})} = 4\sqrt{3}+6,$$

since $(2-\sqrt{3})(2+\sqrt{3}) = 1$. This is the required value.

The speed at the equilibrium position

By symmetry the triangle stays horizontal with its centre below P, so all three particles move vertically with the same speed. No external force other than gravity and the springs does work, so energy is conserved. The elastic energy in a spring of extension x is $kmg\,x^2/(2a)$.

Start ($\theta = \frac{1}{3}\pi$, at rest). Here $\cos\theta = \frac{1}{2}$, so the extension is $x_1 = a$, and the depth below P is $a\tan\frac{1}{3}\pi = \sqrt{3}a$.

Equilibrium ($\theta = \frac{1}{6}\pi$). The extension is

$$x_2 = a\left(\frac{2}{\sqrt{3}} - 1\right) = \frac{a(2-\sqrt{3})}{\sqrt{3}}, \qquad x_2^2 = \frac{a^2(7-4\sqrt{3})}{3},$$

and the depth is $a\tan\frac{1}{6}\pi = \frac{a}{\sqrt{3}}$. So each particle rises by

$$\sqrt{3}a - \frac{a}{\sqrt{3}} = \frac{2a}{\sqrt{3}}.$$

Elastic energy released by the three springs:

$$3\cdot\frac{kmg}{2a}\left(x_1^2-x_2^2\right)=\frac{3kmg}{2a}\cdot\frac{a^2\left(3-7+4\sqrt{3}\right)}{3}$$
$$=2kmga(\sqrt{3}-1).$$

With $k=4\sqrt{3}+6$, $k(\sqrt{3}-1)=12-4\sqrt{3}+6\sqrt{3}-6=6+2\sqrt{3}$, so the elastic energy released is $4mga(3+\sqrt{3})$.

Gravitational energy gained by the three particles: $3mg\cdot\dfrac{2a}{\sqrt{3}}=2\sqrt{3}\,mga$.

The difference becomes kinetic energy of the three particles, each moving with speed V:

$$\tfrac{3}{2}mV^2=4mga(3+\sqrt{3})-2\sqrt{3}\,mga=mga\left(12+2\sqrt{3}\right),$$
$$V^2=\frac{2ag}{3}\left(12+2\sqrt{3}\right)=\frac{4ag}{3}\left(6+\sqrt{3}\right),$$

as required.

Checks

- Signs: at $\theta=\frac{1}{3}\pi$ the vertical pull of each spring is $T\sin\theta=kmg\cdot\frac{\sqrt{3}}{2}\approx 11.2\,mg>mg$, so the triangle does start by moving *up* towards the equilibrium, as the energy equation assumes.
- The answer is positive and has the units of a speed squared (ag).
- Taking $a=1.3\,\text{m}$ and $g=9.81\,\text{m s}^{-2}$, stepping the equation of motion $\ddot{y}=g-\frac{T}{m}\sin\theta$ forward numerically from rest at $\theta=\frac{1}{3}\pi$ gives $V^2\approx 131.48\,\text{m}^2\,\text{s}^{-2}$ at $\theta=\frac{1}{6}\pi$, matching $\frac{4ag}{3}(6+\sqrt{3})$.

Common slips

- Taking the distance from the centre to a vertex to be $\sqrt{3}a$ or $\frac{\sqrt{3}}{2}a$ instead of a.
- Forgetting that there are three springs and three particles: the factor 3 appears in the elastic, gravitational and kinetic terms.
- Getting the direction of motion wrong: the particles *rise* between the two positions, so gravitational energy is gained, not lost.

Beyond STEP

The triangle has only one degree of freedom here because symmetry rules out every other motion. Choosing a single coordinate (here θ, or the depth y) and writing the energy in terms of it is the starting point of Lagrangian mechanics. Expanding the

potential energy about the equilibrium would also give the period of small vertical oscillations. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops energy methods and small oscillations in this spirit.

WORKED SOLUTION

STEP III 2013, Question 12

Probability and Statistics • Indicator Random Variables

STEP III 2013, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s3-q12

The key idea

Each X_k equals 1 exactly when a new block of As starts at position k, so S counts the blocks (runs) of As in the row. The variables are indicators, so every expectation is just the probability that a few named positions hold named letters, and that depends only on how many positions are involved, not on where they are. For the variance we expand S^2 and sort the products X_iX_j into types: equal indices, adjacent indices (which turn out to be impossible) and the rest.

Solution

Throughout, all $\binom{n}{a}$ orderings are equally likely. A useful consequence: for any m *distinct* positions, the probability that they hold a prescribed pattern containing p As and q Bs ($p+q=m$) is obtained by filling the positions one at a time, for example

$$\Pr(\text{positions } r,s \text{ hold } B,A) = \frac{b}{n}\cdot\frac{a}{n-1},$$

whatever the distinct positions r and s are. Since each X_k takes only the values 0 and 1, $\mathrm{E}(X_k) = \Pr(X_k = 1)$, and similarly $\mathrm{E}(X_iX_j) = \Pr(X_i = 1 \text{ and } X_j = 1)$.

Part (i)

For $i = 1$ we need the first letter to be A, so

$$\mathrm{E}(X_1) = \frac{a}{n}.$$

For $2 \le i \le n$ we need a B in position $i-1$ and an A in position i, so

$$\mathrm{E}(X_i) = \frac{b}{n}\cdot\frac{a}{n-1} = \frac{ab}{n(n-1)}.$$

By linearity of expectation, summing over the $n-1$ indices $i \ge 2$,

$$\mathrm{E}(S) = \frac{a}{n} + (n-1)\cdot\frac{ab}{n(n-1)} = \frac{a}{n} + \frac{ab}{n} = \frac{a(b+1)}{n},$$

as required.

Part (ii)

(a) Let $j \geq 3$. Then $X_1X_j = 1$ exactly when position 1 holds A, position $j-1$ holds B and position j holds A. Since $j-1 \geq 2$, these are three distinct positions, holding two As and one B. Filling them in turn,

$$\mathrm{E}(X_1X_j) = \frac{a}{n} \cdot \frac{b}{n-1} \cdot \frac{a-1}{n-2} = \frac{a(a-1)b}{n(n-1)(n-2)}.$$

(b) Now take $2 \leq i$ and $j \geq i+2$. Then $X_iX_j = 1$ exactly when positions $i-1$, i, $j-1$, j hold B, A, B, A respectively. Since $j-1 \geq i+1 > i$, these four positions are distinct, and they hold two As and two Bs, so

$$\mathrm{E}(X_iX_j) = \frac{b}{n} \cdot \frac{a}{n-1} \cdot \frac{b-1}{n-2} \cdot \frac{a-1}{n-3} = \frac{a(a-1)b(b-1)}{n(n-1)(n-2)(n-3)}.$$

(Here $n \geq 4$ because $a, b \geq 2$, so nothing vanishes in the denominator.) This is the same for every pair in the double sum, so we only need to count the pairs. For a given i, j runs from $i+2$ to n, which is $n-i-1$ values; as i runs from 2 to $n-2$ this takes the values $n-3$, $n-4$, …, 1. So the number of pairs is

$$1 + 2 + \cdots + (n-3) = \tfrac{1}{2}(n-3)(n-2),$$

and

$$\sum_{i=2}^{n-2} \left(\sum_{j=i+2}^{n} \mathrm{E}(X_iX_j) \right) = \frac{(n-2)(n-3)}{2} \cdot \frac{a(a-1)b(b-1)}{n(n-1)(n-2)(n-3)} = \frac{a(a-1)b(b-1)}{2n(n-1)}.$$

(c) We use $\mathrm{Var}(S) = \mathrm{E}(S^2) - \big(\mathrm{E}(S)\big)^2$ and expand

$$S^2 = \sum_{i=1}^{n} X_i^2 + 2\sum_{i<j} X_iX_j.$$

Since X_i is 0 or 1, $X_i^2 = X_i$, so the first sum has expectation $\mathrm{E}(S)$.

Ask yourself What about the pairs with $j = i+1$, which part (b) left out?

They contribute nothing: $X_iX_{i+1} = 0$ always. For $i = 1$, $X_1 = 1$ needs an A in position 1 but $X_2 = 1$ needs a B there. For $i \geq 2$, $X_i = 1$ needs an A in position i but $X_{i+1} = 1$ needs a B there. (Two blocks of As can't start in neighbouring positions.)

So the pairs $i < j$ that matter are those with $j \geq i+2$. Those with $i = 1$ have $j = 3, \ldots, n$, which is $n-2$ pairs, each given by part (a); those with $i \geq 2$ are exactly the double sum in part (b). Hence

$$\begin{aligned}\sum_{i<j} \mathrm{E}(X_iX_j) &= (n-2) \cdot \frac{a(a-1)b}{n(n-1)(n-2)} + \frac{a(a-1)b(b-1)}{2n(n-1)} \\ &= \frac{a(a-1)b}{n(n-1)}\left(1 + \frac{b-1}{2}\right) = \frac{a(a-1)b(b+1)}{2n(n-1)},\end{aligned}$$

and therefore

$$\mathrm{E}(S^2) = \frac{a(b+1)}{n} + \frac{a(a-1)b(b+1)}{n(n-1)}.$$

Subtracting $\big(\mathrm{E}(S)\big)^2 = a^2(b+1)^2/n^2$ and taking out the common factor $\dfrac{a(b+1)}{n^2(n-1)}$,

$$\mathrm{Var}(S) = \frac{a(b+1)}{n^2(n-1)}\Big[n(n-1) + n(a-1)b - a(b+1)(n-1)\Big].$$

Expanding the bracket, the terms nab and $-nab$ cancel, leaving

$$\begin{aligned} n^2 - n - nb - na + ab + a &= n(n-a-b) - n + ab + a \\ &= -n + ab + a \qquad \text{since } n = a+b \\ &= ab - b = b(a-1). \end{aligned}$$

So

$$\mathrm{Var}(S) = \frac{a(b+1)}{n^2(n-1)} \cdot b(a-1) = \frac{a(a-1)b(b+1)}{n^2(n-1)},$$

as required.

Checks

Take $a = 3$, $b = 2$, so $n = 5$ and there are 10 orderings. Counting blocks of As, the orderings $AAABB$, $BAAAB$, $BBAAA$ give $S = 1$; $ABABA$ gives $S = 3$; the other six give $S = 2$. So $\mathrm{E}(S) = \frac{18}{10} = \frac{9}{5}$ and $\mathrm{E}(S^2) = \frac{36}{10}$, giving $\mathrm{Var}(S) = \frac{18}{5} - \frac{81}{25} = \frac{9}{25}$. The formulae give $\frac{3\cdot3}{5} = \frac{9}{5}$ and $\frac{3\cdot2\cdot2\cdot3}{25\cdot4} = \frac{9}{25}$. A computer enumeration confirmed every formula in the question exactly for all $2 \le a, b \le 6$.

The formulae also behave sensibly outside the stated range. With $a = 2$, $b = 1$ the orderings AAB, ABA, BAA give $S = 1, 2, 1$, so $\mathrm{E}(S) = \frac{4}{3}$ and $\mathrm{Var}(S) = 2 - \frac{16}{9} = \frac{2}{9}$, matching $\frac{2\cdot2}{3}$ and $\frac{2\cdot1\cdot1\cdot2}{9\cdot2}$. And if $a = 1$ the variance formula gives 0, as it should, since then there is always exactly one block.

Common slips

- Treating X_1 like the others: there is no position 0, so X_1 needs only one letter and $\mathrm{E}(X_1) = a/n$, not $ab/(n(n-1))$.
- Forgetting that X_i and X_{i+1} can't both be 1, or computing $\mathrm{E}(X_iX_{i+1})$ with a non-zero formula.
- Miscounting the pairs in (b): there are $\frac{1}{2}(n-2)(n-3)$, not $\frac{1}{2}n(n-1)$.

Beyond STEP

Writing a count as a sum of indicator variables, then getting the mean by linearity and the variance from pairwise products, is one of the most useful techniques in

undergraduate probability; it is how one analyses runs, matchings, fixed points of random permutations and random graphs. Here S is essentially the statistic of the Wald–Wolfowitz runs test, used to detect non-randomness in a sequence. Blitzstein and Hwang, *Introduction to Probability*, has many examples of the indicator method.

WORKED SOLUTION

STEP III 2013, Question 13

Probability and Statistics • Moments and Distribution Functions

STEP III 2013, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s3-q13

The key idea

The expression $2\mathrm{F}(x)\mathrm{f}(x)$ is the derivative of $\big(\mathrm{F}(x)\big)^2$, because $\mathrm{F}' = \mathrm{f}$. That makes part (i)(b) a single integration by parts, and with the right choice of g it turns every question about Y into a question about X. The bounds then come from replacing f or F by the bigger quantities M or Mx from part (i)(a), each time in the direction that makes the integral larger.

Solution

Throughout, $\mathrm{F}(x) = \int_0^x \mathrm{f}(t)\,\mathrm{d}t$, so $\mathrm{F}'(x) = \mathrm{f}(x)$, with $\mathrm{F}(0) = 0$ and $\mathrm{F}(1) = 1$ since X lies in $[0, 1]$.

Part (i)

(a) For $0 \le x \le 1$, since $0 \le \mathrm{f}(t) \le M$ for every t,

$$0 \le \int_0^x \mathrm{f}(t)\,\mathrm{d}t \le \int_0^x M\,\mathrm{d}t = Mx,$$

which is $0 \le \mathrm{F}(x) \le Mx$.

(b) By the chain rule, $\dfrac{\mathrm{d}}{\mathrm{d}x}\big(\mathrm{F}(x)\big)^2 = 2\mathrm{F}(x)\mathrm{f}(x)$. Integrating by parts with $u = \mathrm{g}(x)$ and $\mathrm{d}v = 2\mathrm{F}(x)\mathrm{f}(x)\,\mathrm{d}x$, so $v = \big(\mathrm{F}(x)\big)^2$,

$$\begin{aligned}\int_0^1 2\mathrm{g}(x)\mathrm{F}(x)\mathrm{f}(x)\,\mathrm{d}x &= \Big[\mathrm{g}(x)\big(\mathrm{F}(x)\big)^2\Big]_0^1 - \int_0^1 \mathrm{g}'(x)\big(\mathrm{F}(x)\big)^2\,\mathrm{d}x \\ &= \mathrm{g}(1) - \int_0^1 \mathrm{g}'(x)\big(\mathrm{F}(x)\big)^2\,\mathrm{d}x,\end{aligned}$$

since $\mathrm{F}(1) = 1$ and $\mathrm{F}(0) = 0$. (We need g differentiable, which the question implicitly assumes.)

Part (ii)

(a) The density of Y must integrate to 1. Taking $g(x) = 1$ in (i)(b), so that $g' = 0$,

$$\int_0^1 2F(y)f(y)\,dy = 1,$$

so $k = 2$. (Equally, $\int_0^1 kFf\,dy = \frac{k}{2}\left[F^2\right]_0^1 = \frac{k}{2}$.) Note also that $kFf \geq 0$, so this really is a density.

(b) We have

$$E(Y^n) = \int_0^1 2y^n\,F(y)f(y)\,dy.$$

Upper bound. By (i)(a), $F(y) \leq My$, and the other factors $2y^n f(y)$ are non-negative, so

$$E(Y^n) \leq \int_0^1 2y^n \cdot My \cdot f(y)\,dy = 2M\int_0^1 y^{n+1}f(y)\,dy = 2M\mu_{n+1}.$$

Lower bound. Now use (i)(b) with $g(y) = y^n$, so $g(1) = 1$ and $g'(y) = ny^{n-1}$:

$$E(Y^n) = 1 - \int_0^1 ny^{n-1}\big(F(y)\big)^2\,dy.$$

(When $n = 0$ the integral is simply 0.) To bound $E(Y^n)$ from below we need to bound this integral from *above*.

? **Ask yourself** Which of the two factors of $\big(F(y)\big)^2$ should we replace by My?

Just one of them. Since $0 \leq F(y) \leq My$, we have $\big(F(y)\big)^2 \leq My\,F(y)$, and $ny^{n-1} \geq 0$, so

$$\int_0^1 ny^{n-1}\big(F(y)\big)^2\,dy \leq nM\int_0^1 y^n F(y)\,dy.$$

Replacing both factors would leave $\int y^{n+1}\,dy$, which says nothing about X; keeping one F lets us integrate by parts once more and produce a moment of X. Integrating by parts with $u = F(y)$ and $dv = y^n\,dy$,

$$\int_0^1 y^n F(y)\,dy = \Big[\frac{y^{n+1}}{n+1}F(y)\Big]_0^1 - \int_0^1 \frac{y^{n+1}}{n+1}f(y)\,dy = \frac{1-\mu_{n+1}}{n+1}.$$

Putting this together,

$$E(Y^n) \geq 1 - \frac{nM}{n+1}(1-\mu_{n+1}) = 1 + \frac{nM}{n+1}\mu_{n+1} - \frac{nM}{n+1},$$

which, with the upper bound, is the required pair of inequalities.

(c) Combining the two ends of (b), for $n \geq 0$,

$$1 + \frac{nM}{n+1}\mu_{n+1} - \frac{nM}{n+1} \leq 2M\mu_{n+1},$$

and collecting the μ_{n+1} terms on the right,

$$1 - \frac{nM}{n+1} \leq \left(2M - \frac{nM}{n+1}\right)\mu_{n+1} = \frac{(n+2)M}{n+1}\,\mu_{n+1}.$$

The coefficient $\frac{(n+2)M}{n+1}$ is positive ($M > 0$, since f integrates to 1), so we may divide by it without reversing the inequality:

$$\mu_{n+1} \geq \frac{n+1}{(n+2)M} - \frac{n}{n+2}.$$

This holds for every $n \geq 0$. Writing $m = n+1$, so that $m \geq 1$, it becomes

$$\mu_m \geq \frac{m}{(m+1)M} - \frac{m-1}{m+1} \qquad (m \geq 1),$$

which is the required result (with m renamed n).

Checks

- Uniform X: $\mathrm{f} = 1$, $M = 1$, $\mu_n = \frac{1}{n+1}$. The bound in (c) is $\frac{n}{n+1} - \frac{n-1}{n+1} = \frac{1}{n+1}$, so it holds with equality. Here Y has density $2y$ and $\mathrm{E}(Y^n) = \frac{2}{n+2}$, and both bounds in (b) also equal $\frac{2}{n+2}$. This makes sense: every inequality we used, $\mathrm{F}(y) \leq My$, is an equality for the uniform distribution.
- $\mathrm{f}(x) = 3x^2$, $M = 3$, $n = 2$: $\mu_2 = \frac{3}{5}$ and the bound is $\frac{2}{9} - \frac{1}{3} < 0$, comfortably true. For this density, $\mathrm{E}(Y^2) = \frac{3}{4}$ and $\mu_3 = \frac{1}{2}$, so the bounds in (b) are $1 + 2 \cdot \frac{1}{2} - 2 = 0$ and $2 \cdot 3 \cdot \frac{1}{2} = 3$, and $\frac{3}{4}$ lies between them.
- The identity in (i)(b) and all the inequalities were also confirmed by numerical integration for several densities and several choices of g.

Common slips

- Bounding $\left(\mathrm{F}\right)^2$ by M^2y^2: that is true but leads nowhere, since it loses all information about X.
- Forgetting that $\mathrm{E}(Y^n)$ appears with a minus sign in front of the integral, so a lower bound for $\mathrm{E}(Y^n)$ needs an *upper* bound for the integral.
- Stopping at the bound for μ_{n+1} without shifting the index to get the stated form for $n \geq 1$.

Beyond STEP

The density $2\mathrm{F}(y)\mathrm{f}(y)$ is no accident: it is the density of $\max(X_1, X_2)$ for two independent copies of X, since $\mathrm{P}(\max \leq y) = \big(\mathrm{F}(y)\big)^2$. More generally, the rth smallest of n independent observations has a density built from powers of F and $1 - \mathrm{F}$; these *order statistics* are central to non-parametric statistics. Blitzstein and Hwang, *Introduction to Probability*, has a clear treatment of order statistics.

Beyond STEP

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